

AIRFIELD PAVEMENT CONDITION INDEX SURVEY REPORT



MAXWELL AIR FORCE BASE, AL March 2019

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**June 2020
Final Report**

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Contract Number GS00Q14OADU104

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PREFACE

This report documents a March 2019 assessment of the pavement condition of the airfield pavements network at Maxwell Air Force Base (AFB), AL performed by Applied Research Associates, Inc. (ARA). This report provides information for:

- Planning and programming for pavement maintenance, repairs, and rehabilitation
- Prioritization of pavement facilities for maintenance and rehabilitation (M&R) and construction projects
- Allocation of funds for annual M&R
- The validation of the Real Property pavement inventory

This report meets the requirements for pavement condition inspection established in Air Force Instruction (AFI) 32-1041, “Air Force Pavement Evaluation Program.”

The Air Force Civil Engineer Center (AFCEC) provided funding for this airfield pavements network Pavement Condition Index (PCI) survey. Mrs. Giselle Concepcion, Headquarters Air Force Civil Engineer Center (HQ AFCEC)/COAT, was the technical contact. The prime contractor was Applied Research Associates, Inc., under Contract No. GS00Q14OADU104.

Frank C. Manning, AETC 42 CES/CEY, provided valuable support and information for the survey.

The field survey team consisted of Grady Holly, Victor Parker, Paul Rohrer, and Joshua Turner. Geographic Information System (GIS) drawings were provided by Grady Holley. Victor Parker, Paul Rohrer, and Joshua Turner prepared this report.

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EXECUTIVE SUMMARY

A pavement condition survey for the airfield network at Maxwell AFB, AL, was performed by ARA in March 2019, and a PAVER database for the airfield pavements was updated. The goals of the assessment were to:

- Determine the PCI, between zero (failed) and 100 (no distresses), of both Portland cement concrete (PCC) and asphalt concrete (AC) airfield pavements
- Update a PAVER pavement management system (PMS) to identify M&R requirements and conduct cost planning
- Validate the Real Property pavement inventory

The objectives of the pavement condition survey were to:

- Acquire pertinent data to determine the current condition of the airfield pavements
- Provide the Base Civil Engineer (BCE) with the PAVER database and associated PCI Survey Report for use in making pavement management decisions
- Develop 10-year work plans for M&R cost planning for the airfield pavements

PAVEMENT CONDITION INDEX

PAVER pavement management software was used to perform PCI calculations and analyses. Table ES.1 summarizes the condition survey results by branch use, and Figure ES.1 graphically compares the 2015 and 2019 PCI survey results by branch use. Figure ES.2 is the 7-color PCI by section map of the airfield network.

Table ES.1. Current Condition Summary.

Branch Use	Area-Weighted Average		Area-Weighted Average		Condition Category	Pavement		
	2015 PCI	2015 Age	2019 PCI	2019 Age		Area (ft²)	Area (%)	Sections
Apron	85	25	86	25	Good	2,533,470	45	15
					Fair	84,955	2	2
					Poor	36,205	1	1
Overrun	89	8	78	6	Good	378,463	7	6
					Fair	--	--	--
					Poor	--	--	--
Runway	85	20	91	15	Good	1,429,389	25	18
					Fair	--	--	--
					Poor	--	--	--
Taxiway	93	20	86	20	Good	1,109,882	20	17
					Fair	50,117	<1	1
					Poor	--	--	--
Overall Airfield	87	18	87	20	Good	5,622,481	100	60
<i>Shoulder</i>	--	--	--	--	Fair	1,714,359	--	6

SIGNIFICANT ISSUES FOR BASE PAVEMENTS

Joint-seal damage was present throughout the PCC pavements and requires replacement in many areas. Linear cracking and spalling were also prevalent. The majority of the AC pavements were in Good condition with the exception of Runway 009/189 Turnaround (T06A). Longitudinal and Transverse (L&T) cracking has formed along paving lanes throughout. Localized crack sealing is recommended to mitigate deterioration.

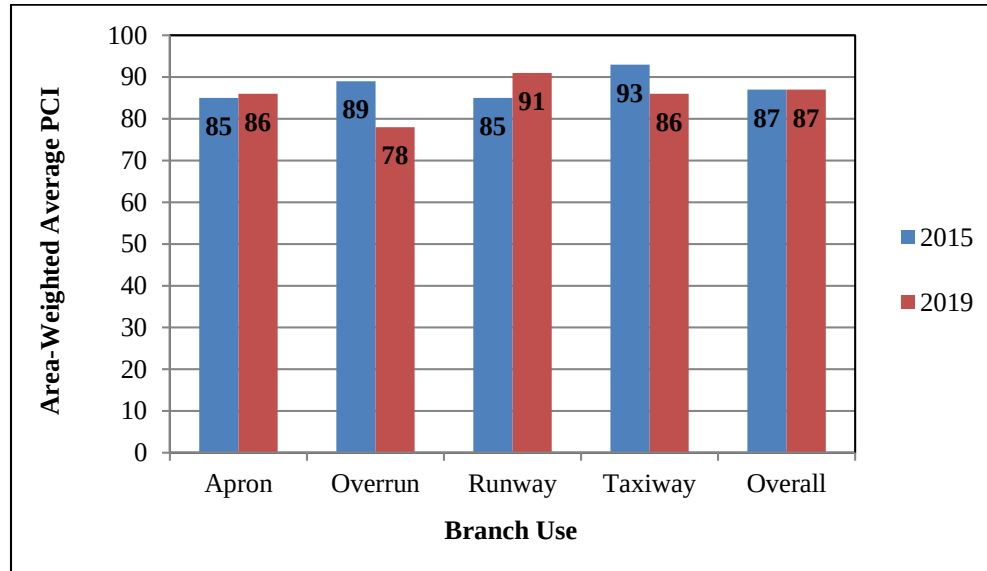


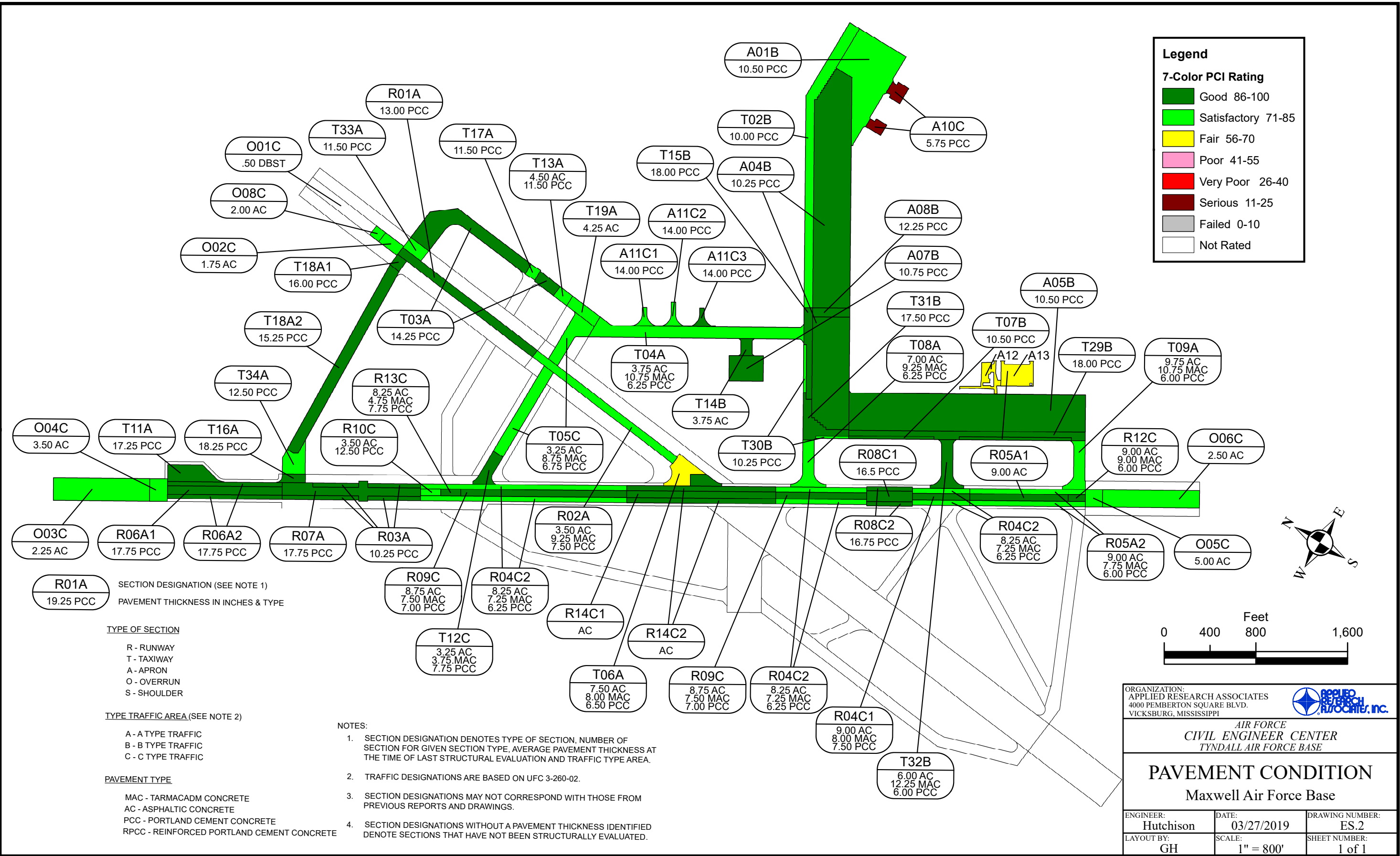
Figure ES.1. Comparison of Area-Weighted Average PCI for 2015 and 2019.

RECOMMENDED BUDGET SCENARIO

All work plans were assumed to start at the beginning of fiscal year (FY) 2020 and extend through FY 2029. The backlog elimination budget was chosen as the optimum budget because it repaired all pavements below their critical PCI value and left no backlog. This budget costs \$7.6 million over the ten-year plan. Table ES.2 shows the yearly costs of the recommended budget. A full project list is included in Chapter 4. Closed pavements and shoulders were excluded from the budget scenario.

Table ES.2. PAVER Recommended Budget Summary.

Plan FY	M&R Policy Type				Total	Backlog After Plan Year	PCI	
	Stopgap	Preventive	Global	Major			Before	After
2020	\$10,927	\$534,558	\$0	\$72,889	\$618,374	\$965,377	86	87
2021	\$0	\$552,298	\$0	\$965,377	\$1,517,675	\$0	86	87
2022	\$0	\$572,147	\$0	\$0	\$572,147	\$0	86	86
2023	\$0	\$594,665	\$0	\$0	\$594,665	\$0	85	85
2024	\$0	\$599,067	\$67,555	\$130,740	\$797,362	\$0	84	86
2025	\$0	\$607,807	\$86,607	\$0	\$694,413	\$0	85	85
2026	\$0	\$622,609	\$8,683	\$0	\$631,291	\$0	84	85
2027	\$0	\$646,062	\$0	\$0	\$646,062	\$0	84	84
2028	\$0	\$662,872	\$9,145	\$146,688	\$818,704	\$0	83	84
2029	\$0	\$687,176	\$1,806	\$14,292	\$703,273	\$0	83	83
Total	\$10,927	\$6,079,261	\$173,795	\$1,329,985	\$7,593,967		86	83



REFERENCES

Below is a list of the references used during the composition of this report.

- ¹ *2019 RSMeans Heavy Construction Cost Data*. Kingston, MA, 2019.
- ² *Airfield Pavement Evaluation, Maxwell Air Force Base, AL*, January 2016, Air Force Civil Engineer Center, Tyndall Air Force Base, Florida.
- ³ American Society for Testing and Materials [ASTM] D 5340-12. *Standard Test Method for Airfield Pavement Condition Index Surveys*. West Conshohocken, PA, 2015.
- ⁴ Engineering Technical Letter (ETL) 04-9. *Pavement Engineering Assessment (EA) Standards*, April 2004.
- ⁵ Maxwell AFB, <https://www.maxwell.af.mil/About-Us/Fact-Sheets/Display/Article/1004960/maxwell-air-force-base/>, June 10, 2020
- ⁶ Shahin, M.Y. *Pavement Management For Airports, Roads, and Parking Lots*. New York, NY, 2005.
- ⁷ United Facility Criteria (UFC) 3-260-02. *Pavement Design for Airfields*, June 2001.
- ⁸ United Facility Criteria (UFC) 3-260-03. *Airfield Pavement Evaluation*, April 2001.
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- ¹⁰ United Facility Criteria (UFC) 3-270-05. *O&M: PAVER Concrete Surfaced Airfields Pavement Condition Index (PCI)*, March 2001.
- ¹¹ United Facility Criteria (UFC) 3-270-06. *O&M: PAVER Asphalt Surfaced Airfields Pavement Condition Index (PCI)*, March 2001.
- ¹² United Facility Criteria (UFC) 3-270-08. *Pavement Maintenance Management*, January 2004.

¹³ United Facility Criteria (UFC) 3-701-01. *DoD Facilities Pricing Guide For FY 2011*, March 2011.

¹⁴ US Air Force Instruction 32-1041. *Pavement Evaluation Program*. February 2017.

¹⁵ US Army Corps of Engineers. *Asphalt Surfaced Airfields-PAVER Distress Identification Manual*. June 2009.

¹⁶ US Army Corps of Engineers. *Concrete Surfaced Airfields-PAVER Distress Identification Manual*. June 2009.

DISTRIBUTION LIST

Table ES.3 is the distribution list for Dyess AFB.

Table ES.3. Distribution List.

Agency	Draft Report	CD	Final Report
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Installation Contact 2 Attn: Mr. Don Brown Vectrus Lead Engineer 400 Cannon Street, Bldg 1060 Maxwell AFB, AL 36112-5978 Phone: (334) 306-6562	1	1	1
Contracting Officer Attn: Mr. Ian Batchelor 772 ESS/PKH 139 Barnes Drive, Suite 1, Bldg 1120 Tyndall AFB, FL 32403-5319 Phone (850) 283-6570	NA	1	NA
Contract Administrator Attn: Ms. Ann Galloway 772 ESS/PKH 139 Barnes Drive, Suite 1, Bldg 1120 Tyndall AFB, FL 32403-5319 Phone (850) 283-4278	NA	1	NA
Technical Contact and COR: Attn: Mrs. Giselle Concepcion HQ AFCEC/COAP 139 Barnes Drive, Suite 1, Bldg 1120 Tyndall AFB, FL 32403-5319 Phone (850) 283-6783	1	1	1

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE AND SCOPE

This report describes the condition of the airfield pavements as determined by the March 2019 PCI survey. A subsequent update of a PAVER-based PMS for the airfield is presented in this report. The scope of this assessment included all airfield pavements in current use. The PAVER PMS provides AFCEC and other organizations with a means to identify and prioritize pavement M&R requirements. The overall goals of this survey were to determine the PCI of both rigid and flexible pavements, to update the existing PAVER PMS, to identify M&R requirements and estimate the cost of recommended M&R, and to validate the Real Property pavement inventory.

The following tasks support the project goals:

- Interview personnel, and review as-built drawings, construction records, and aerial photography.
- Define the pavement network and divide it into branches and sections conforming to AFCEC structural evaluation reports.
- Conduct a condition survey of the airfield pavements in accordance with "Standard Test Methods for Airport Pavement Condition Index Surveys," American Society for Testing and Materials (ASTM) D5340-12 and UFC 3-260-16FA.
- Enter condition survey information into PAVER 7.0 (32-byte).
- Compute PCI values for each pavement section.
- Develop ten-year M&R work plans for the pavement network using PAVER.
- Prepare a CADD map of the airfield branches, sections, and sample units.
- Prepare a geographic information system (GIS) dataset of the airfield sections.
- Report the findings of the study to installation pavement engineer, the BCE, and AFCEC.

1.2 DESCRIPTION OF TASKS PERFORMED

1.2.1 PCI Survey

The PCI survey is a visual inspection of pavements to determine their present surface condition. The condition survey consists of inspecting the pavement surfaces for specific types of distresses, determining the severity level of each distress, and measuring the quantity of each distress. These distresses are combined to determine the PCI value of the pavement, a number between zero (failed) and 100 (no distresses) that reflects the current condition. ASTM D5340-12 defines distress types and severities, and defines survey methods and calculations to determine the PCI.

1.2.2 Validation of Real Property Pavement Inventory

During the field survey, meticulous care is taken to document all airfield pavements at the installation. Real Property attributes such as the Facility ID, RPUID, and CATCODE are populated in the PAVER database. This enables a detailed inventory of the pavement to be

developed and compared against the Real Property records after the conclusion of the field survey. Inconsistencies between the Real Property records, AFI 32-1041, and the SOW are identified in Appendix B.

1.2.3 CADD/GIS Mapping

Prior to the field investigation, a pavement network map was developed using available aerial photography and the common installation plan (CIP) provided by base personnel. The map was divided into branches, sections, and sample units. It is provided in AutoDesk drawing (.dwg) format. As surveyors inspect the pavement, they are mindful to ensure that the pre-survey airfield map depicts the actual pavement, otherwise known as a “ground-truth” survey. Noticeable differences between what was present at the installation and what was displayed on the maps provided by the base were adjusted by a CADD technician. Small systematic errors in the digitization of the installation by base personnel may not have been detectable by inspectors and therefore not amended in the maps provided by ARA. The maps are geo-referenced to be compatible with existing datasets such as GeoBase. A section-level map that is linked to PAVER is provided. This map is also provided in ESRI Shapefile format and is suitable for linking to an existing geodatabase.

1.2.4 PAVER Implementation

The network definition, construction history, Real Property information, and data from the condition survey were entered into PAVER.

After the data were entered and the PCI determined, family curves were developed to model the change in pavement condition over time. These curves are used by PAVER to estimate future pavement condition. Several curves are typically developed, with separate curves defined for different combinations of pavement usage, e.g., apron, overrun, runway, or taxiway; surface type, e.g., AC or PCC; and pavement rank, e.g., primary, secondary, or tertiary. Engineering judgment is used to determine the number and types of deterioration curves that are appropriate for a particular installation.

The database "Maxwell_Air_2019," which contains the construction history, survey data, deterioration curves, M&R policies, and budgets, is provided in PAVER 7.0.9 (32 byte) format on the accompanying CD-ROM. Specific PAVER operating details are located in "The PAVER User's Guide" under the “Help” tab.

1.2.5 PAVER Work Plan Development

After the data were entered and the deterioration curves were created, the following work plan analyses were performed using standardized Air Force M&R policies and costs:

- Cost per year to eliminate the Major M&R backlog over 10 years
- Cost per year to maintain the current area-weighted average PCI for 10 years
- Cost for M&R to only deal with flight safety issues over 10 years

These analyses were then used to determine the M&R needs and optimum funding levels over a ten-year analysis period. PAVER determines Major M&R needs by using the concept of “Critical PCI.” The critical PCI is the PCI value at which Major M&R is to be considered for the section. Primary airfield pavements are assigned a critical PCI of 70, while secondary and tertiary pavements are assigned a value of 55.

1.2.6 Written Report

The following report includes background information on Maxwell AFB, PCI results and recommendations, and M&R budget scenarios. In this report, Appendix A contains the airfield maps. Appendix B contains the real property summary, PAVER/RPAD comparison, and pavement inventory. Appendix C contains PCI deterioration rate summary. Appendix D is reserved. In Appendix E contains a work history report. Appendix F contains PAVER reports. Appendix G, H, I contain respectively the FOD Potential, Structural Index, and Friction Index values. Appendix J contains the Engineering Assessment (EA). Appendix K contains activity management and preventative maintenance plan. Appendix L contains the budget details for the work plans. Appendix M contains the family models for condition deterioration prediction.

1.2.7 Photographs

Photographs depicting typical pavement condition and distress types observed at Maxwell AFB are included throughout Chapter 2 of this report. Global Positioning System (GPS) coordinates of each photograph are recorded in the lower right corner of the photograph. Photograph 1.1 is a sample image of the GPS tagged photographs. Additional photographs may be found on the CD-ROM attached to this report.



Photograph 1.1. GPS Coordinates on Photographs.

1.3 BACKGROUND INFORMATION

1.3.1 Location

Maxwell AFB is located in Montgomery County, AL. The city has a population of over 199,000. Maxwell AFB is situated in the east gulf coastal plain of south central AL approximately 92 miles southeast of Birmingham, AL.

1.3.2 Design and Construction History

Occupying the original site of the first Wright brothers civilian flying school, Maxwell AFB was named Maxwell Field in 1922 in honor of Second Lieutenant William C. Maxwell. Lieutenant Maxwell lost his life in the Philippines in 1920 while avoiding a group of playing children as he attempted to land. Maxwell AFB is hosted by the 42nd Air Base Wing and is home

to Air University. The 908th Airlift Wing of the Air Force Reserve Command is a tenant unit and the only operational flying unit at Maxwell AFB.

Recent construction projects include the repair of AC Runway 15/33 and Assault Strip 009/189 in 2012, the mill and overlay of Runway 15/33 and 009/189 in 2013, and the mill and overlay of Taxiways A, B, C, D, and E in 2014.

Every effort was made to collect the most complete possible construction history for the airfield network at Maxwell AFB. First, the base point of contact (POC) was requested to provide the latest construction dates from electronic records (GeoBase, Automated Civil Engineering System “ACES”, Real Property records, etc.). During the field survey, inspectors noted recent construction, Localized M&R, and Global M&R. As a last measure, time-lapse aerial photography was used to estimate the last construction dates for some areas of pavement.

1.3.3 Current Traffic Data

The primary aircraft operating at Maxwell AFB include the C-130H, T1A, and various light aircraft via the Aero Club.

1.3.4 Previous Reports

Airfield Pavement Evaluation, Maxwell AFB, Alabama, January 2016, AFCEC, Tyndall AFB, Florida

1.4 SYNOPSIS OF MAJOR PROJECTS SINCE LAST EVALUATION

Major M&R projects since the June 2015 inspection are listed below in Table 1.1 and can also be found in the complete Work History Report in Appendix E.

Table 1.1. Major M&R Since Previous Inspection.

Branch Name	Branch ID	Section ID	Major M&R	Year
Runway 15/33	RW15/33	R14C1	Mill and Overlay	2016
Runway 15/33	RW15/33	R12A	Mill and Overlay	2016
Runway 15/33	RW15/33	R14C2	Mill and Overlay	2016
Runway 15/33	RW15/33	R13C	Mill and Overlay	2016
Runway 15/33	RW15/33	R04C1	Mill and Overlay	2016
Runway 15/33	RW15/33	R05A1	Mill and Overlay	2016
Runway 15/33	RW15/33	R09C	Mill and Overlay	2016
Runway 009/189 Turn Around	TW009/189T	T35A	New Construction - AC	2016

CHAPTER 2

PCI RESULTS AND RECOMMENDATIONS

2.1 PCI SURVEY PROCEDURE

The PCI survey was performed in accordance with ASTM and Air Force guidelines. The following paragraphs outline the procedures used to evaluate the pavement, the implementation of a PMS, and information about the life cycle of pavement.

2.1.1 PCI Survey Process

The pavement network to be surveyed was defined and divided into branches, sections, and sample units. A branch is defined as a complete unit of the airfield that is used for a particular type of operation. Branches are typically named pieces of pavement, e.g., Runway 15/33, Taxiway C, or West Apron. After branches are defined, they are divided into sections based on pavement type, structure, and usage. Branch and section identifications used in this report are in general agreement with and correspond to feature designations in the January 2016 AFCEC structural evaluation report.

Sections are divided into sample units as prescribed by ASTM D5340-12. A sample unit is a subdivision of a section used exclusively to aid in the inspection process and reduce the effort needed to determine distress quantities and the PCI. The specified sample unit size for an AC pavement is 5,000 ft² ± 2,000 ft². Sample units on PCC pavements contain 20 ± 8 slabs. A small number of sample units did not meet the size criteria due to odd-size sections.

Inspecting all of the sample units within a section may consume considerable resources; therefore, ARA uses a statistical sampling technique to determine the number of sample units to be inspected. Sample units are inspected in accordance with ASTM D5340-12 to provide a 95% confidence level, with an acceptable error of ± 5 PCI points. Figure 2.1 presents an example of the sampling method employed by ARA. Sample units with grey shading represent sample units to be surveyed. A pavement surveyor visually inspects the required sample units of each section. Distresses are identified, counted or measured, and recorded.

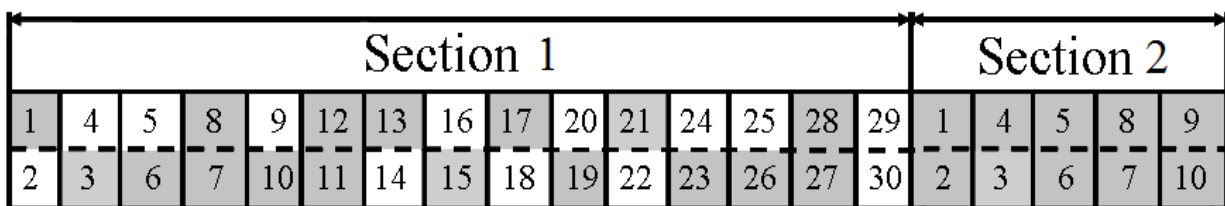


Figure 2.1. Systematic Random Selection of Sample Units to Survey.

2.1.2 PAVER Implementation and Report Products

A PMS is an organized method of data management combined with predictive techniques to assist decision makers in finding optimum strategies for providing and maintaining pavements in a serviceable condition over a given period of time. The function of a PMS is to improve the efficiency of decision-making, provide feedback on the consequences of decisions, facilitate the coordination of activities, and ensure the consistency of decisions.

PAVER is a management system that has the capability of

- being updated and/or modified as new information and better models become available,
- considering alternative strategies,
- identifying the optimum alternative,
- basing decisions on a rational procedure with quantified attributes, criteria, and constraints,
- making predictions of pavement deterioration,
- selecting appropriate maintenance and repair methods to maximize performance, and
- using feedback information regarding the consequences of decisions.

PAVER is a menu-driven computer program that provides the user a practical approach for identifying cost-effective maintenance strategies. It includes such features as data storage and retrieval, pavement network identification, PCI rating, determination of needs for maintenance and repair, performance of economic analyses, and budget planning.

The PAVER database storage capability includes historical construction data, pavement structure and materials information, condition ratings (stores previous and current condition data), and non-destructive testing results (e.g., falling weight deflectometer deflection data). Data from the current survey has been input into PAVER 7.0 and provided on the accompanying CD-ROM.

After the conclusion of the pavement assessment, the PCI distress data were input in the PAVER database. PAVER was used to perform PCI calculations, develop pavement deterioration curves and generate pavement work plans.

2.1.3 Life Cycle Graph

Figure 2.2 shows a typical pavement life cycle curve, including the effects of M&R. As the PCI of pavement decreases, applicable M&R treatments become more involved and therefore more costly. This concept is important because pavements often remain in serviceable condition for a long period of time, followed by a period of rapid deterioration.

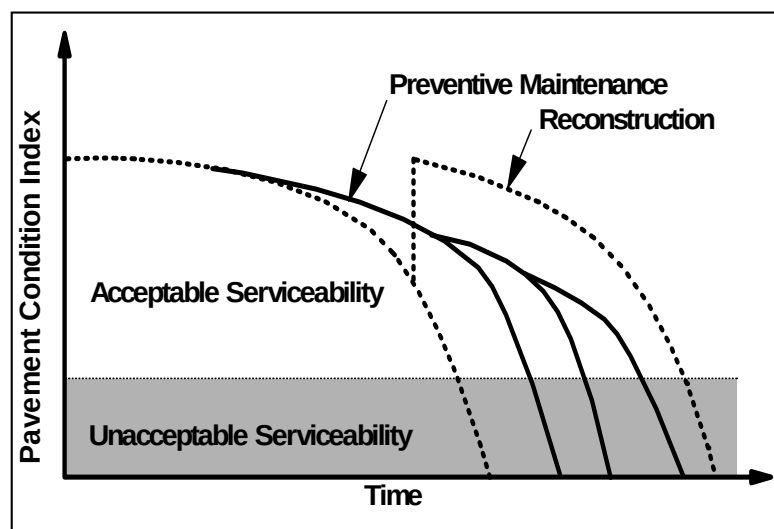


Figure 2.2. Pavement Life Cycle Showing the Effects of M&R.

2.2 OVERVIEW OF PCI SURVEY RESULTS

A total of 5,622,481 ft² of pavement was surveyed as part of this assessment. The airfield pavements consisted of 63.7% PCC and 36.3% AC. The area-weighted average PCI for all pavements surveyed was 87-Good. A total of 56 out of 60 surveyed sections had a PCI greater than 70. Of the four pavement sections below 70, two sections of pavement are at or below their respective critical PCI value. These two pavement sections are spread throughout two branches over the airfield and are in need of Major M&R. The airfield had 1,714,359 ft² of paved shoulders that were generally considered to be in Fair condition. Appendix A provides the airfield network layouts showing the pavement sections and color-coded results of the survey. Table 2.1 provides a definition of PCI rating and pavement distribution.

Table 2.1. Definition of PCI Ratings and Pavement Distribution.

Simplified PCI Rating	PCI Rating	PCI Range	Definition	Pavement Area (ft ²)	Pavement Area (%)
Good	Good	86-100	GOOD: Pavement has minor or no distresses and will require only routine maintenance.	3,503,138	62.3
	Satisfactory	71-85	SATISFACTORY: Pavement has scattered low-severity distresses that need only routine maintenance.	1,948,066	34.7
Fair	Fair	56-70	FAIR: Pavement has a combination of generally low- and medium-severity distresses. M&R needs are routine to major in the near future.	135,072	2.4
Poor	Poor	41-55	POOR: Pavement has low-, medium-, and high-severity distresses that probably cause some operational problems. Near-term maintenance and repair needs may range from routine up to a requirement for reconstruction.	--	--
Poor < 40	Very Poor	26-40	VERY POOR: Pavement has predominantly medium- and high-severity distresses that cause considerable maintenance and operational problems. Near-time maintenance and repair needs will be intensive in nature.	--	--
	Serious	11-25	SERIOUS: Pavement has mainly high-severity distresses that cause operational restrictions; immediate repairs are needed.	36,205	0.6
	Failed	0-10	FAILED: Pavement deterioration has progressed to the point that safe operations are no longer possible; complete reconstruction is required.	--	--

2.2.1 PCI Results by Branch Use and Pavement Surface Type

Table 2.2 summarizes the PCI survey results by branch use and pavement surface type. Figure 2.3 is a pie-chart illustrating the current condition of the pavement by surface type and Figure 2.4 shows a PCI distribution graph comparing the 2015 and 2019 surveys by pavement surface type.

Table 2.2. Current Condition Summary.

Branch Use	Surface Type	Pavement Area (ft ²)	Pavement Area (%)	Area-Weighted Average PCI
Apron	AC	100,523	1.8	69
	PCC	2,554,107	45.4	87

(Page 1 of 2)

Table 2.2. Current Condition Summary (concluded).

Branch Use	Surface Type	Pavement Area (ft ²)	Pavement Area (%)	Area-Weighted Average PCI
Overrun	AC	378,463	6.7	79
	PCC	--	--	--
Runway	AC	92,001	1.6	97
	APC	825,551	14.7	87
	PCC	511,837	9.1	95
Taxiway	AC	30,173	0.5	79
	APC	611,797	10.9	82
	PCC	518,029	9.2	90
Overall	AC	601,160	10.7	79
	APC	1,437,348	25.6	85
	PCC	3,583,973	63.7	89
	ALL	5,622,481	100.0	87
(Page 2 of 2)				

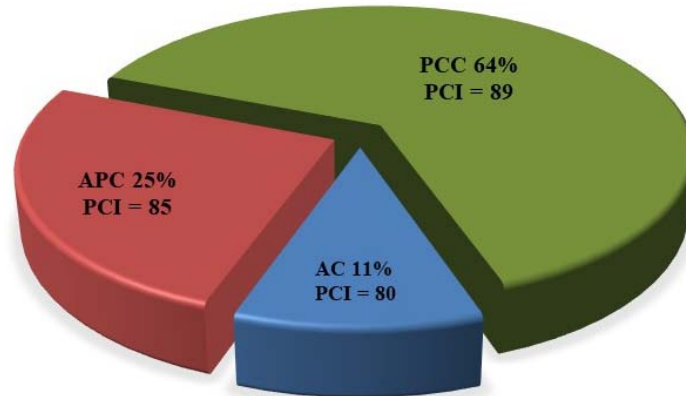


Figure 2.3. Current Condition by Surface Type.

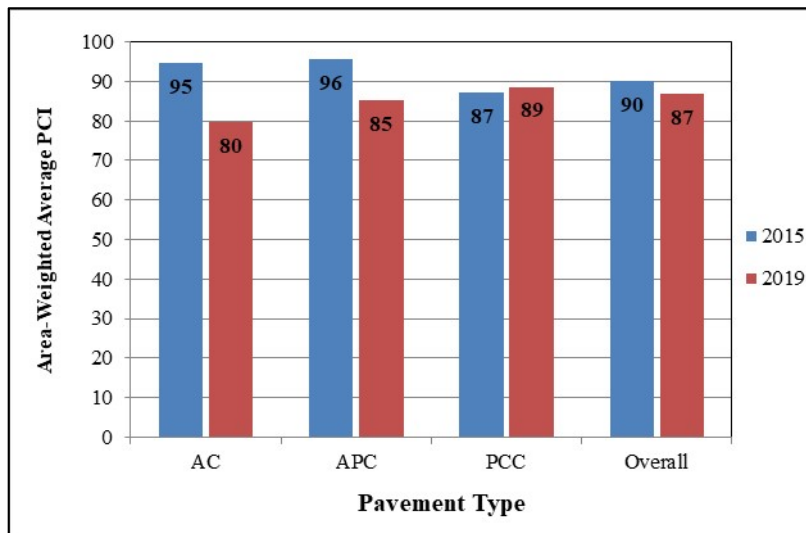


Figure 2.4. PCI by Pavement Surface Type for 2015 and 2019 PCI Surveys.

2.2.2 Pavement Issues

The predominant AC pavement distresses were weathering and L&T cracking. Weathering and L&T cracking are both weather-related distresses. The majority of the AC pavement was in Good condition, with the exceptions of the Assault Strip Turnaround (T06A) and the AGE Storage Lot (A12 and A13).

Joint seal damage, linear cracks, joint spalls and corner spalls were the predominant distresses in the PCC pavements. Joint seal damage was the significant weather-related distresses. Linear cracking was the significant load related distress. The majority of the PCC pavement was in Good condition, with the exception of the Aero Club Apron (A10C).

The primary AC pavements at Maxwell AFB are deteriorating at an average rate of 0.69 PCI points/year. The secondary AC pavements are deteriorating at a rate of 2.85 PCI points/year. The tertiary AC pavements are deteriorating at a rate of 2.93 PCI points/year. The primary PCC pavements are deteriorating at a rate of 0.33 PCI points/year. The secondary and tertiary PCC pavements are deteriorating at a rate of 0.97 PCI points/year. A more detailed analysis showing plots and containing further explanations may be found in Appendix M.

2.3 SURVEY RESULTS

In the following paragraphs, the pavements at Maxwell AFB are discussed by branch and further broken into sections in some instances. The recommended M&R actions for each branch and section may differ from the PAVER recommended actions based on engineering judgment. These recommendations are based on the PCI and pavement type as well as the type, severity, and quantity of the distresses present. A surface treatment is recommended for AC pavements with medium- to high-severity weathering in order to slow further surface deterioration. Any localized maintenance should be conducted prior to the application of surface treatments.

2.3.1 Aprons

The aprons at Maxwell AFB consisted of 6 branches containing 18 sections of PCC and AC pavement. The total area of apron pavements was 2,654,630 ft² and had an area-weighted average PCI of 86-Good. These branches will be discussed more in the following paragraphs. The most common PCC distresses were joint seal damage, linear cracks, large patches, and small patches. The most common AC distresses were L&T cracking, patching and weathering. The distribution of pavement area and sections by PCI range are shown in Table 2.3.

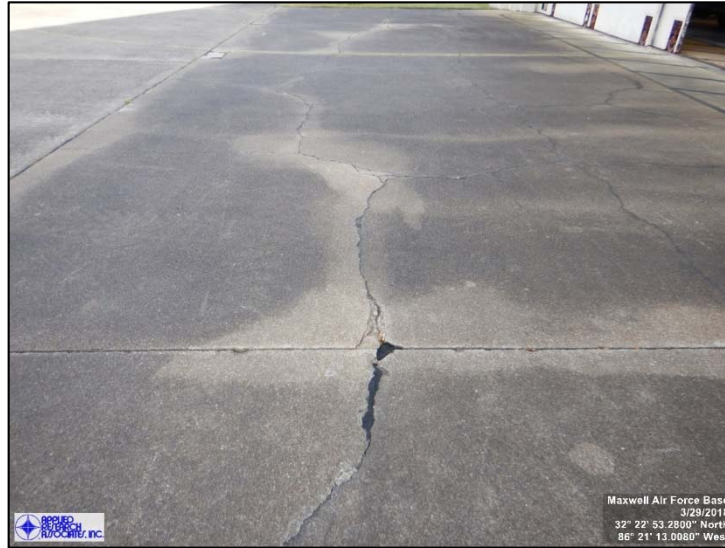
Table 2.3. Apron Pavement Area and Sections by PCI Range.

PCI Range	Rating	Number of Sections	Pavement Area (ft ²)	Pavement Area (%)
86-100	Good	10	2,006,963	75.6
71-85		5	526,507	19.8
56-70	Fair	2	84,955	3.2
0-55	Poor	1	36,205	1.4
Total	Good	18	2,654,630	100.0

2.3.1.1 Aero Club Apron

Aero Club Apron consisted of one section of PCC pavement. Section A10C had an area of 36,205 ft² and a PCI of 14-Poor. Distresses with the highest deduct were medium-severity shattered slabs, low-severity shattered slabs and medium-severity linear cracking. No stopgap maintenance is recommended at this time. A complete reconstruction is recommended in FY

2021. Photograph 2.1 below shows a medium-severity shattered slab in Section A10C of Aero Club Apron.



Photograph 2.1. Aero Club Apron, Section A10C, Medium-Severity Shattered Slab.

2.3.1.2 AGE Storage Lot

AGE Storage Lot consisted of two sections of AC pavement. The area-weighted average PCI of the 84,955-ft² branch was 66-Fair.

- AC Section A12 had a PCI of 65-Fair and an area of 30,392 ft². Distresses with the highest deduct were medium-severity weathering, medium-severity L&T cracking and medium-severity patching. Localized preventive maintenance actions include crack sealing and deep patching. A mill and overlay is recommended in FY 2024.
- AC Section A13 had a PCI of 66-Fair and an area of 54,563 ft². Distresses with the highest deduct were medium-severity L&T cracking and medium-severity weathering. Crack sealing is the only preventive maintenance recommended. A mill and overlay is recommended in FY 2024. Photograph 2.2 on the following page shows medium-severity L&T cracking in Section A13 of the AGE Storage Lot.



Photograph 2.2. AGE Storage Lot, Section A13, Medium-Severity L&T Cracking.

2.3.1.3 ***Engine Run-up Apron***

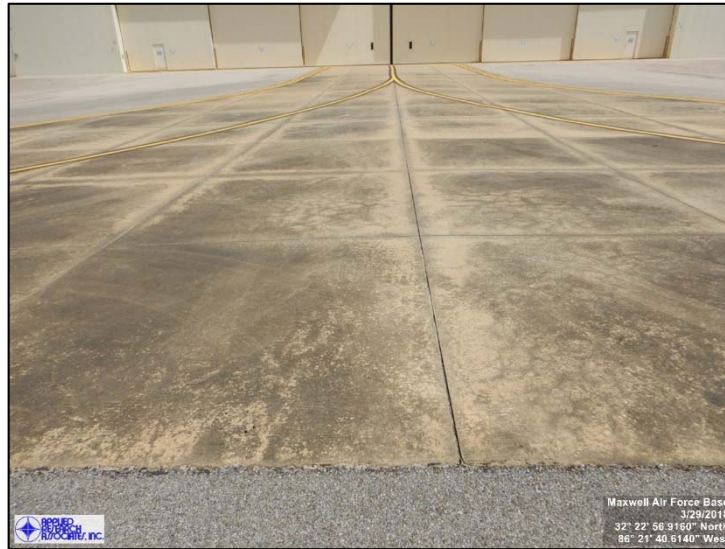
Engine Run-up Apron consisted of one PCC section and one AC section. The area-weighted average PCI of the 82,601-ft² branch was 91-Good.

- PCC Section A07B had a PCI of 91-Good and an area of 67,033 ft². Distresses with the highest deduct were medium-and high-severity joint seal damage, and low-severity joint spalls. Localized preventive maintenance actions include crack sealing and joint seal replacement.
- AC Section T14B had a PCI of 90-Good and an area of 15,568 ft². Distresses with the highest deduct low-severity L&T cracking and low-severity weathering. No maintenance is recommended at this time.

2.3.1.4 ***Maintenance Hangar Access Apron***

Maintenance Hangar Access Apron consisted of three sections of PCC pavement. The area-weighted average PCI of the 35,995-ft² branch was 85-Good.

- PCC Section A11C1 had a PCI of 83-Good and an area of 11,253 ft². Distresses with the highest deduction include high-severity joint seal damage, medium-severity joint seal damage, and medium-severity linear cracks. Shattered slabs were not observed during the 2019 inspection. Localized preventive maintenance actions include crack sealing, joint seal replacement, and partial-depth patching.
- PCC Section A11C2 had a PCI of 79-Good and an area of 13,344 ft². Distresses with the highest deduct include low-severity linear cracks, medium-severity joint seal damage, and low-severity corner spalls. Localized preventive maintenance actions include crack sealing, joint seal replacement, and partial-depth patching.
- PCC Section A11C3 had a PCI of 93-Good and an area of 11,398 ft². The only distress was medium-severity joint seal damage. High-severity joint seal damage was not observed in the 2019 inspection. Joint seal replacement is the only preventive maintenance recommended at this time. Photograph 2.3 below shows an overview of section A11C3 of the Maintenance Hangar Access Apron.



Photograph 2.3. Maintenance Hangar Access Apron, Section A11C3, Overview.

2.3.1.5 *North Apron*

North Apron consisted of seven sections of PCC pavement. The area-weighted average PCI of the 1,589,398-ft² branch was 87-Good.

- PCC Section A01B had a PCI of 84-Good and an area of 274,393 ft². Distresses with the highest deduction include medium-severity joint seal damage, medium-severity linear cracks, and low-severity large patches. Localized preventive maintenance actions include crack sealing, joint seal replacement, and partial-depth patching. Photograph 2.4 below shows a medium-severity linear crack in Section A01B of the North Apron.



Photograph 2.4. North Apron, Section A01B, Medium-severity Linear Crack.

- PCC Section A04B had a PCI of 88-Good and an area of 957,323 ft². Distresses with the highest deduct include medium-severity joint seal damage, low-severity large patches, and low-severity linear cracks. Corner breaks, scaling and shrinkage cracks were not

observed during the 2019 inspection. Localized preventive maintenance actions include crack sealing, grinding, joint seal replacement, and partial-depth patching.

- PCC Section A08B had a PCI of 99-Good and an area of 26,000 ft². The only two distresses were low-severity small patches and shrinkage cracks. No maintenance is recommended at this time.
- PCC Section T02B had a PCI of 85-Good and an area of 211,267 ft². Distresses with the highest deduct include high-severity joint seal damage, medium-severity joint seal damage, and medium-severity linear cracks. There was a decrease in the quantity of corner breaks in section T02B since the previous inspection. Localized preventive maintenance actions include crack sealing, grinding, joint seal replacement, and partial-depth patching. Photograph 2.5 below shows high-severity joint seal damage in Section T02B of the North Apron.



Photograph 2.5. North Apron, Section T02B, High-severity Joint Seal Damage.

- PCC Section T15B had a PCI of 93-Good and an area of 48,000 ft². Distresses with the highest deduct include medium-severity joint seal damage, low-severity joint seal damage, and low-severity joint spalls. Localized preventive maintenance actions include crack sealing and joint seal replacement.
- PCC Section T30B had a PCI of 81-Good and an area of 16,250 ft². Distresses with the highest deduct include medium-severity linear cracks, medium-severity joint seal damage, and low-severity large patches. Localized preventive maintenance actions include crack sealing and joint seal replacement.
- PCC Section T31B had a PCI of 94-Good and an area of 56,165 ft². Distresses with the highest deduct include medium-severity joint seal damage, low-severity joint seal damage, and shrinkage cracks. Joint seal replacement is the only preventive maintenance recommended at this time.

2.3.1.6 West Apron

West Apron consisted of three sections of PCC pavement. The area-weighted average PCI of the 825,476-ft² branch was 89-Good.

- PCC Section A05B had a PCI of 89-Good and an area of 670,734 ft². Distresses with the highest deduction include medium-severity joint seal damage, low-severity large patches,

and medium-severity large patches. Faulting, pumping and scaling were not observed in the 2019 inspection. Localized preventive maintenance actions include crack sealing, joint seal replacement, full-depth patching, and partial-depth patching. Photograph 2.6 below shows a medium-severity large patch in Section A05B of the West Apron.



Photograph 2.6. West Apron, Section A05B, Medium-severity Large Patch.

- PCC Section T07B had a PCI of 88-Good and an area of 43,093 ft². Distresses with the highest deduct include medium-severity joint seal damage, medium-severity linear cracks, and low-severity large patches. Localized preventive maintenance actions include crack sealing, joint seal replacement, and partial-depth patching.
- PCC Section T29B had a PCI of 90-Good and an area of 111,649 ft². Distresses with the highest deduct include medium-severity joint seal damage, shrinkage cracks, and low-severity joint seal damage. Localized preventive maintenance actions include crack sealing and joint seal replacement.

2.3.2 Overruns

The overruns at Maxwell AFB consisted of two branches containing 6 sections of AC pavement. The total area of overrun pavements was 378,463 ft² and had an area-weighted average PCI of 78-Good. These branches will be discussed more in the following paragraphs. The most common AC distresses were L&T cracking and weathering. The distribution of pavement area and sections by PCI range are shown in Table 2.4.

Table 2.4. Overrun Pavement Area and Sections by PCI Range.

PCI Range	Rating	Number of Sections	Pavement Area (ft ²)	Pavement Area (%)
86-100	Good	--	--	--
71-85		6	378,463	100.0
56-70	Fair	--	--	--
0-55	Poor	--	--	--
Total	Good	6	378,463	100.0

2.3.2.1 *Overrun Runway 009/189*

Overrun Runway 009/189 consisted of two sections of AC pavement. This branch contained 29,925 ft² of pavement and had an area-weighted average PCI of 80-Good.

- AC Section O02C had a PCI of 79-Good and an area of 19,810 ft². Distresses with the highest deduction include low-severity L&T cracking, weathering, and swelling. Section O02C has experienced a higher than average deterioration rate since the last inspection due to an increase in quantity of L&T cracking and the development of weathering. A surface treatment is recommended in FY 2024. Photograph 2.7 below shows an overview of Section O02C of the Overrun Runway 009/189.



Photograph 2.7. Overrun Runway 009/189, Section O02C, Overview.

- AC Section O08C had a PCI of 83-Good and an area of 10,115 ft². Distresses with the highest deduction include low-severity weathering and low-severity L&T cracking. Section O08C has experienced a higher than average deterioration rate since the last inspection due to development of these distresses. A surface treatment is recommended in FY 2024.

2.3.2.2 *Overrun Runway 15/33*

Overrun Runway 15/33 consisted of four sections of AC pavement. This branch contained 348,538 ft² of pavement and had an area-weighted average PCI of 78-Good.

- Section O03C had a PCI of 79-Good and an area of 168,000 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section O03C has experienced a higher than average deterioration rate since the last inspection due to the development of these distresses. A surface treatment is recommended in FY 2024.
- Section O04C had a PCI of 79-Good and an area of 31,100 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section O04C has experienced a higher than average deterioration rate since the last inspection due to the development of these distresses. A surface treatment is recommended in FY 2024.

- Section O05C had a PCI of 78-Good and an area of 22,508 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section O05C has experienced a higher than average deterioration rate since the last inspection due to the development of these distresses. A surface treatment is recommended in FY 2024. Photograph 2.8 below shows an overview of Section O05C of the Overrun Runway 15/33.



Photograph 2.8. Overrun Runway 15/33, Section O05C, Overview.

- Section O06C had a PCI of 77-Good and an area of 126,930 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section O06C has experienced a higher than average deterioration rate since the last inspection due to the development of these distresses. A surface treatment is recommended in FY 2024.

2.3.3 Runways

The runways at Maxwell AFB consisted of two branches containing 18 sections of PCC and AC pavement. The total area of runway pavements was 1,429,389 ft² and had an area-weighted average PCI of 91-Good. These branches will be discussed more in the following paragraphs. The most common PCC distresses were joint spalls, linear cracks, small patches and large patches. The most common AC distresses were L&T cracking and weathering. The distribution of pavement area and sections by PCI range are shown in Table 2.5.

Table 2.5. Runway Pavement Area and Sections by PCI Range.

PCI Range	Rating	Number of Sections	Pavement Area (ft ²)	Pavement Area (%)
86-100	Good	14	942,321	65.9
71-85		4	487,068	34.1
56-70	Fair	--	--	--
0-55	Poor	--	--	--
Total	Good	18	1,429,389	100.0

2.3.3.1 Runway 009/189 (Assault Strip)

Runway 009/189 (Assault Strip) consisted of one section of PCC pavement and one section of AC pavement. The area-weighted average PCI of this branch was 84-Good. The total area of the branch was 218,434 ft².

- PCC Section R01A had a PCI of 93-Good and an area of 110,851 ft². Distresses with the highest deduction include medium-severity large patches, low-severity large patches, and medium-severity faulting. Medium-and high-severity joint seal damage were not observed in the 2019 inspection. Localized preventive maintenance actions include crack sealing, grinding, and partial-depth patching. Photograph 2.9 below shows a medium-severity large patch in Section R01A of Runway 009/189.



Photograph 2.9. Runway 009/189, Section R01A, Medium-severity Large Patch.

- AC Section R02A had a PCI of 75-Good and an area of 107,583 ft². Distresses with the highest deduction include low-severity L&T cracking, medium-severity L&T cracking, and low-severity weathering. Crack sealing is the only preventive maintenance recommended at this time. A mill and overlay is recommended in FY 2028. Photograph 2.10 on the following page shows an overview of Section R02A of the Runway 009/189.

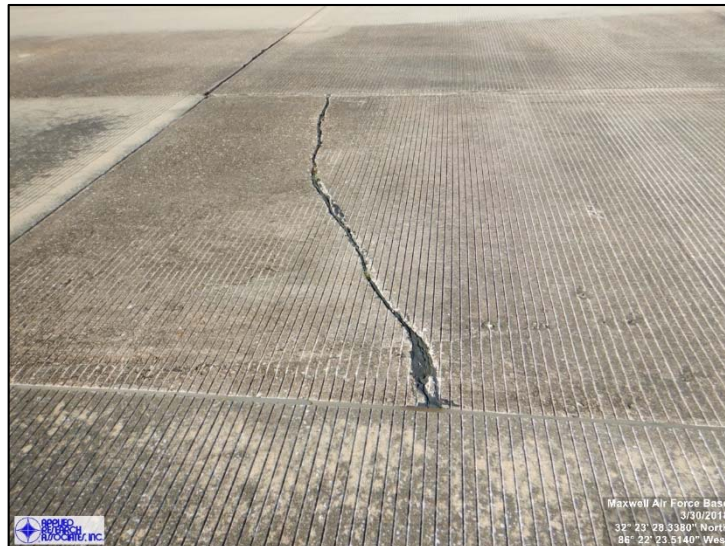


Photograph 2.10. Runway 009/189, Section R02A, Overview.

2.3.3.2 Runway 15/33

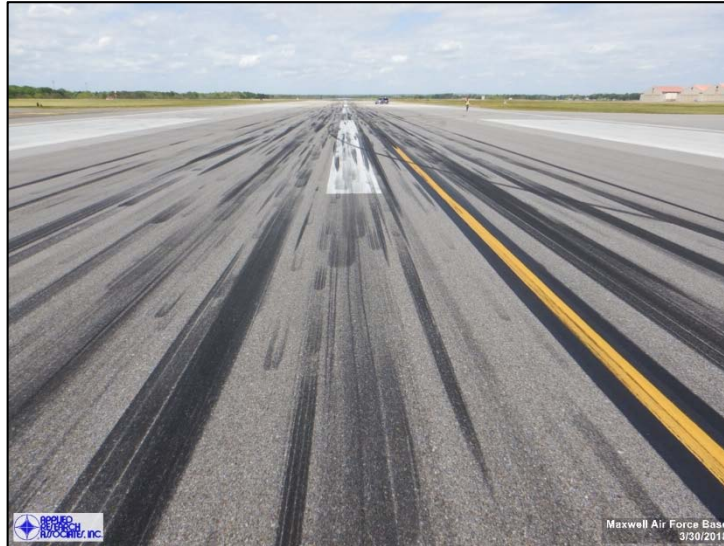
Runway 15/33 consisted of 6 sections of PCC pavement and 10 sections of AC pavement. The area-weighted average PCI of this branch was 92-Good. The total area of the branch was 1,210,955 ft².

- PCC Section R03A had a PCI of 86-Good and an area of 74,530 ft². Distresses with the highest deduction include medium-severity linear cracks, low-severity linear cracks and low-severity corner breaks. Localized preventive maintenance actions include crack sealing, full-depth patching and partial-depth patching. Photograph 2.11 below shows a medium-severity linear crack in Section R03A of Runway 15/33.



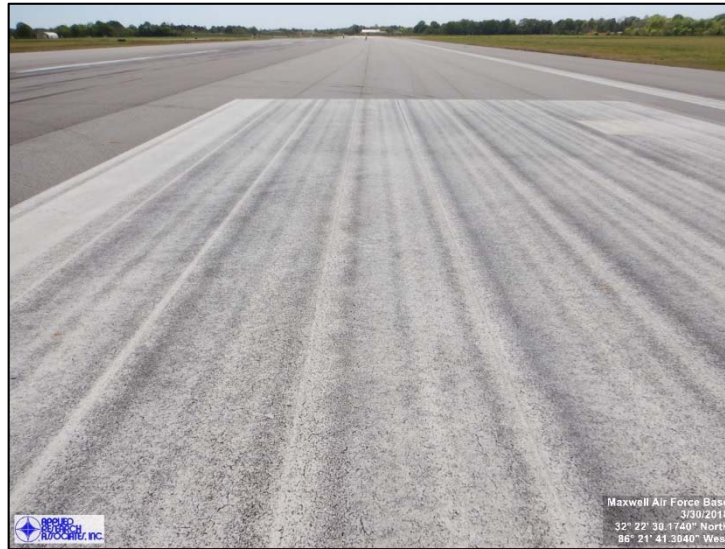
Photograph 2.11. Runway 15/33, Section R03A, Medium-severity Linear Crack.

- AC Section R04C1 had a PCI of 99-Good and an area of 30,000 ft². The only distress was low-severity L&T cracking. Section R04C1 has received Major M&R since the previous inspection. No further maintenance is recommended at this time. Photograph 2.12 below shows an overview of Section R04C1 of Runway 15/33.



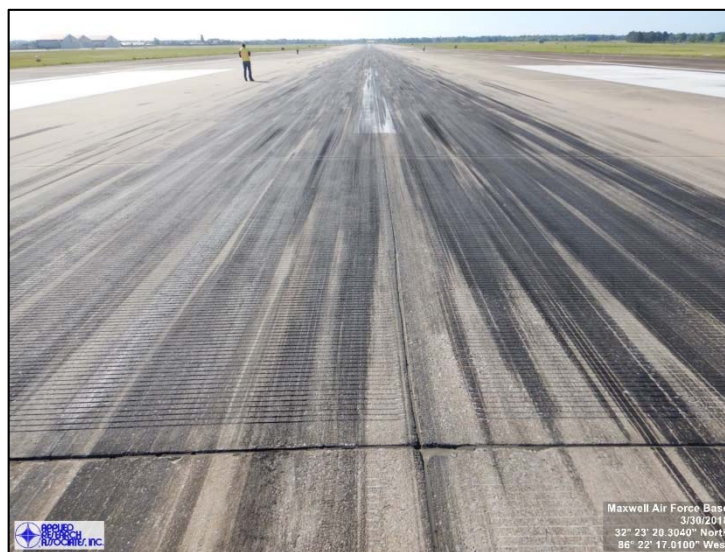
Photograph 2.12. Runway 15/33, Section R04C1, Overview.

- AC Section R04C2 had a PCI of 84-Good and an area of 278,222 ft². Distresses with the highest deduction include low-severity L&T cracking, weathering and patching. No maintenance is recommended at this time.
- AC Section R05A1 had a PCI of 100-Good and an area of 51,501 ft². There were no distresses. Section R05A1 has received Major M&R since the previous inspection. No further maintenance is recommended at this time.
- AC Section R05A2 had a PCI of 80-Good and an area of 90,763 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section R05A2 has experienced a higher than average deterioration rate since the previous inspection due to the development of low-severity L&T cracking. No maintenance is recommended at this time. Photograph 2.13 on the following page shows an overview of Section R05A2 of Runway 15/33.



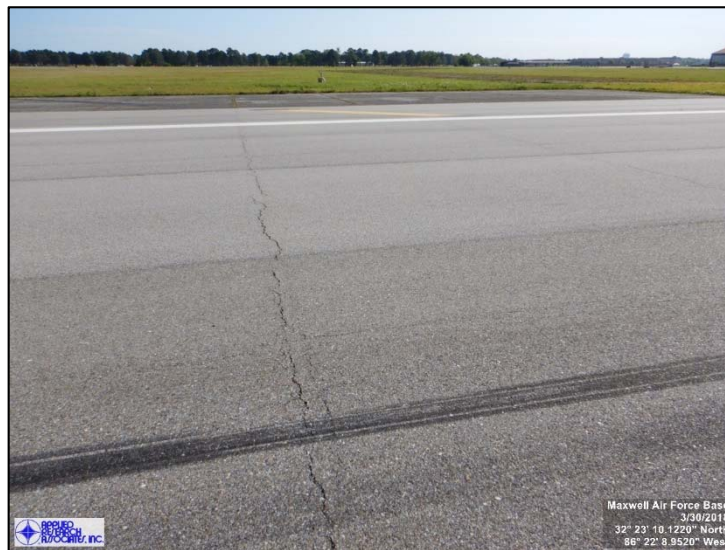
Photograph 2.13. Runway 15/33, Section R05A2, Overview.

- PCC Section R06A1 had a PCI of 99-Good and an area of 75,062 ft². The only distresses were low-severity joint spalls and low-severity joint seal damage. Corner spalls, large patches, shrinkage cracks, and small patches were not observed during the 2019 inspection. Crack sealing is the only preventive maintenance recommended at this time.
- PCC Section R06A2 had a PCI of 97-Good and an area of 75,062 ft². Distresses with the highest deduction include medium-severity large patches, low-severity joint seal damage and low-severity joint spalls. Corner spalls were not recorded during the 2019 inspection. Localized preventive maintenance actions include crack sealing and partial-depth patching.
- PCC Section R07A had a PCI of 99-Good and an area of 108,832 ft². Low-severity joint spalls were the only distress. Corner spalls and joint seal damage were not recorded in the 2019 inspection. Crack sealing is the only preventive maintenance recommended at this time. Photograph 2.14 below shows an overview of Section R07A of Runway 15/33.



Photograph 2.14. Runway 15/33, Section R07A, Overview.

- PCC Section R08C1 had a PCI of 93-Good and an area of 37,500 ft². Distresses with the highest deduction include low-severity linear cracks, shrinkage cracks and medium-severity joint spalls. Localized preventive maintenance actions include crack sealing and partial-depth patching.
- PCC Section R08C2 had a PCI of 98-Good and an area of 30,000 ft². The only distresses were low-severity large patches and low-severity joint spalls. Crack sealing is the only preventive maintenance recommended at this time.
- AC Section R09C had a PCI of 90-Good and an area of 134,482 ft². The only distresses were low-severity weathering and low-severity L&T cracking. Section R09C has received Major M&R since the previous inspection. No further maintenance is recommended at this time.
- AC Section R10C had a PCI of 76-Good and an area of 10,500 ft². Distresses with the highest deduction include low-severity L&T cracking, weathering and joint reflection cracking. Section R10C has experienced a higher than average deterioration rate since construction due to the development of low-severity L&T cracking and weathering. No preventive maintenance is recommended at this time. A mill and overlay is recommended in FY 2029. Photograph 2.15 below shows low-severity L&T cracking in Section R10C of Runway 15/33.



Photograph 2.15. Runway 15/33, Section R10C, Low-severity L&T cracking.

- AC Section R12A had a PCI of 100-Good and an area of 9,003 ft². There were no distresses. Section R12A has received Major M&R since the previous inspection. No further maintenance is recommended at this time.
- AC Section R13C had a PCI of 92-Good and an area of 10,500 ft². The only distresses were low-severity weathering and low-severity L&T cracking. R13C has received Major M&R since the previous inspection. No further maintenance is recommended at this time.
- AC Section R14C1 had a PCI of 100-Good and an area of 78,000 ft². There were no distresses recorded. R14C1 has received Major M&R since the previous inspection. No further maintenance is recommended at this time.
- AC Section R14C2 had a PCI of 100-Good and an area of 116,998 ft². There were no distresses recorded. R14C2 has received Major M&R since the previous inspection. No further maintenance is recommended at this time.

2.3.4 Taxiways

The taxiways at Maxwell AFB consisted of 9 branches containing 18 sections of PCC and AC pavement. The total area of taxiway pavements was 1,159,999 ft² and had an area-weighted average PCI of 86-Good. These branches will be discussed more in the following paragraphs. The most common PCC distresses were joint seal damage, linear cracks, shrinkage cracks and joint spalls. The most common AC distresses were L&T cracking, block cracking and weathering. The distribution of pavement area and sections by PCI range are shown in Table 2.6.

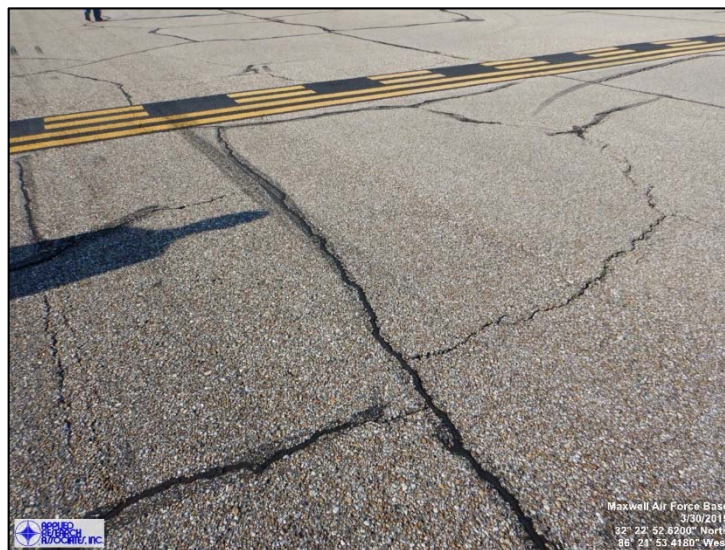
Table 2.6. Taxiway Pavement Area and Sections by PCI Range.

PCI Range	Rating	Number of Sections	Pavement Area (ft ²)	Pavement Area (%)
86-100	Good	8	553,854	47.7
71-85		9	556,028	47.9
56-70	Fair	1	50,117	4.3
0-55	Poor	--	--	--
Total	Good	18	1,159,999	100.0

2.3.4.1 Runway 009/189 Turn Around

Runway 009/189 Turn Around consisted of two sections of AC pavement with an area-weighted average PCI of 70-Fair. The total area of the branch was 72,468 ft².

- AC Section T06A had a PCI of 59-Fair and an area of 50,117 ft². Distresses with the highest deduction include low-severity block cracking, medium-severity weathering, and low-severity weathering. No local maintenance is recommended at this time. A mill and overlay is recommended in FY 2020. Photograph 2.16 below shows low-severity block cracking in Section T06A of the Runway 009/189 Turn Around.



Photograph 2.16. Runway 009/189 Turn Around, Section T06A, Low-severity Block Cracking.

- AC Section T35A had a PCI of 96-Good and an area of 22,351 ft². The only distresses recorded were low-severity patching and low-severity L&T cracking. No local maintenance is recommended at this time.

2.3.4.2 Runway 15/33 Turn Around

Runway 15/33 Turn Around consisted of one section of PCC pavement. Section T11A had an area of 57,425 ft² and a PCI of 99-Good. The only distresses recorded were low-severity joint spalls and shrinkage cracks. Joint seal replacement has occurred since the previous inspection. Crack sealing is the only preventive maintenance recommended at this time.

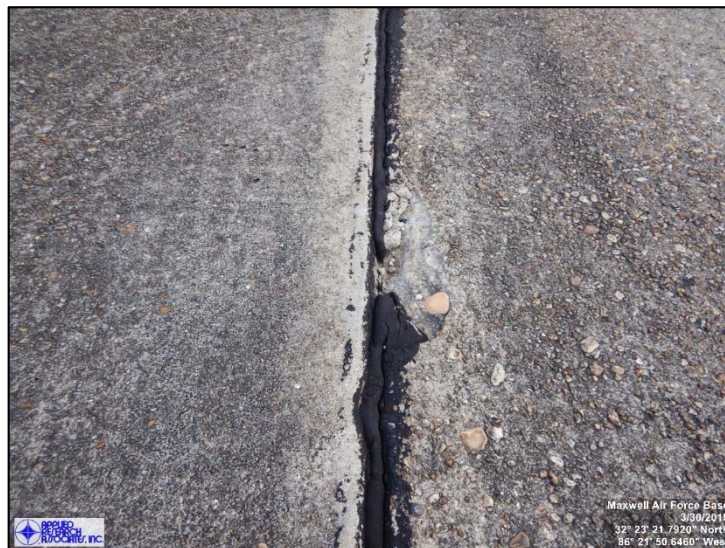
2.3.4.3 Taxiway A

Taxiway A consisted of one section of AC pavement. Section T04A had an area of 191,559 ft² and a PCI of 82-Good. The only distresses recorded were low-severity L&T cracking and low-severity weathering. Section T04A has experienced a higher than average deterioration rate since the last inspection due to these distresses. No local maintenance is recommended at this time. A surface treatment is recommended in FY 2025.

2.3.4.4 Taxiway A North

Taxiway A North consisted of four sections of PCC pavement. The area-weighted average PCI of the branch was 89-Good. The total area was 251,647 ft².

- PCC Section T16A had a PCI of 95-Good and an area of 16,046 ft². The only distresses recorded were low-severity linear cracks, shrinkage cracks and low-severity corner spalls. Joint seal replacement has taken place since the previous inspection. Crack sealing is the only preventive maintenance recommended.
- PCC Section T18A1 had a PCI of 87-Good and an area of 11,215 ft². Distresses with the highest deduction include medium-severity joint seal damage, low-severity linear cracks and medium-severity joint spalls. Large patches and high-severity joint seal damage were not observed in the 2019 inspection. Localized preventive maintenance actions include crack sealing, joint seal replacement and partial-depth patching. Photograph 2.17 below shows a medium-severity joint spall in Section T18A of Taxiway A North.



Photograph 2.17. Taxiway A North, Section T18A, Medium-severity Joint Spalls.

- PCC Section T18A2 had a PCI of 90-Good and an area of 183,486 ft². Distresses with the highest deduction include high-severity joint seal damage, medium-severity joint seal

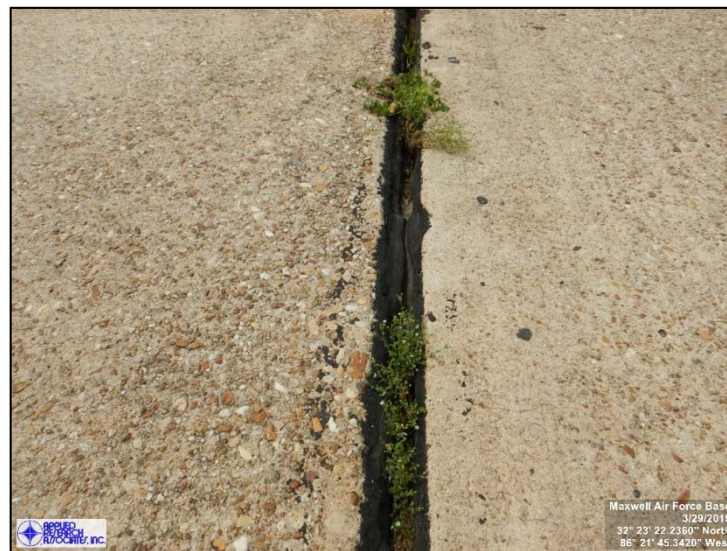
damage, and low-severity linear cracks. Localized preventive maintenance actions include crack sealing and joint seal replacement.

- PCC Section T34A had a PCI of 85-Good and an area of 40,900 ft². Distresses with the highest deduction include medium-severity joint seal damage, low-severity large patches, and low-severity linear cracks. Durability cracking, faulting, joint spalls and shrinkage cracks were not observed during the 2019 inspection. Localized preventive maintenance actions include crack sealing and joint seal replacement.

2.3.4.5 Taxiway A Northwest

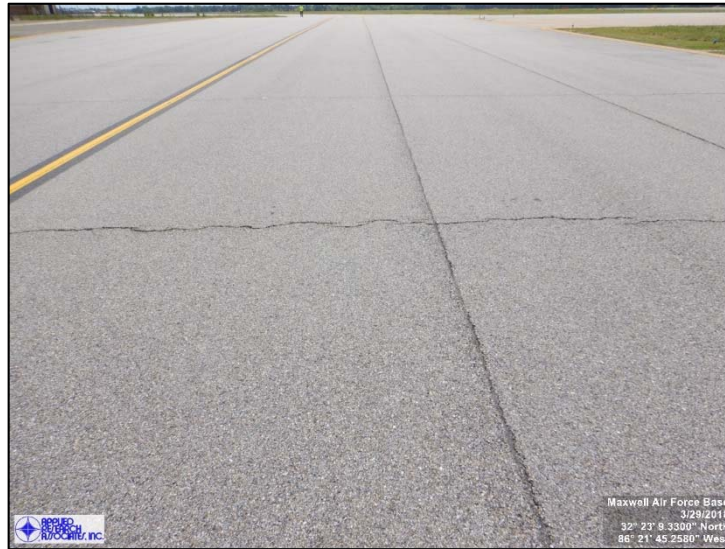
Taxiway A Northwest consisted of 3 sections of PCC pavement and two sections of AC pavement. The area-weighted average PCI of this branch was 87-Good. The total area of the branch was 253,257 ft².

- PCC Section T03A had a PCI of 89-Good and an area of 176,681 ft². Distresses with the highest deduction include high-severity joint seal damage, low-severity blow-ups and medium-severity linear cracks. Localized preventive maintenance actions include crack sealing, joint seal replacement, full-depth patching and partial depth patching. Photograph 2.18 below shows high-severity joint seal damage in Section T03A of Taxiway A Northwest.



Photograph 2.18. Taxiway A Northwest, Section T03A, High-severity Joint Seal Damage.

- AC Section T13A had a PCI of 81-Good and an area of 14,127 ft². Distresses with the highest deduction include low-severity joint reflection cracking and low-severity weathering. Section T13A has experienced a higher than average deterioration rate since the previous inspection due to the development of these distresses. No preventive maintenance is recommended at this time. A surface treatment is recommended in FY 2025. Photograph 2.19 on the following page shows an overview of section T13A of Taxiway A Northwest.



Photograph 2.19. Taxiway A Northwest, Section T13A, Overview.

- PCC Section T17A had a PCI of 82-Good and an area of 10,000 ft². Distresses with the highest deduction include medium-severity large patches, low-severity corner breaks, and shrinkage cracks. Localized preventive maintenance actions include crack sealing and partial-depth patching.
- AC Section T19A had a PCI of 79-Good and an area of 30,173 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section T19A has experienced a higher than average deterioration rate since the last inspection due to the development of these distresses. No preventive maintenance is recommended at this time. A surface treatment is recommended in FY 2025.
- PCC Section T33A had a PCI of 85-Good and an area of 22,276 ft². Distresses with the highest deduction include high-severity joint seal damage, medium-severity joint seal damage and low-severity linear cracks. Section T33A has experienced a higher than average deterioration rate since the previous inspection due to the development of high-severity joint seal damage and an increase in the quantity of low-severity linear cracks. Localized preventive maintenance actions include crack sealing, joint seal replacement and partial-depth patching. Photograph 2.20 on the following page shows a medium-severity linear crack in Section T33A of Taxiway A Northwest.



Photograph 2.20. Taxiway A Northwest, Section T33A, Medium-severity Linear Crack.

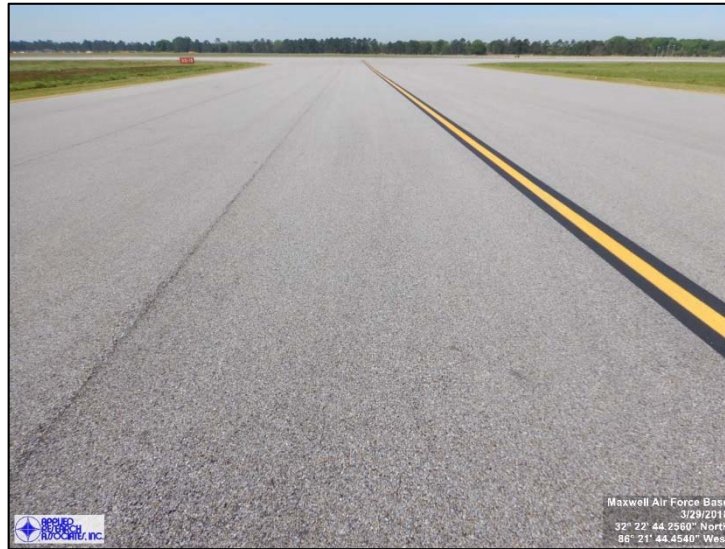
2.3.4.6 *Taxiway B*

Taxiway B consisted of two sections of AC pavement. This branch contained 180,724 ft² of pavement and had an area-weighted average PCI of 84-Good.

- AC Section T05C had a PCI of 83-Good and an area of 147,120 ft². Distresses with the highest deduction include low-severity L&T cracking and low-severity weathering. Section T05C has experienced a higher than average deterioration rate since the last inspection due to the development of these distresses. No Localized preventive maintenance actions are recommended at this time. A surface treatment is recommended in FY 2025.
- AC Section T12C had a PCI of 89-Good and an area of 33,604 ft². Distresses with the highest deduction include low-severity L&T cracking, weathering and patching. No preventive maintenance is recommended at this time. A surface treatment is recommended in FY 2025.

2.3.4.7 *Taxiway C*

Taxiway C consisted of one section of AC pavement. Section T08A had an area of 51,231 ft² and a PCI of 85-Good. Distresses with the highest deduct include low-severity L&T cracking and low-severity weathering. Section T08A has experienced a higher than average deterioration rate since the previous inspection due to the development of these distresses. No preventive maintenance is recommended at this time. A surface treatment is recommended in FY 2027. Photograph 2.21 on the following page shows an overview of Section T08A of Taxiway C.



Photograph 2.21. Taxiway C, Section T08A, Overview.

2.3.4.8 Taxiway D

Taxiway D consisted of one section of AC pavement. Section T32B had an area of 53,046 ft² and a PCI of 89-Good. Distresses with the highest deduct include low-severity L&T cracking and low-severity weathering. No preventive maintenance is recommended at this time. A surface treatment is recommended in FY 2025.

2.3.4.9 Taxiway E

Taxiway E consisted of one section of AC pavement. Section T09A had an area of 48,642 ft² and a PCI of 84-Good. Distresses with the highest deduct include low-severity L&T cracking and low-severity weathering. Section T09A has experienced a higher than average deterioration rate since the last inspection due to these distresses. No preventive maintenance is recommended at this time. A surface treatment is recommended in FY 2026.

2.3.5 Projected PCI

The projected PCI for the next ten years is shown in Table 2.7. PCI values were predicted using the deterioration rates found in Appendix M. These deterioration rates assume that no work is performed. The first future year is calculated at the beginning of the next fiscal year. The PCI is calculated as shown below:

$$PCI_{\text{Future Year}} = PCI_{\text{Current}} - (\text{Det. Rate}) * (\text{Time since last evaluation})$$

$$PCI_{\text{Future Year}} = \text{the PCI at the beginning of each fiscal year}$$

$$PCI_{\text{Current}} = \text{the current PCI at the time of evaluation}$$

$$\text{Det. Rate} = \text{the specific deterioration rate for a section of a given rank and surface type}$$

$$\text{Time since last evaluation} = \text{time between last evaluation and the beginning of each fiscal year}$$

Table 2.7. Ten-year Projected PCI.

Branch Name	Section ID	Pavement Surface Type	Pavement Rank	Current PCI	PCI Deterioration Rate	FY 2020 PCI Before M&R	FY 2021 PCI Before M&R	FY 2022 PCI Before M&R	FY 2023 PCI Before M&R	FY 2024 PCI Before M&R	FY 2025 PCI Before M&R	FY 2026 PCI Before M&R	FY 2027 PCI Before M&R	FY 2028 PCI Before M&R	FY 2029 PCI Before M&R
Aprons															
Aero Club Apron	A10C	PCC	S	14	0.97	14	13	12	11	10	9	8	7	6	5
AGE Storage Lot	A12	AC	S	65	2.93	63	61	58	55	52	49	46	43	40	37
AGE Storage Lot	A13	AC	S	66	2.93	64	62	59	56	53	50	47	44	41	38
Engine Run-up Apron	A07B	PCC	S	91	0.97	91	90	89	88	87	86	85	84	83	82
Engine Run-up Apron	T14B	AC	S	90	2.85	89	86	83	80	77	74	71	69	66	63
Maintenance Hangar Access Apron	A11C1	PCC	S	83	0.97	83	82	81	80	79	78	77	76	75	74
Maintenance Hangar Access Apron	A11C2	PCC	S	79	0.97	79	78	77	76	75	74	73	72	71	70
Maintenance Hangar Access Apron	A11C3	PCC	S	93	0.97	93	92	91	90	89	88	87	86	85	84
North Apron	A01B	PCC	S	84	0.97	84	83	82	81	80	79	78	77	76	75
North Apron	A04B	PCC	P	88	0.33	88	88	87	87	87	86	86	86	85	85
North Apron	A08B	PCC	P	99	0.33	99	99	98	98	98	97	97	97	96	96
North Apron	T02B	PCC	P	85	0.33	85	85	84	84	84	83	83	83	82	82
North Apron	T15B	PCC	P	93	0.33	93	93	92	92	92	91	91	91	90	90
North Apron	T30B	PCC	P	81	0.33	81	81	80	80	80	79	79	79	78	78
North Apron	T31B	PCC	P	94	0.33	94	94	93	93	93	92	92	92	91	91
West Apron	A05B	PCC	P	89	0.33	89	89	88	88	88	87	87	87	86	86
West Apron	T07B	PCC	P	88	0.33	88	88	87	87	87	86	86	86	85	85
West Apron	T29B	PCC	P	90	0.33	90	90	89	89	89	88	88	88	87	87
Overruns															
Overrun Runway 009/189	O02C	AAC	T	79	2.93	77	75	72	69	66	63	60	57	54	51
Overrun Runway 009/189	O08C	AC	T	83	2.93	81	79	76	73	70	67	64	61	58	55
Overrun Runway 15/33	O03C	AC	T	79	2.93	77	75	72	69	66	63	60	57	54	51
Overrun Runway 15/33	O04C	AAC	T	79	2.93	77	75	72	69	66	63	60	57	54	51
Overrun Runway 15/33	O05C	AAC	T	78	2.93	76	74	71	68	65	62	59	56	53	50
Overrun Runway 15/33	O06C	AAC	T	77	2.93	75	73	70	67	64	61	58	55	52	49
Runways															
Runway 009/189 (Assault Strip)	R01A	PCC	P	93	0.33	93	93	92	92	92	91	91	91	90	90
Runway 009/189 (Assault Strip)	R02A	APC	P	75	0.69	75	74	73	73	72	71	70	70	69	68
Runway 15/33	R03A	PCC	P	86	0.33	86	86	85	85	85	84	84	84	83	83
Runway 15/33	R04C1	AAC	P	99	0.69	99	98	97	97	96	95	94	94	93	92
Runway 15/33	R04C2	APC	P	84	0.69	84	83	82	82	81	80	79	79	78	77
Runway 15/33	R05A1	AC	P	100	0.69	100	99	98	98	97	96	95	95	94	93
Runway 15/33	R05A2	APC	P	80	0.69	80	79	78	78	77	76	75	75	74	73
Runway 15/33	R06A1	PCC	P	99	0.33	99	99	98	98	98	97	97	97	96	96
Runway 15/33	R06A2	PCC	P	97	0.33	97	97	96	96	96	95	95	95	94	94
Runway 15/33	R07A	PCC	P	99	0.33	99	99	98	98	98	97	97	97	96	96
Runway 15/33	R08C1	PCC	P	93	0.33	93	93	92	92	92	91	91	91	90	90
(Page 1 of 2)															

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Table 2.7. Ten-year Projected PCI (concluded).

Branch Name	Section ID	Pavement Surface Type	Pavement Rank	Current PCI	PCI Deterioration Rate	FY 2020 PCI Before M&R	FY 2021 PCI Before M&R	FY 2022 PCI Before M&R	FY 2023 PCI Before M&R	FY 2024 PCI Before M&R	FY 2025 PCI Before M&R	FY 2026 PCI Before M&R	FY 2027 PCI Before M&R	FY 2028 PCI Before M&R	FY 2029 PCI Before M&R
Runways (concluded)															
Runway 15/33	R08C2	PCC	P	98	0.33	98	98	97	97	97	96	96	96	95	95
Runway 15/33	R09C	APC	P	90	0.69	90	89	88	88	87	86	85	85	84	83
Runway 15/33	R10C	AC	P	76	0.69	76	75	74	74	73	72	71	71	70	69
Runway 15/33	R12A	APC	P	100	0.69	100	99	98	98	97	96	95	95	94	93
Runway 15/33	R13C	APC	P	92	0.69	92	91	90	90	89	88	87	87	86	85
Runway 15/33	R14C1	APC	P	100	0.69	100	99	98	98	97	96	95	95	94	93
Runway 15/33	R14C2	APC	P	100	0.69	100	99	98	98	97	96	95	95	94	93
Taxiways															
Runway 009/189 Turn Around	T06A	APC	P	59	0.69	59	58	57	57	56	55	54	54	53	52
Runway 009/189 Turn Around	T35A	APC	P	96	0.69	96	95	94	94	93	92	91	91	90	89
Runway 15/33 Turn Around	T11A	PCC	P	99	0.33	99	99	98	98	98	97	97	97	96	96
Taxiway A	T04A	APC	P	82	0.69	82	81	80	80	79	78	77	77	76	75
Taxiway A North	T16A	PCC	P	95	0.33	95	95	94	94	94	93	93	93	92	92
Taxiway A North	T18A1	PCC	P	87	0.33	87	87	86	86	86	85	85	85	84	84
Taxiway A North	T18A2	PCC	P	90	0.33	90	90	89	89	89	88	88	88	87	87
Taxiway A North	T34A	PCC	P	85	0.33	85	85	84	84	84	83	83	83	82	82
Taxiway A Northwest	T03A	PCC	P	89	0.33	89	89	88	88	88	87	87	87	86	86
Taxiway A Northwest	T13A	APC	P	81	0.69	81	80	79	79	78	77	76	76	75	74
Taxiway A Northwest	T17A	PCC	P	82	0.33	82	82	81	81	81	80	80	80	79	79
Taxiway A Northwest	T19A	AC	P	79	0.69	79	78	77	77	76	75	74	74	73	72
Taxiway A Northwest	T33A	PCC	P	85	0.33	85	85	84	84	84	83	83	83	82	82
Taxiway B	T05C	APC	S	83	2.85	82	79	76	73	70	67	64	62	59	56
Taxiway B	T12C	APC	S	89	2.85	88	85	82	79	76	73	70	68	65	62
Taxiway C	T08A	APC	P	85	0.69	85	84	83	83	82	81	80	80	79	78
Taxiway D	T32B	APC	S	89	2.85	88	85	82	79	76	73	70	68	65	62
Taxiway E	T09A	APC	P	84	0.69	84	83	82	82	81	80	79	79	78	77
(Page 2 of 2)															

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2.4 SHOULDERS

All shoulders were constructed of either AC or PCC pavement and contained a total of 1,714,359 ft² of pavement. The shoulders were generally rated as “Fair” after a brief visual inspection. The predominant AC distresses observed were block cracking, L&T cracking, raveling and weathering. The predominant PCC distress observed was joint seal damage. Table 2.9 provides general shoulder information.

Table 2.9. Shoulder Location and Pavement Area.

Name	Section	Surface	Area (ft ²)	Rating
Shoulder Pavement				
Runway 009/189 Shoulders	S02D	PCC	299,890	Fair
Runway 009/189 Shoulders	S03D	AC	386,333	Fair
Runway 15/33 Shoulders	S01D	AC	517,290	Good
Runway Overrun 009/189 Shoulders	S06D	AC	199,682	Good
Runway Overrun 15/33 Shoulders	S05D	AC	149,333	Poor
Taxiway B Shoulders	S04D	AC	161,831	Good
All Shoulder Pavement			1,714,359	Fair

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CHAPTER 3

ENGINEERING ASSESSMENT PROCESS AND CRITERIA

3.1 ENGINEERING ASSESSMENT RATING DESCRIPTION

The EA is a measure of the adequacy of airfield pavements as defined by AFI-32-1041. The four ratings used to determine the EA of a pavement are PCI, FOD Potential, Structural Index, and Friction Index (runways only). Each of these indices has an associated rating defined in AFI-32-1041. The EA for a section is defined as the lowest rating of the area-weighted average of any of the four indices. Appendices F, G, H, and I present the tables showing data and calculations used to determine the EA for each of the four indices. Table 3.1 summarizes the criteria for determination of the EA.

Table 3.1. EA Rating Criteria.

EA (Color Code)	Criteria			
	Area-Weighted PCI (Rating)	FOD Potential (Rating)	Structural Index (Rating)	Friction Index* (Rating)
Adequate (Green)	$70 < \text{PCI} \leq 100$ (Good)	$\text{FOD} \leq 45$ (Good)	$\text{ACN/PCN} < 1.10$ (Good)	Grip Tester® > 0.49 (Good)
Degraded (Yellow)	$55 < \text{PCI} \leq 70$ (Fair)	FOD 46-60 (Fair)	$1.10 \leq \text{ACN/PCN} \leq 1.40$ (Fair)	Grip Tester® 0.34-0.49 (Fair)
Unsatisfactory (Red)	$\text{PCI} \leq 55$ (Poor)	$\text{FOD} \geq 61$ (Poor)	$\text{ACN/PCN} > 1.40$ (Poor)	Grip Tester® < 0.34 (Poor)
*Applies to Grip Tester® at 40 mph only.				

3.1.1 Area-Weighted Average PCI

The PCI as determined by this report and described in Section 2.1 is used to determine the EA rating.

3.1.2 FOD Index and Potential

FOD Index is based on pavement distresses from the PCI survey that result in loose pieces of pavement. The FOD Potential rating is dependent upon the type of aircraft using the pavement, the type of pavement surface, and the FOD Index. Aircraft with low-slung engine intakes are more susceptible to FOD than those with high-mount engines. Three standard aircraft categories are used to calculate FOD Potential: the F-16, the KC-135, and the C-17. Typically, the aircraft that is most susceptible to FOD that uses a given pavement section is used as the controlling aircraft. The FOD Potential for pavements at Maxwell AFB was calculated using the C-17 or KC-135 FOD Potential aircraft category depending on the critical mission aircraft for a given section. Figure A.5 graphically represents the results.

3.1.3 Structural Index

Aircraft Classification Number (ACN) expresses the load imparted by an aircraft in terms of a standardized single wheel load. Pavement Classification Number (PCN) expresses the structural capacity of a pavement in terms of the same standardized single wheel load. The ratio between an ACN and PCN reflects the potential damage a given aircraft is likely to cause to a given pavement. The PCN values were obtained from the January 2016 structural evaluation report by AFCEC. The primary mission aircraft used for the ACN/PCN values were

the C-130H except for the Aero Club, which used the T-1A. Figure A-6 graphically presents these data. Sections not applicable are listed as “NA” and are assumed to have a structural rating of “Good” when calculating the EA rating. Further explanation of the Structural Index is provided in Appendix H.

3.1.4 Friction Index

Friction data were used from the January 2016 evaluation completed by AFCEC as it is the most recent available (see Table I.1 and Table I.2). Significant mill and overlay projects have occurred on the Runway since the evaluation was completed, therefore, the data may not be applicable.

3.2 EA RATINGS

Table 3.2 presents the EA ratings by section. Controlling criteria are in **boldface** type. Friction Index ratings apply only to runway keel sections. Figure A.7 graphically presents the EA by section.

Table 3.2. EA Ratings by Section.

Branch Name	Section ID	PCI		FOD		Structural		Friction Rating		EA Rating
		Index	Rating	Potential	Rating	Index	Rating	Index	Rating	
Aero Club Apron	A10C	14	Poor	98	Poor	0.40	Good	NA	NA	Unsatisfactory
AGE Storage Lot	A12	65	Fair	35	Good	ND	Good	NA	NA	Degraded
	A13	66	Fair	34	Good	ND	Good	NA	NA	Degraded
Engine Run-up Apron	A07B	91	Good	32	Good	0.83	Good	NA	NA	Adequate
	T14B	90	Good	10	Good	0.68	Good	NA	NA	Adequate
Maintenance Hangar Access Apron	A11C1	83	Good	36	Good	0.49	Good	NA	NA	Adequate
	A11C2	79	Good	39	Good	0.49	Good	NA	NA	Adequate
	A11C3	93	Good	28	Good	0.49	Good	NA	NA	Adequate
North Apron	A01B	84	Good	34	Good	1.06	Good	NA	NA	Adequate
	A04B	88	Good	18	Good	1.03	Good	NA	NA	Adequate
	A08B	99	Good	0	Good	0.67	Good	NA	NA	Adequate
	T02B	85	Good	23	Good	1.00	Good	NA	NA	Adequate
	T15B	93	Good	27	Good	0.45	Good	NA	NA	Adequate
	T30B	81	Good	24	Good	0.43	Good	NA	NA	Adequate
	T31B	94	Good	22	Good	0.64	Good	NA	NA	Adequate
West Apron	A05B	89	Good	15	Good	0.93	Good	NA	NA	Adequate
	T07B	88	Good	32	Good	0.97	Good	NA	NA	Adequate
	T29B	90	Good	19	Good	0.48	Good	NA	NA	Adequate
Overrun Runway 009/189	O02C	79	Good	20	Good	NA	NA	NA	NA	Adequate
	O08C	83	Good	17	Good	NA	NA	NA	NA	Adequate
Overrun Runway 15/33	O03C	79	Good	21	Good	NA	NA	NA	NA	Adequate
	O04C	79	Good	21	Good	NA	NA	NA	NA	Adequate
	O05C	78	Good	22	Good	NA	NA	NA	NA	Adequate
	O06C	77	Good	23	Good	NA	NA	NA	NA	Adequate
Runway 009/189 (Assault Strip)	R01A	93	Good	6	Good	0.61	Good	0.71	Good	Adequate
	R02A	75	Good	25	Good	0.22	Good	0.83	Good	Adequate
Runway 15/33	R03A	86	Good	13	Good	1.03	Good	NA	NA	Adequate

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Table 3.2. EA Ratings by Section (concluded).

Branch Name	Section ID	PCI		FOD		Structural		Friction Rating		EA Rating
		Index	Rating	Potential	Rating	Index	Rating	Index	Rating	
Runway 15/33	R04C1	99	Good	1	Good	0.33	Good	0.73	Good	Adequate
	R04C2	84	Good	16	Good	0.36	Good	NA	NA	Adequate
	R05A1	100	Good	0	Good	0.18	Good	0.77	Good	Adequate
	R05A2	80	Good	20	Good	0.18	Good	NA	NA	Adequate
	R06A1	99	Good	2	Good	0.52	Good	0.73	Good	Adequate
	R06A2	97	Good	3	Good	0.52	Good	NA	NA	Adequate
	R07A	99	Good	1	Good	0.48	Good	0.76	Good	Adequate
	R08C1	93	Good	5	Good	0.39	Good	0.73	Good	Adequate
	R08C2	98	Good	2	Good	0.43	Good	NA	NA	Adequate
	R09C	90	Good	10	Good	0.21	Good	0.8	Good	Adequate
	R10C	76	Good	24	Good	0.18	Good	0.85	Good	Adequate
	R12A	100	Good	0	Good	0.18	Good	0.82	Good	Adequate
	R13C	92	Good	8	Good	0.21	Good	0.73	Good	Adequate
	R14C1	100	Good	0	Good	0.33	Good	NA	NA	Adequate
	R14C2	100	Good	0	Good	0.36	Good	NA	NA	Adequate
Runway 009/189 Turn Around	T06A	59	Fair	41	Good	0.18	Good	NA	NA	Degraded
	T35A	96	Good	4	Good	ND	Good	NA	NA	Adequate
Runway 15/33 Turn Around	T11A	99	Good	0	Good	0.48	Good	NA	NA	Adequate
Taxiway A	T04A	82	Good	18	Good	0.40	Good	NA	NA	Adequate
Taxiway A North	T16A	95	Good	3	Good	0.37	Good	NA	NA	Adequate
	T18A1	87	Good	33	Good	0.64	Good	NA	NA	Adequate
	T18A2	90	Good	37	Good	0.62	Good	NA	NA	Adequate
	T34A	85	Good	21	Good	0.76	Good	NA	NA	Adequate
Taxiway A Northwest	T03A	89	Good	28	Good	0.60	Good	NA	NA	Adequate
	T13A	81	Good	19	Good	0.37	Good	NA	NA	Adequate
	T17A	82	Good	19	Good	1.00	Good	NA	NA	Adequate
	T19A	79	Good	21	Good	1.45	Poor	NA	NA	Unsatisfactory
	T33A	85	Good	40	Good	0.95	Good	NA	NA	Adequate
Taxiway B	T05C	83	Good	17	Good	0.23	Good	NA	NA	Adequate
	T12C	89	Good	11	Good	0.30	Good	NA	NA	Adequate
Taxiway C	T08A	85	Good	15	Good	0.33	Good	NA	NA	Adequate
Taxiway D	T32B	89	Good	11	Good	0.23	Good	NA	NA	Adequate
Taxiway E	T09A	84	Good	16	Good	0.19	Good	NA	NA	Adequate
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CHAPTER 4

MAINTENANCE AND REPAIR BUDGET SCENARIOS

4.1 BUDGET SCENARIO PROCESS OVERVIEW

Several funding scenarios were modeled using PAVER work plans. The work plans were assumed to start in FY 2020 and extend through FY 2029 (years 1 through 10). The costs for each work plan are based on a cost-by-condition method employed by PAVER. For pavements below their critical PCI value, cost-by-condition curves were created by using installation specific prices for the appropriate Major M&R treatment such as a mill and overlay. Localized Stopgap and Preventive M&R cost-by-condition curves were developed for all pavements by applying installation specific prices for repairs, such as patching, to a historical database of distresses. Where localized costs were not provided, a local cost factor was applied to the *2019 RSMeans Heavy Construction Cost Data Handbook*. These cost-by-condition curves were then used to generate work plans in PAVER. Appendix L provides in-depth details about the cost-by-condition curves that outline the basis of cost estimations.

The PCI for each year of the ten-year period is shown in Figure 4.1 for each budget scenario. PCI values reflect the condition of the pavement after the recommended work has been accomplished. Work plans were developed for the following budget scenarios:

- Cost per year to maintain current area-weighted average PCI for ten years
- Cost per year to eliminate the Major M&R backlog over ten years
- Cost for M&R to only deal with safety issues and the associated drop in PCI

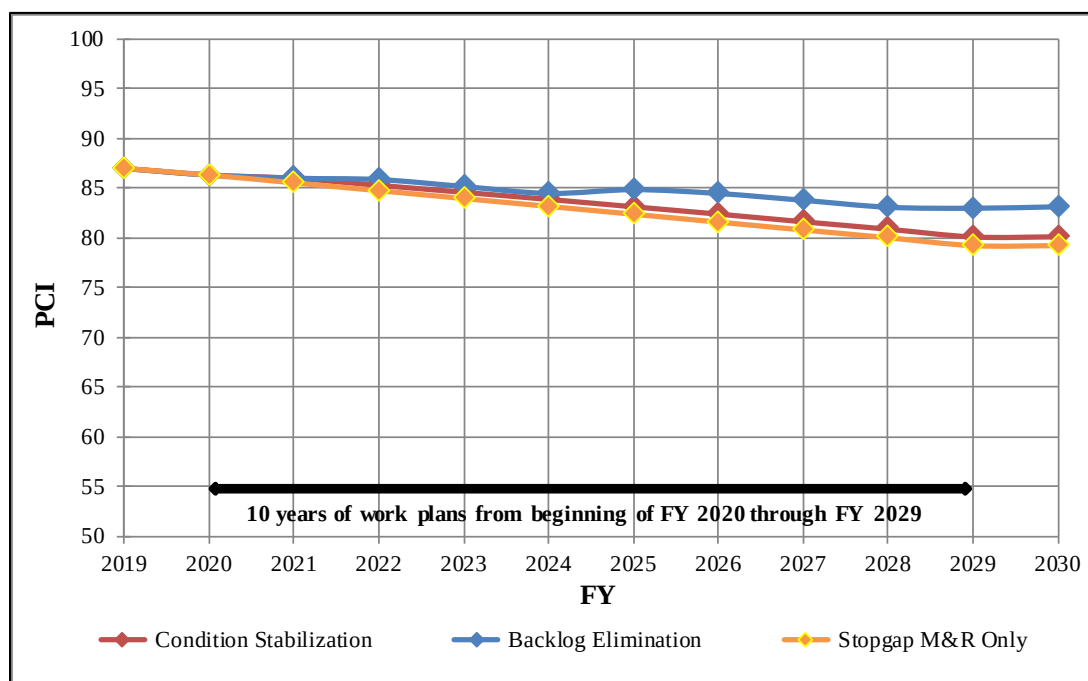


Figure 4.1. Summary of Budget Scenarios.

Figure 4.2 shows graphically the funding required for the budget scenarios by plan year. Figure 4.3 graphically shows the amount of backlog remaining for the budget scenarios

after each plan year. The backlog, also referred to as the unfunded amount, remaining after each year is the cost to fund all the M&R needs not addressed in that year of the work plan. The ten-year budget scenarios are detailed in Table 4.1. The work plans in this table are separated by M&R policy type and also show the backlog remaining after each plan year.

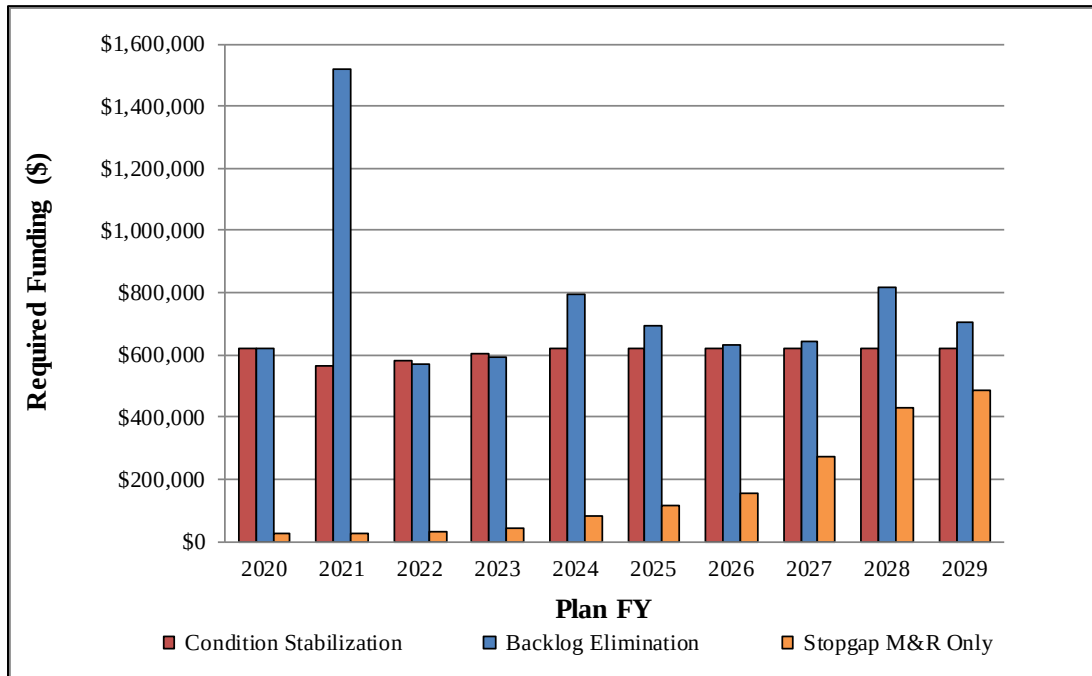


Figure 4.2. Summary of Funding Required for Budget Scenarios.

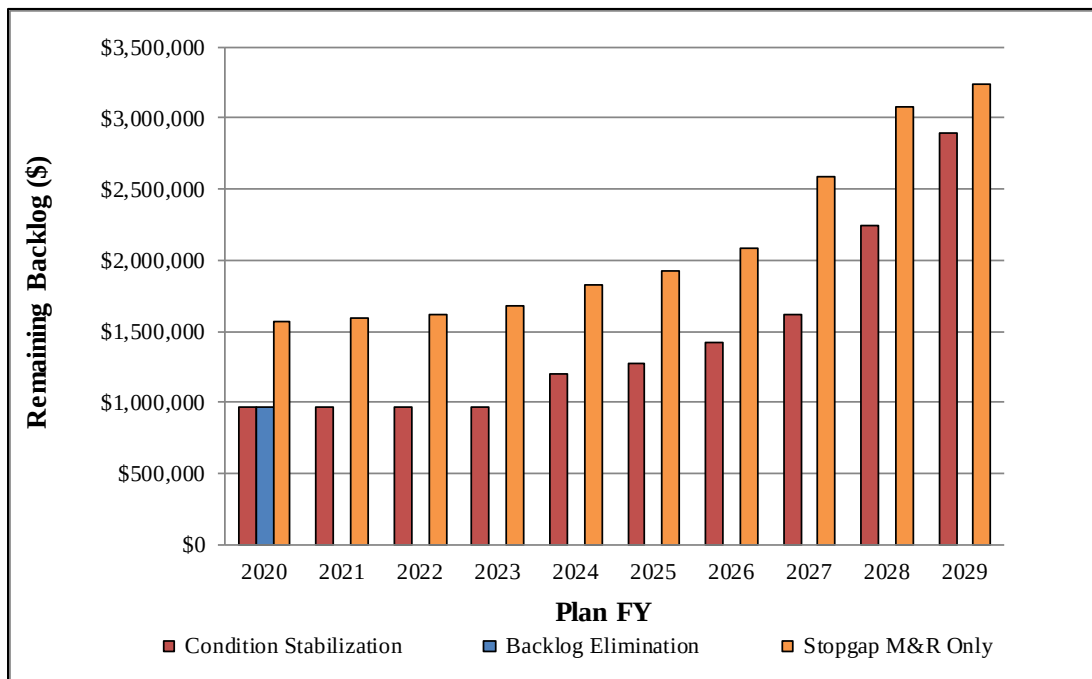


Figure 4.3. Summary of Backlog Remaining for Budget Scenarios.

Table 4.1. Summary of PAVER Work Plan Results.

Plan FY	M&R Policy Type					Total	Backlog After Plan Year	PCI	
	Stopgap	Preventive	Global	Major Below Critical PCI	Major Above Critical PCI			Before	After
Condition Stabilization									
2020	\$10,927	\$534,558	\$0	\$72,889	\$0	\$618,374	\$965,377	86	87
2021	\$11,227	\$552,298	\$0	\$0	\$0	\$563,524	\$965,377	86	86
2022	\$11,526	\$571,670	\$0	\$0	\$0	\$583,196	\$965,377	85	85
2023	\$11,826	\$593,761	\$0	\$0	\$0	\$605,586	\$965,377	85	85
2024	\$58,518	\$564,425	\$0	\$0	\$0	\$622,942	\$1,196,982	84	84
2025	\$89,350	\$530,330	\$2,522	\$0	\$0	\$622,202	\$1,277,543	83	83
2026	\$124,419	\$498,456	\$0	\$0	\$0	\$622,875	\$1,423,097	82	82
2027	\$163,290	\$457,751	\$0	\$0	\$0	\$621,041	\$1,618,379	82	82
2028	\$286,774	\$335,172	\$0	\$0	\$0	\$621,946	\$2,240,260	81	81
2029	\$446,174	\$176,145	\$0	\$0	\$0	\$622,319	\$2,897,808	80	80
Total	\$1,214,031	\$4,814,565	\$2,522	\$72,889	\$0	\$6,104,006		86	80
Condition Stabilization Gross Funding Requirement							\$9,001,814		
Backlog Elimination									
2020	\$10,927	\$534,558	\$0	\$72,889	\$0	\$618,374	\$965,377	86	87
2021	\$0	\$552,298	\$0	\$965,377	\$0	\$1,517,675	\$0	86	87
2022	\$0	\$572,147	\$0	\$0	\$0	\$572,147	\$0	86	86
2023	\$0	\$594,665	\$0	\$0	\$0	\$594,665	\$0	85	85
2024	\$0	\$599,067	\$67,555	\$130,740	\$0	\$797,362	\$0	84	86
2025	\$0	\$607,807	\$86,607	\$0	\$0	\$694,413	\$0	85	85
2026	\$0	\$622,609	\$8,683	\$0	\$0	\$631,291	\$0	84	85
2027	\$0	\$646,062	\$0	\$0	\$0	\$646,062	\$0	84	84
2028	\$0	\$662,872	\$9,145	\$146,688	\$0	\$818,704	\$0	83	84
2029	\$0	\$687,176	\$1,806	\$14,292	\$0	\$703,273	\$0	83	83
Total	\$10,927	\$6,079,261	\$173,795	\$1,329,985	\$0	\$7,593,967		86	83
Backlog Elimination Gross Funding Requirement							\$7,593,967		
Stopgap M&R Only									
2020	\$27,468	\$0	\$0	\$0	\$0	\$27,468	\$1,572,824	86	86
2021	\$28,589	\$0	\$0	\$0	\$0	\$28,589	\$1,593,356	86	86
2022	\$29,707	\$0	\$0	\$0	\$0	\$29,707	\$1,616,707	85	85
2023	\$43,732	\$0	\$0	\$0	\$0	\$43,732	\$1,683,013	84	84
2024	\$81,741	\$0	\$0	\$0	\$0	\$81,741	\$1,823,742	83	83
2025	\$114,902	\$0	\$0	\$0	\$0	\$114,902	\$1,928,642	82	82
2026	\$154,607	\$0	\$0	\$0	\$0	\$154,607	\$2,088,340	82	82
2027	\$275,598	\$0	\$0	\$0	\$0	\$275,598	\$2,590,764	81	81
2028	\$431,163	\$0	\$0	\$0	\$0	\$431,163	\$3,080,033	80	80
2029	\$488,790	\$0	\$0	\$0	\$0	\$488,790	\$3,238,490	79	79
Total	\$1,676,298	\$0	\$0	\$0	\$0	\$1,676,298		86	79
Stopgap M&R Only Gross Funding Requirement							\$4,914,788		

4.2 BUDGET SCENARIOS OVERVIEW

4.2.1 Condition Stabilization

The goal of the condition stabilization scenario is to maintain the network PCI over ten years. The annual budget ranges from \$563,524 to \$622,942. Due to the limited number of sections requiring Major M&R, PAVER was unable to maintain the condition. The PCI decreases 6 points from the start of FY 2020 to the conclusion of FY 2029.

Only one of the six sections requiring Major M&R is funded under this scenario. After the initial Major M&R project, the required Stopgap funding increases each year as more sections fall below their respective critical PCI value. Preventive M&R accounts for 79% of the total budget. Only one section is funded for Global M&R. Figure 4.4 shows the funded amounts and resulting PCI for the Condition Stabilization scenario.

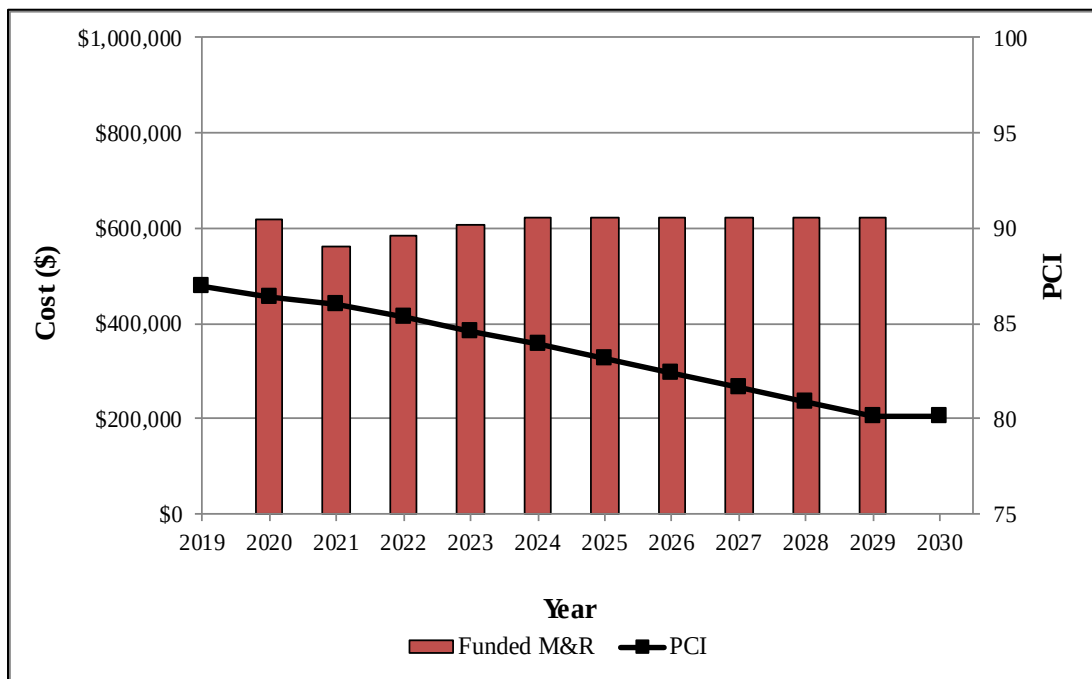


Figure 4.4. Funded M&R Costs to Maintain Current Condition.

4.2.2 Major M&R Backlog Elimination over Ten Years

The funding scenario to eliminate the Major M&R backlog over ten years assumes that funding is available for each of the ten years to repair all major pavement distresses with no unfunded Major M&R amount after the tenth year. Due to the limited number of sections requiring Major M&R, the PCI will still decrease from 86 to 83. A total of approximately \$1.3 million of major M&R funding is required throughout the ten year period. The backlog of unfunded major M&R is eliminated after the second year.

Major M&R accounts for 17% of the total budget. Global M&R was recommended for 10 sections of AC pavement for a total of approximately \$173,795 to mitigate deterioration due to climatic distresses. Stopgap funding is only required in the first year. Figure 4.5 shows the funded amounts and resulting PCI for the Backlog Elimination scenario.

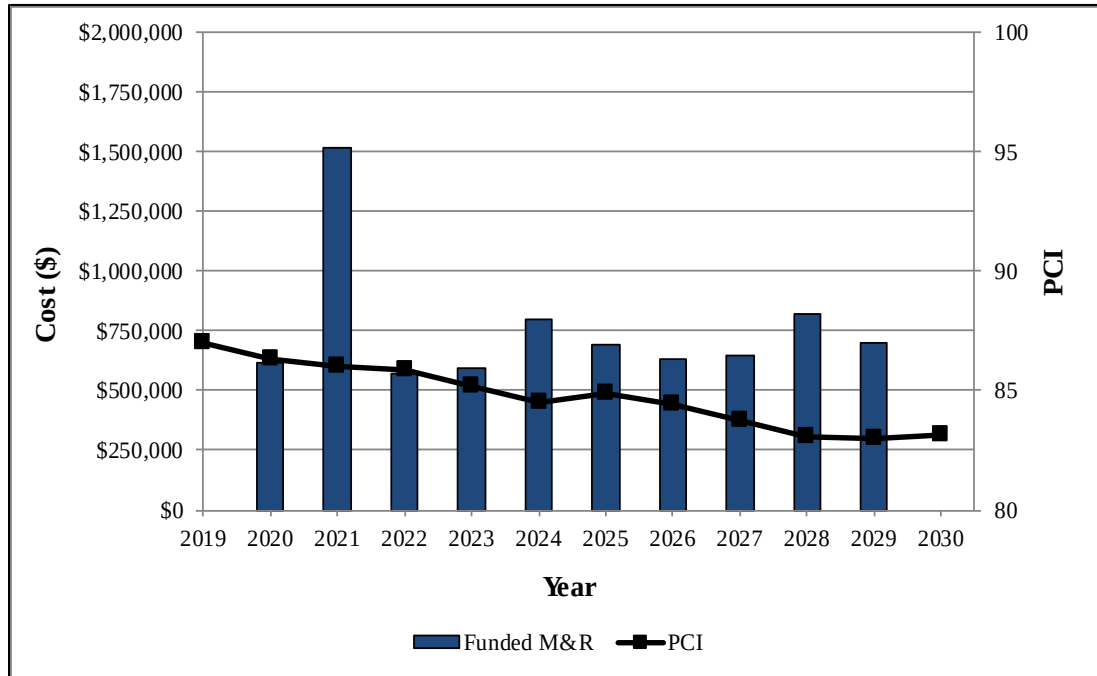


Figure 4.5. Funded M&R Costs to Eliminate the Major M&R Backlog.

4.2.3 Stopgap M&R Only (safety issues)

The Stopgap M&R only funding scenario presumes that only Stopgap M&R will be performed on the pavement. This scenario determines the minimum amount of money necessary to maintain useable pavements. It is presumed that the overall PCI does not improve over ten years. Table 4.1 shows the area-weighted average PCI at the end of ten years will decrease from 86-Good to 79-Good. A total of approximately \$1.7 million in funding is required over the ten-year plan. The backlog of unfunded major M&R increases to approximately \$4.9 million. Figure 4.6 shows the funded amounts and resulting PCI for the localized Stopgap M&R Only scenario.

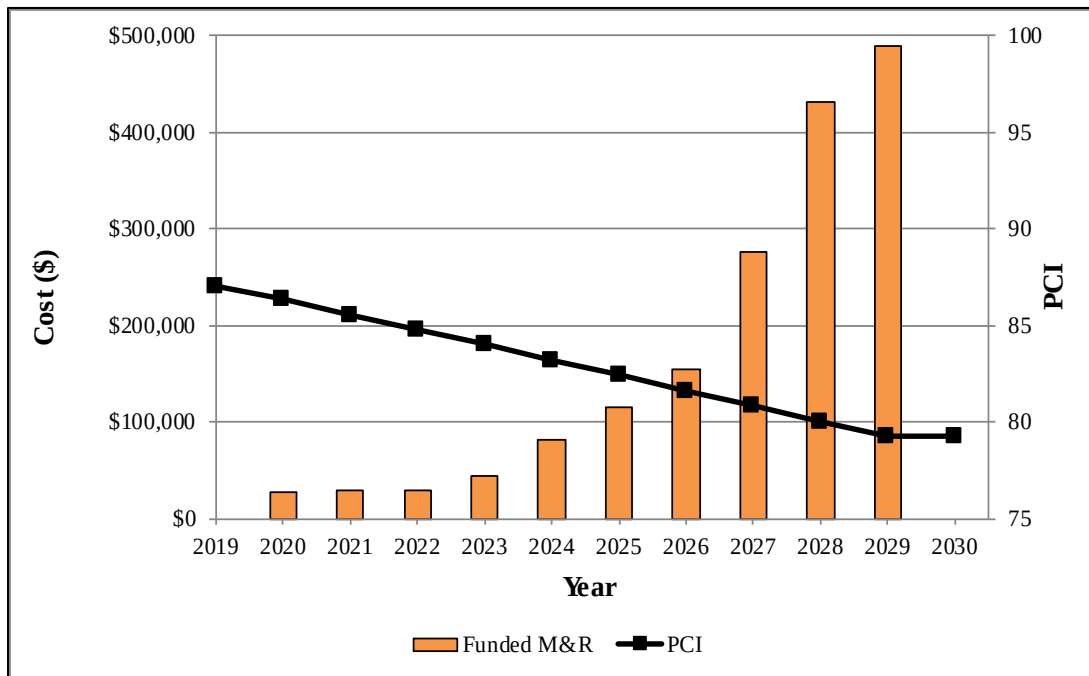


Figure 4.6. Funded M&R Costs for the Localized Stopgap Budget.

4.3 OPTIMUM BUDGET SELECTED

The Backlog Elimination scenario was selected as the recommended optimum budget scenario because all sections requiring major M&R are funded. Engineering judgement was used to apply M&R based on cost by work type instead of cost by condition.

4.4 REPAIR RECOMMENDATIONS AND COST

4.4.1 Major M&R Projects

The Major M&R Backlog Elimination in ten years scenario, selected as the optimum budget, was used to develop the Major M&R plan shown in Table 4.2. Projects are listed by the PAVER funding year. The PCI listed is the predicted PCI prior to repairs.

Recommended repair is based on cost-by-work-type and the applied localized cost factor of 0.85 from the *2019 RSMeans Heavy Construction Cost Data*.

Table 4.2. PAVER Recommended Major M&R by Year.

Project Locations		Rank	PCI	Surface Type	Area (ft²)	Recommended Repair	Estimated Cost
Branch Name	Section						
FY 2020							
Runway 009/189 Turn Around	T06A	P	59	APC	50,117	Mill and Overlay	\$72,889
1st Plan FY Total							\$72,889
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Table 4.2. PAVER Recommended Major M&R by Year (concluded).

Project Locations		Rank	PCI	Surface Type	Area (ft²)	Recommended Repair	Estimated Cost
Branch Name	Section						
FY 2021							
Aero Club Apron	A10C	S	13	PCC	36,205	Complete Reconstruction	\$965,377
2nd Plan FY Total							\$965,377
FY 2022							
No Major M&R Recommended in FY 2022							
FY 2023							
No Major M&R recommended in FY 2023							
FY 2024							
AGE Storage Lot	A12	S	54	AC	30,392	Mill and Overlay	\$47,153
AGE Storage Lot	A13	S	55	AC	54,563	Mill and Overlay	\$83,586
5th Plan FY Total							\$83,586
FY 2025							
No Major M&R recommended in FY 2025							
FY 2026							
No Major M&R recommended in FY 2026							
FY 2027							
No Major M&R recommended in FY 2027							
FY 2028							
Runway 009/189 (Assault Strip)	R02A	P	69	APC	107,583	Mill and Overlay	\$146,688
9th Plan FY Total							\$146,688
FY 2029							
Runway 15/33	R10C	P	70	AC	10,500	Mill and Overlay	\$14,292
10th Plan FY Total							\$14,292
(Page 2 of 2)							

4.4.2 Consequence of Localized Distress Maintenance

The first year work requirements, quantity, and cost as determined by a PAVER consequence analysis are presented in Table 4.3. Unlike the ten-year work plans, the consequence analysis does not consider current or future Major M&R needs and is instead based solely on current pavement distresses. Therefore, PAVER will calculate the cost and quantity of work based on the Preventive and Stopgap M&R policies outlined in Tables L.3 to L.6 for specific distresses that were recorded during the current PCI survey at Maxwell AFB.

Localized M&R costs for the consequence analysis are based on a more detailed analysis of project scope including actual costs for specific activities, such as slab replacement or crack sealing, to repair the corresponding distress. The derivation of cost-by-work type M&R is explained in Section L.6.1. Localized Stopgap and Preventive M&R funding costs for the first year are \$453,280 based on the 2019 *RSM* means costs with an assigned cost factor of 0.85.

The cost-by-work type recommendations below are separate from and not reflected in the ten-year work plans previously presented. The work costs are based on extrapolated

quantities of the distresses recorded during the current survey. Table 4.3 shows the recommended maintenance plan details for localized preventive and stopgap M&R. It should be noted that the work quantities presented in Table 4.3 are extrapolated from distresses recorded in March 2019. Work quantities should be expected to increase the longer the recommended maintenance is delayed.

Table 4.3. Consequence of Localized Distress Maintenance.

Branch	Section	Distress	Severity	Quantity	Work Description	Work Quantity	Unit Cost	Work Cost
AGE Storage Lot	A12	L & T Cr	Medium	541 ft	Crack Sealing - AC	541 ft	\$0.99	\$538
AGE Storage Lot	A12	Patching	Medium	608 ft ²	Patching - AC Deep	712 ft ²	\$12.43	\$8,836
AGE Storage Lot	A13	L & T Cr	Medium	3,459 ft	Crack Sealing - AC	3,459 ft	\$0.99	\$3,440
Engine Run-up Apron	A07B	Joint Spall	Low	5 Slabs	Crack Sealing - PCC	9 ft	\$3.57	\$30
Engine Run-up Apron	A07B	Jt Seal Dmg	Medium	154 Slabs	Joint Seal (Localized)	5,505 ft	\$2.97	\$16,378
Engine Run-up Apron	A07B	Jt Seal Dmg	High	26 Slabs	Joint Seal (Localized)	918 ft	\$2.97	\$2,730
Maintenance Hangar Access Apron	A11C1	Corner Spall	Low	6 Slabs	Crack Sealing - PCC	10 ft	\$3.57	\$35
Maintenance Hangar Access Apron	A11C1	Joint Spall	Low	7 Slabs	Crack Sealing - PCC	12 ft	\$3.57	\$41
Maintenance Hangar Access Apron	A11C1	Linear Cr	Medium	2 Slabs	Crack Sealing - PCC	20 ft	\$3.57	\$72
Maintenance Hangar Access Apron	A11C1	Jt Seal Dmg	Medium	70 Slabs	Joint Seal (Localized)	1,190 ft	\$2.97	\$3,539
Maintenance Hangar Access Apron	A11C1	Jt Seal Dmg	High	24 Slabs	Joint Seal (Localized)	414 ft	\$2.97	\$1,231
Maintenance Hangar Access Apron	A11C1	Corner Spall	Medium	6 Slabs	Patching - PCC Partial Depth	16 ft ²	\$8.92	\$145
Maintenance Hangar Access Apron	A11C2	Corner Spall	Low	17 Slabs	Crack Sealing - PCC	28 ft	\$3.57	\$101
Maintenance Hangar Access Apron	A11C2	Joint Spall	Low	8 Slabs	Crack Sealing - PCC	13 ft	\$3.57	\$44
Maintenance Hangar Access Apron	A11C2	Jt Seal Dmg	Medium	133 Slabs	Joint Seal (Localized)	2,291 ft	\$2.97	\$6,816
Maintenance Hangar Access Apron	A11C2	Joint Spall	Medium	1 Slabs	Patching - PCC Partial Depth	7 ft ²	\$8.92	\$62
Maintenance Hangar Access Apron	A11C3	Jt Seal Dmg	Medium	114 Slabs	Joint Seal (Localized)	1,948 ft	\$2.97	\$5,795
North Apron	A01B	Joint Spall	Low	8 Slabs	Crack Sealing - PCC	13 ft	\$3.57	\$46
North Apron	A01B	Linear Cr	Medium	24 Slabs	Crack Sealing - PCC	387 ft	\$3.57	\$1,381
North Apron	A01B	Jt Seal Dmg	Medium	1,045 Slabs	Joint Seal (Localized)	32,980 ft	\$2.97	\$98,116
North Apron	A01B	Joint Spall	Medium	5 Slabs	Patching - PCC Partial Depth	34 ft ²	\$8.92	\$305
North Apron	A01B	Small Patch	Medium	8 Slabs	Patching - PCC Partial Depth	22 ft ²	\$8.92	\$190
North Apron	A04B	Corner Spall	Low	20 Slabs	Crack Sealing - PCC	33 ft	\$3.57	\$116
North Apron	A04B	Joint Spall	Low	20 Slabs	Crack Sealing - PCC	33 ft	\$3.57	\$116
North Apron	A04B	Linear Cr	Medium	27 Slabs	Crack Sealing - PCC	431 ft	\$3.57	\$1,537
North Apron	A04B	Faulting	Medium	7 Slabs	Grinding (Localized)	83 ft	\$4.39	\$363

(Page 1 of 4)

Table 4.3. Consequence of Localized Distress Maintenance (continued).

Branch	Section	Distress	Severity	Quantity	Work Description	Work Quantity	Unit Cost	Work Cost
North Apron	A04B	Jt Seal Dmg	Medium	821 Slabs	Joint Seal (Localized)	26,001 ft	\$2.97	\$77,352
North Apron	A04B	Corner Spall	Medium	13 Slabs	Patching - PCC Partial Depth	36 ft²	\$8.92	\$318
North Apron	A04B	Large Patch	Medium	13 Slabs	Patching - PCC Partial Depth	815 ft²	\$8.92	\$7,267
North Apron	A04B	Small Patch	Medium	13 Slabs	Patching - PCC Partial Depth	36 ft²	\$8.92	\$318
North Apron	T02B	Corner Break	Low	2 Slabs	Crack Sealing - PCC	17 ft	\$3.57	\$62
North Apron	T02B	Corner Spall	Low	9 Slabs	Crack Sealing - PCC	14 ft	\$3.57	\$50
North Apron	T02B	Joint Spall	Low	2 Slabs	Crack Sealing - PCC	4 ft	\$3.57	\$13
North Apron	T02B	Linear Cr	Medium	15 Slabs	Crack Sealing - PCC	243 ft	\$3.57	\$867
North Apron	T02B	Faulting	Medium	4 Slabs	Grinding (Localized)	54 ft	\$4.39	\$234
North Apron	T02B	Jt Seal Dmg	Medium	171 Slabs	Joint Seal (Localized)	4,964 ft	\$2.97	\$14,769
North Apron	T02B	Jt Seal Dmg	High	51 Slabs	Joint Seal (Localized)	1,489 ft	\$2.97	\$4,431
North Apron	T02B	Large Patch	Medium	2 Slabs	Patching - PCC Partial Depth	131 ft²	\$8.92	\$1,170
North Apron	T02B	Linear Cr	High	2 Slabs	Patching - PCC Partial Depth	114 ft²	\$8.92	\$1,014
North Apron	T02B	Small Patch	Medium	4 Slabs	Patching - PCC Partial Depth	12 ft²	\$8.92	\$102
North Apron	T15B	Corner Spall	Low	0 Slabs	Crack Sealing - PCC	1 ft	\$3.57	\$3
North Apron	T15B	Joint Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$8
North Apron	T15B	Jt Seal Dmg	Medium	58 Slabs	Joint Seal (Localized)	1,658 ft	\$2.97	\$4,933
North Apron	T30B	Corner Break	Low	1 Slabs	Crack Sealing - PCC	8 ft	\$3.57	\$26
North Apron	T30B	Linear Cr	Medium	3 Slabs	Crack Sealing - PCC	44 ft	\$3.57	\$157
North Apron	T30B	Jt Seal Dmg	Medium	22 Slabs	Joint Seal (Localized)	479 ft	\$2.97	\$1,426
North Apron	T31B	Jt Seal Dmg	Medium	159 Slabs	Joint Seal (Localized)	4,591 ft	\$2.97	\$13,657
West Apron	A05B	Corner Spall	Low	5 Slabs	Crack Sealing - PCC	8 ft	\$3.57	\$27
West Apron	A05B	Joint Spall	Low	5 Slabs	Crack Sealing - PCC	8 ft	\$3.57	\$27
West Apron	A05B	Jt Seal Dmg	Medium	111 Slabs	Joint Seal (Localized)	3,497 ft	\$2.97	\$10,405
West Apron	A05B	Corner Break	Medium	5 Slabs	Patching - PCC Full Depth	149 ft²	\$52.41	\$7,802
West Apron	A05B	Corner Spall	Medium	5 Slabs	Patching - PCC Partial Depth	13 ft²	\$8.92	\$111
West Apron	A05B	Joint Spall	Medium	5 Slabs	Patching - PCC Partial Depth	30 ft²	\$8.92	\$265
West Apron	A05B	Large Patch	Medium	18 Slabs	Patching - PCC Partial Depth	1,135 ft²	\$8.92	\$10,114
West Apron	A05B	Small Patch	Medium	23 Slabs	Patching - PCC Partial Depth	62 ft²	\$8.92	\$553
West Apron	T07B	Corner Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$6
West Apron	T07B	Linear Cr	Medium	1 Slabs	Crack Sealing - PCC	16 ft	\$3.57	\$58

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Table 4.3. Consequence of Localized Distress Maintenance (continued).

Branch	Section	Distress	Severity	Quantity	Work Description	Work Quantity	Unit Cost	Work Cost
West Apron	T07B	Jt Seal Dmg	Medium	172 Slabs	Joint Seal (Localized)	3,854 ft	\$2.97	\$11,466
West Apron	T07B	Small Patch	Medium	1 Slabs	Patching - PCC Partial Depth	2 ft ²	\$8.92	\$24
West Apron	T29B	Corner Break	Low	6 Slabs	Crack Sealing - PCC	51 ft	\$3.57	\$181
West Apron	T29B	Corner Spall	Low	12 Slabs	Crack Sealing - PCC	20 ft	\$3.57	\$72
West Apron	T29B	Joint Spall	Low	2 Slabs	Crack Sealing - PCC	3 ft	\$3.57	\$9
West Apron	T29B	Jt Seal Dmg	Medium	191 Slabs	Joint Seal (Localized)	5,230 ft	\$2.97	\$15,559
Runway 009/189 (Assault Strip)	R01A	Corner Break	Low	1 Slabs	Crack Sealing - PCC	12 ft	\$3.57	\$42
Runway 009/189 (Assault Strip)	R01A	Corner Spall	Low	6 Slabs	Crack Sealing - PCC	10 ft	\$3.57	\$34
Runway 009/189 (Assault Strip)	R01A	Joint Spall	Low	7 Slabs	Crack Sealing - PCC	12 ft	\$3.57	\$42
Runway 009/189 (Assault Strip)	R01A	Faulting	Medium	1 Slabs	Grinding (Localized)	18 ft	\$4.39	\$79
Runway 009/189 (Assault Strip)	R01A	Corner Spall	Medium	1 Slabs	Patching - PCC Partial Depth	4 ft ²	\$8.92	\$35
Runway 009/189 (Assault Strip)	R01A	Large Patch	Medium	9 Slabs	Patching - PCC Partial Depth	534 ft ²	\$8.92	\$4,764
Runway 009/189 (Assault Strip)	R01A	Scaling	Medium	1 Slabs	Patching - PCC Partial Depth	74 ft ²	\$8.92	\$662
Runway 009/189 (Assault Strip)	R02A	L & T Cr	Medium	510 ft	Crack Sealing - AC	510 ft	\$0.99	\$507
Runway 15/33	R03A	Corner Break	Low	7 Slabs	Crack Sealing - PCC	57 ft	\$3.57	\$205
Runway 15/33	R03A	Corner Spall	Low	2 Slabs	Crack Sealing - PCC	4 ft	\$3.57	\$14
Runway 15/33	R03A	Joint Spall	Low	8 Slabs	Crack Sealing - PCC	14 ft	\$3.57	\$48
Runway 15/33	R03A	Linear Cr	Medium	6 Slabs	Crack Sealing - PCC	95 ft	\$3.57	\$339
Runway 15/33	R03A	Corner Break	Medium	1 Slabs	Patching - PCC Full Depth	38 ft ²	\$52.41	\$1,978
Runway 15/33	R03A	Corner Spall	Medium	1 Slabs	Patching - PCC Partial Depth	3 ft ²	\$8.92	\$28
Runway 15/33	R03A	Joint Spall	Medium	2 Slabs	Patching - PCC Partial Depth	15 ft ²	\$8.92	\$135
Runway 15/33	R06A1	Joint Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$7
Runway 15/33	R06A2	Joint Spall	Low	7 Slabs	Crack Sealing - PCC	12 ft	\$3.57	\$40
Runway 15/33	R06A2	Joint Spall	Medium	1 Slabs	Patching - PCC Partial Depth	8 ft ²	\$8.92	\$66
Runway 15/33	R06A2	Large Patch	Medium	1 Slabs	Patching - PCC Partial Depth	107 ft ²	\$8.92	\$947
Runway 15/33	R07A	Joint Spall	Low	12 Slabs	Crack Sealing - PCC	20 ft	\$3.57	\$71
Runway 15/33	R08C1	Joint Spall	Low	3 Slabs	Crack Sealing - PCC	5 ft	\$3.57	\$18
Runway 15/33	R08C1	Joint Spall	Medium	2 Slabs	Patching - PCC Partial Depth	13 ft ²	\$8.92	\$115
Runway 15/33	R08C1	Small Patch	Medium	1 Slabs	Patching - PCC Partial Depth	2 ft ²	\$8.92	\$24
Runway 15/33	R08C2	Joint Spall	Low	2 Slabs	Crack Sealing - PCC	3 ft	\$3.57	\$12
Runway 15/33 Turn Around	T11A	Joint Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$7

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Table 4.3. Consequence of Localized Distress Maintenance (concluded).

Branch	Section	Distress	Severity	Quantity	Work Description	Work Quantity	Unit Cost	Work Cost
Taxiway A North	T16A	Corner Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$9
Taxiway A North	T18A1	Joint Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$6
Taxiway A North	T18A1	Jt Seal Dmg	Medium	45 Slabs	Joint Seal (Localized)	1,244 ft	\$2.97	\$3,701
Taxiway A North	T18A1	Joint Spall	Medium	1 Slabs	Patching - PCC Partial Depth	7 ft ²	\$8.92	\$58
Taxiway A North	T18A2	Joint Spall	Low	2 Slabs	Crack Sealing - PCC	3 ft	\$3.57	\$11
Taxiway A North	T18A2	Jt Seal Dmg	Medium	257 Slabs	Joint Seal (Localized)	7,672 ft	\$2.97	\$22,824
Taxiway A North	T18A2	Jt Seal Dmg	High	404 Slabs	Joint Seal (Localized)	12,056 ft	\$2.97	\$35,866
Taxiway A North	T34A	Corner Break	Low	5 Slabs	Crack Sealing - PCC	37 ft	\$3.57	\$132
Taxiway A North	T34A	Corner Spall	Low	2 Slabs	Crack Sealing - PCC	3 ft	\$3.57	\$9
Taxiway A North	T34A	Jt Seal Dmg	Medium	54 Slabs	Joint Seal (Localized)	1,618 ft	\$2.97	\$4,815
Taxiway A Northwest	T03A	Corner Break	Low	2 Slabs	Crack Sealing - PCC	15 ft	\$3.57	\$53
Taxiway A Northwest	T03A	Corner Spall	Low	4 Slabs	Crack Sealing - PCC	6 ft	\$3.57	\$21
Taxiway A Northwest	T03A	Joint Spall	Low	14 Slabs	Crack Sealing - PCC	24 ft	\$3.57	\$84
Taxiway A Northwest	T03A	Linear Cr	Medium	4 Slabs	Crack Sealing - PCC	58 ft	\$3.57	\$208
Taxiway A Northwest	T03A	Jt Seal Dmg	High	284 Slabs	Joint Seal (Localized)	8,463 ft	\$2.97	\$25,177
Taxiway A Northwest	T03A	Corner Break	Medium	2 Slabs	Patching - PCC Full Depth	58 ft ²	\$52.41	\$3,037
Taxiway A Northwest	T03A	Blow-Up	Low	2 Slabs	Patching - PCC Partial Depth	73 ft ²	\$8.92	\$656
Taxiway A Northwest	T03A	Large Patch	Medium	2 Slabs	Patching - PCC Partial Depth	111 ft ²	\$8.92	\$984
Taxiway A Northwest	T17A	Corner Break	Low	2 Slabs	Crack Sealing - PCC	16 ft	\$3.57	\$59
Taxiway A Northwest	T17A	Corner Spall	Low	1 Slabs	Crack Sealing - PCC	2 ft	\$3.57	\$6
Taxiway A Northwest	T17A	Joint Spall	Low	2 Slabs	Crack Sealing - PCC	3 ft	\$3.57	\$12
Taxiway A Northwest	T17A	Joint Spall	Medium	1 Slabs	Patching - PCC Partial Depth	7 ft ²	\$8.92	\$58
Taxiway A Northwest	T17A	Large Patch	Medium	1 Slabs	Patching - PCC Partial Depth	61 ft ²	\$8.92	\$549
Taxiway A Northwest	T33A	Joint Spall	Low	2 Slabs	Crack Sealing - PCC	3 ft	\$3.57	\$12
Taxiway A Northwest	T33A	Linear Cr	Medium	1 Slabs	Crack Sealing - PCC	16 ft	\$3.57	\$58
Taxiway A Northwest	T33A	Jt Seal Dmg	High	30 Slabs	Joint Seal (Localized)	867 ft	\$2.97	\$2,580
Taxiway A Northwest	T33A	Jt Seal Dmg	Medium	60 Slabs	Joint Seal (Localized)	1,734 ft	\$2.97	\$5,159
Taxiway A Northwest	T33A	Joint Spall	Medium	1 Slabs	Patching - PCC Partial Depth	7 ft ²	\$8.92	\$58

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4.4.3 Global Maintenance Plan

Global preventive maintenance is a preventive maintenance application such as a surface treatment applied to an entire section to slightly increase the pavement condition and decrease the rate of deterioration. PAVER requires that certain parameters be met in order to recommend Global funding. These parameters require pavements to be ten years of age or

older, be above their respective critical PCI, and have no structural distresses. A total of 166 sections receive global preventive M&R for a total of approximately \$1.7 million with the recommended maintenance plan. Table 4.4 provides a list of sections recommended for global preventive maintenance by fiscal year.

Table 4.4. Recommended Global Maintenance.

Branch Name	Section ID	Area (ft ²)	PCI Before	PCI After	Recommendations	Cost
FY 2020						
No Global M&R Recommended						
FY 2021						
No Global M&R Recommended						
FY 2022						
No Global M&R Recommended						
FY 2023						
No Global M&R Recommended						
FY 2024						
Overrun Runway 009/189	O02C	19,810	68	74	Surface Seal - Fog Seal	\$3,536
Overrun Runway 009/189	O08C	10,115	72	78	Surface Seal - Fog Seal	\$1,806
Overrun Runway 15/33	O03C	168,000	68	74	Surface Seal - Fog Seal	\$29,988
Overrun Runway 15/33	O04C	31,100	68	74	Surface Seal - Fog Seal	\$5,551
Overrun Runway 15/33	O05C	22,508	67	73	Surface Seal - Fog Seal	\$4,018
Overrun Runway 15/33	O06C	126,930	66	72	Surface Seal - Fog Seal	\$22,657
FY 2024 Total						\$67,555
FY 2025						
Engine Run-up Apron	T14B	15,568	77	83	Surface Seal - Fog Seal	\$2,778.86
Taxiway A	T04A	191,559	78	80	Surface Seal - Fog Seal	\$34,192.96
Taxiway A Northwest	T13A	14,127	77	79	Surface Seal - Fog Seal	\$2,521.65
Taxiway A Northwest	T19A	30,173	75	77	Surface Seal - Fog Seal	\$5,385.83
Taxiway B	T05C	147,120	70	76	Surface Seal - Fog Seal	\$26,260.67
Taxiway B	T12C	33,604	76	82	Surface Seal - Fog Seal	\$5,998.26
Taxiway D	T32B	53,046	76	82	Surface Seal - Fog Seal	\$9,468.62
FY 2025 Total						\$86,607
FY 2026						
Taxiway E	T09A	48,642	80	81	Surface Seal - Fog Seal	\$8,683
FY 2026 Total						\$8,683
FY 2027						
No Global M&R Recommended						
FY 2028						
Taxiway C	T08A	51,231	79	81	Surface Seal - Fog Seal	\$9,144.65
FY 2028 Total						\$9,145
No Global M&R Recommended						
FY 2029						
Overrun Runway 009/189	O08C	10,115	65	72	Surface Seal - Fog Seal	\$1,806
FY 2026 Total						\$1,806

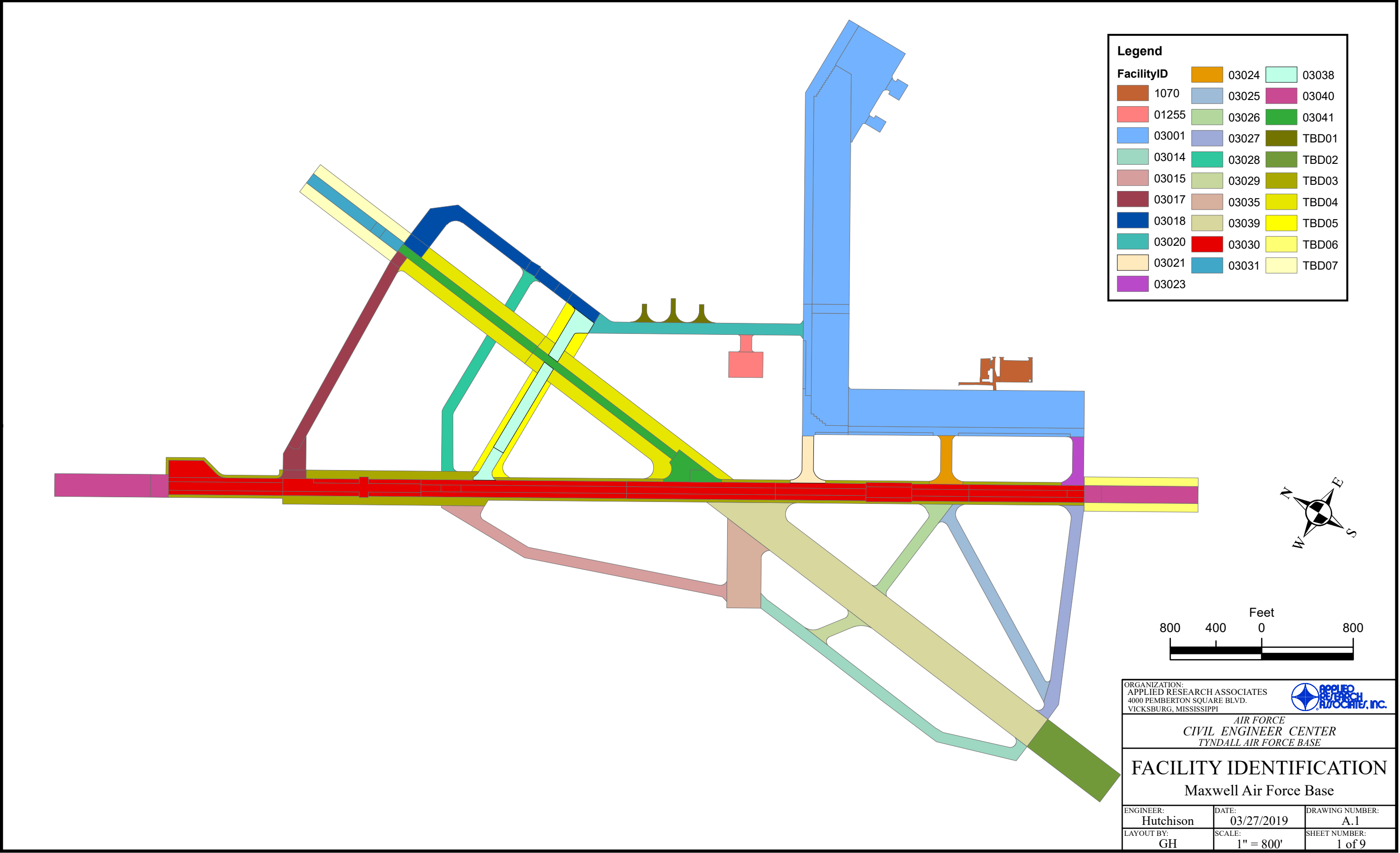
4.5 SECTIONS EXCLUDED FROM PAVER WORK PLANS

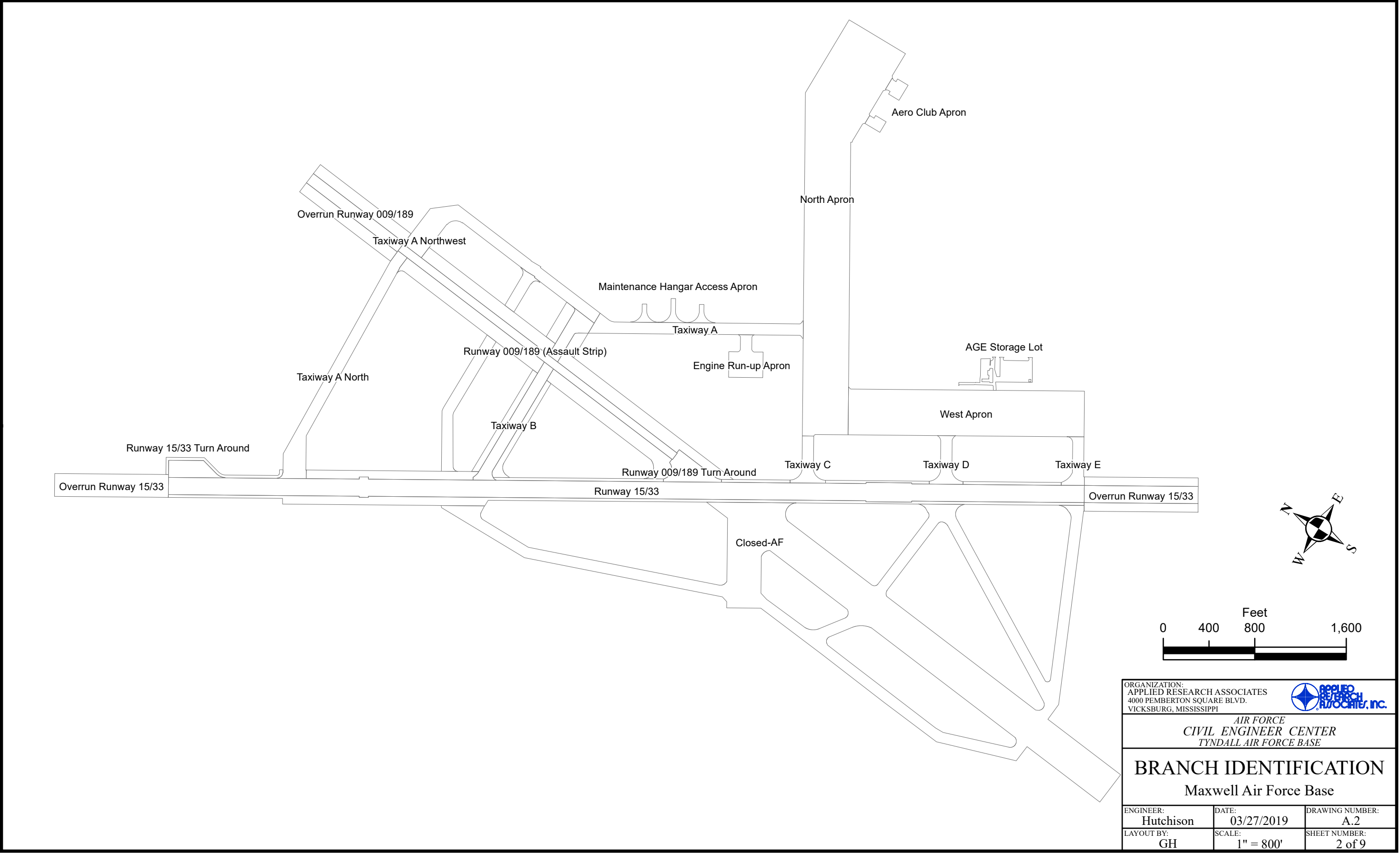
PAVER analyses and work plans included inspected airfield pavements maintained by Maxwell AFB. Closed pavements and shoulders were excluded.

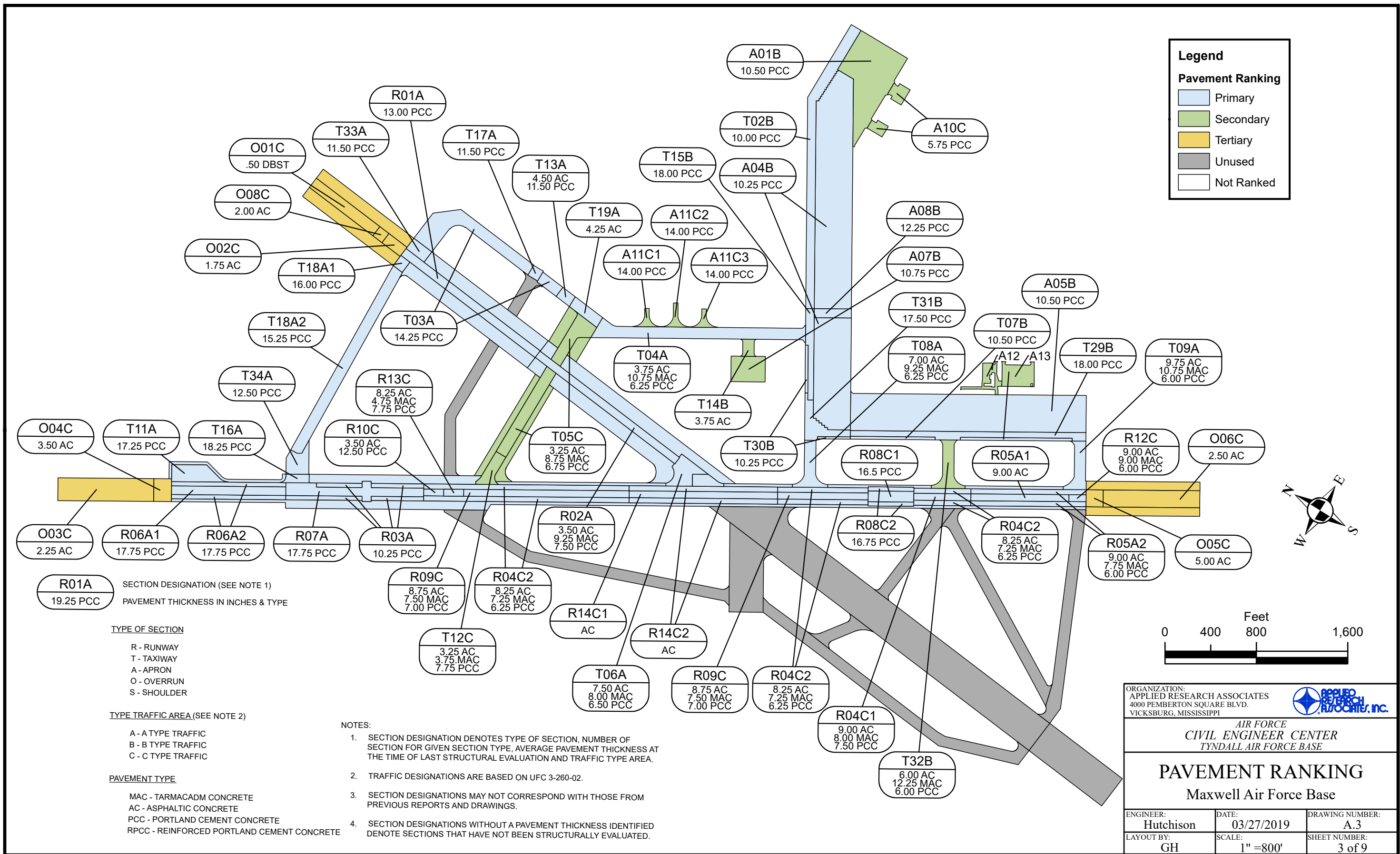
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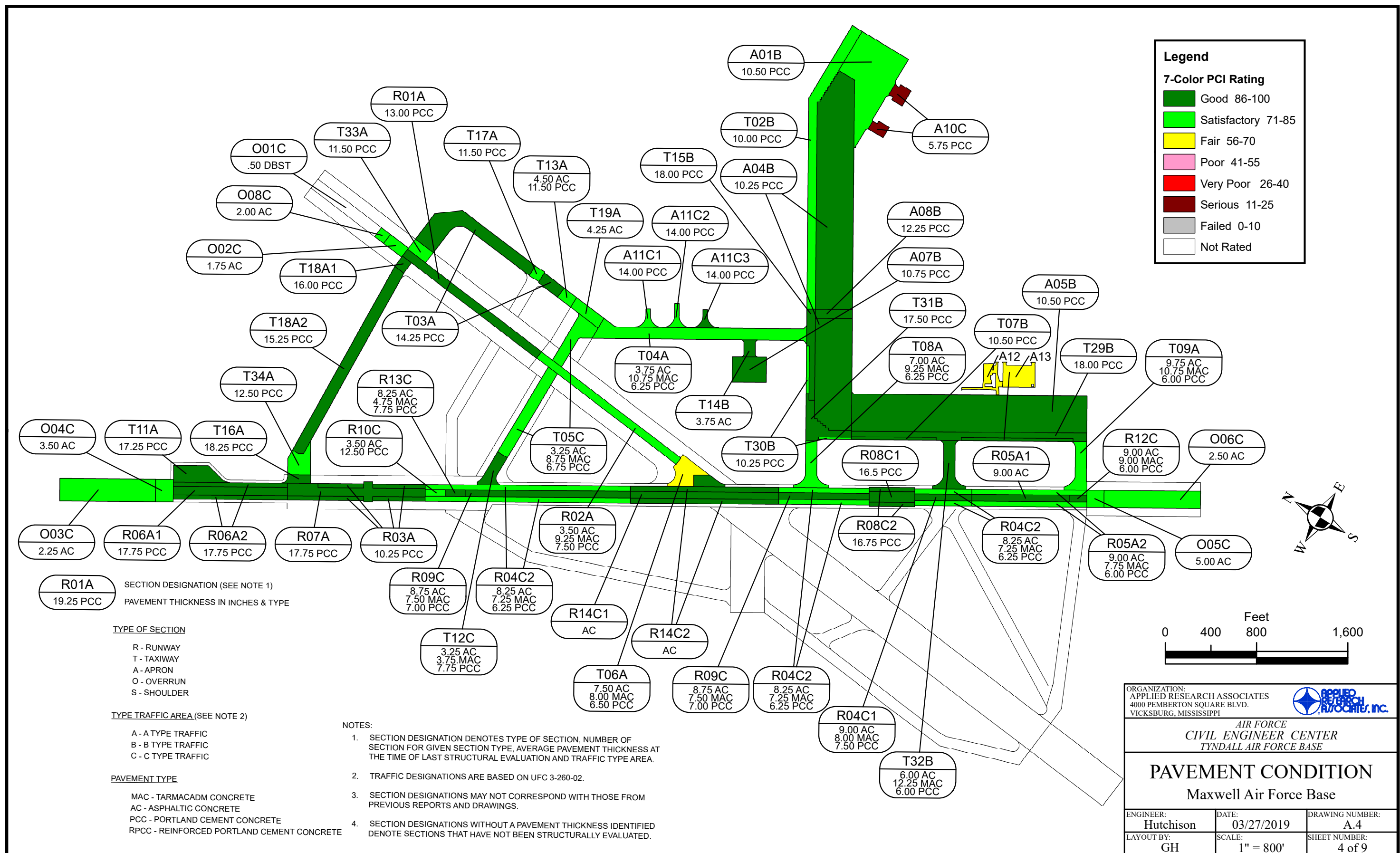
APPENDIX A
AIRFIELD MAPS

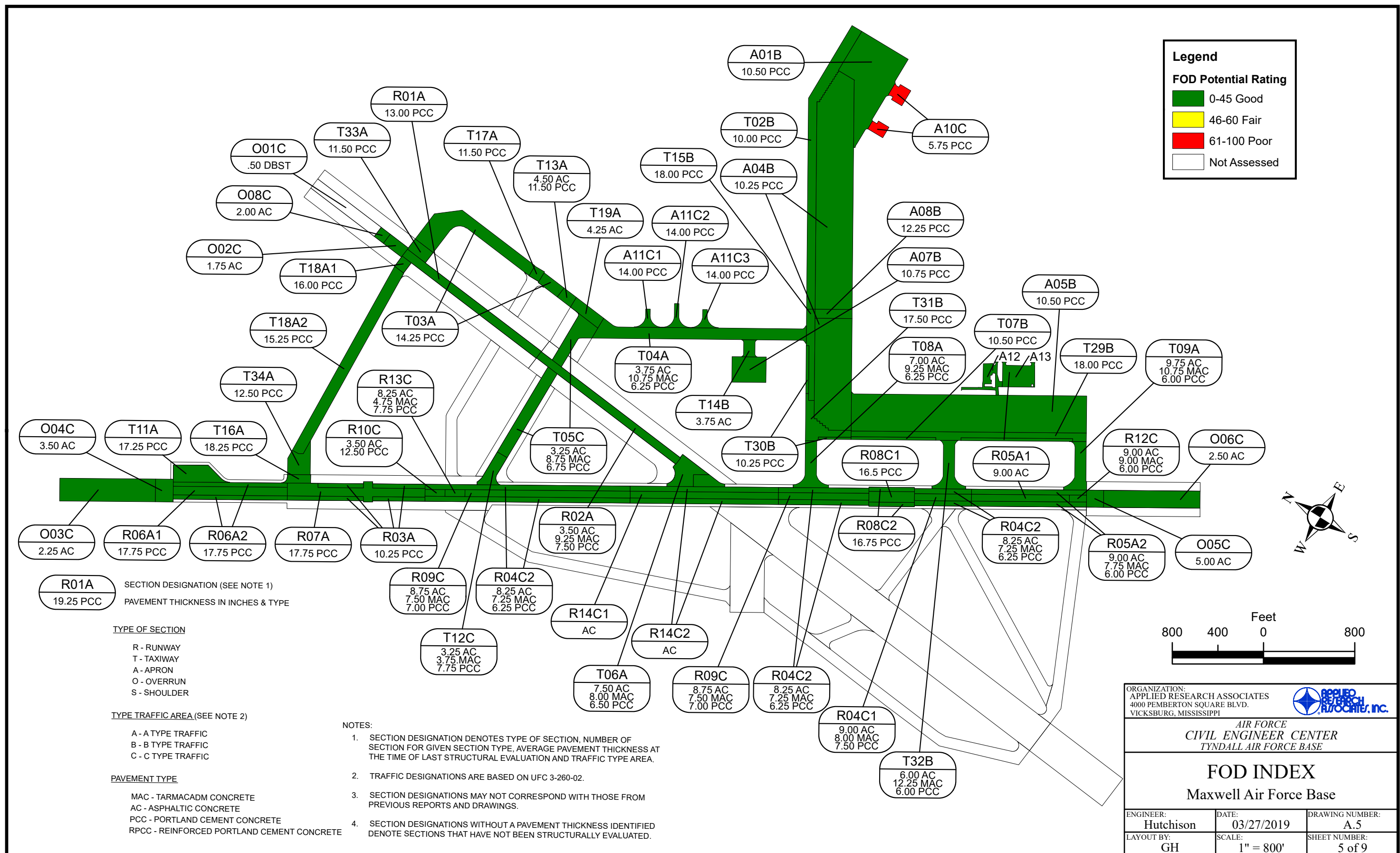
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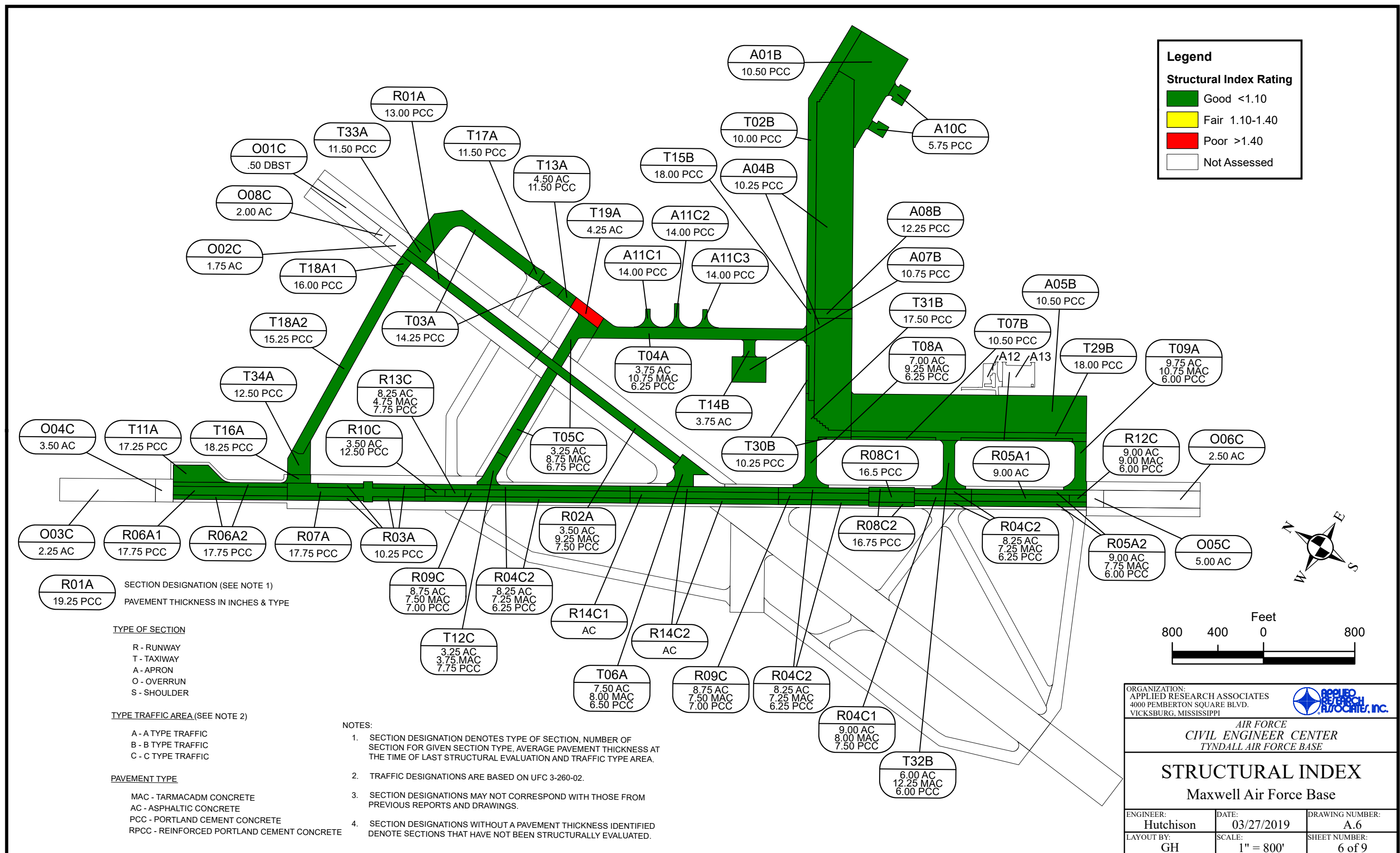


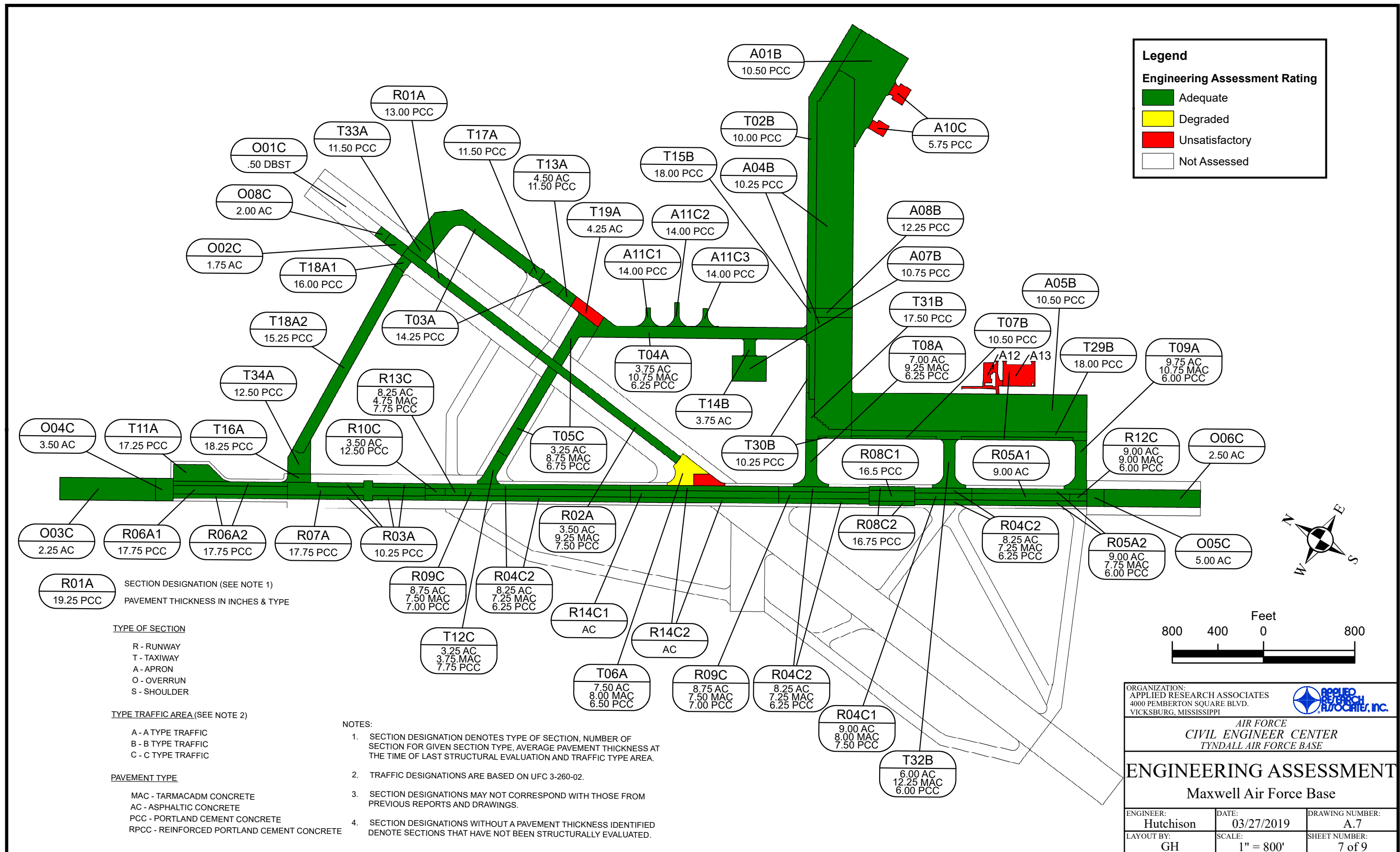


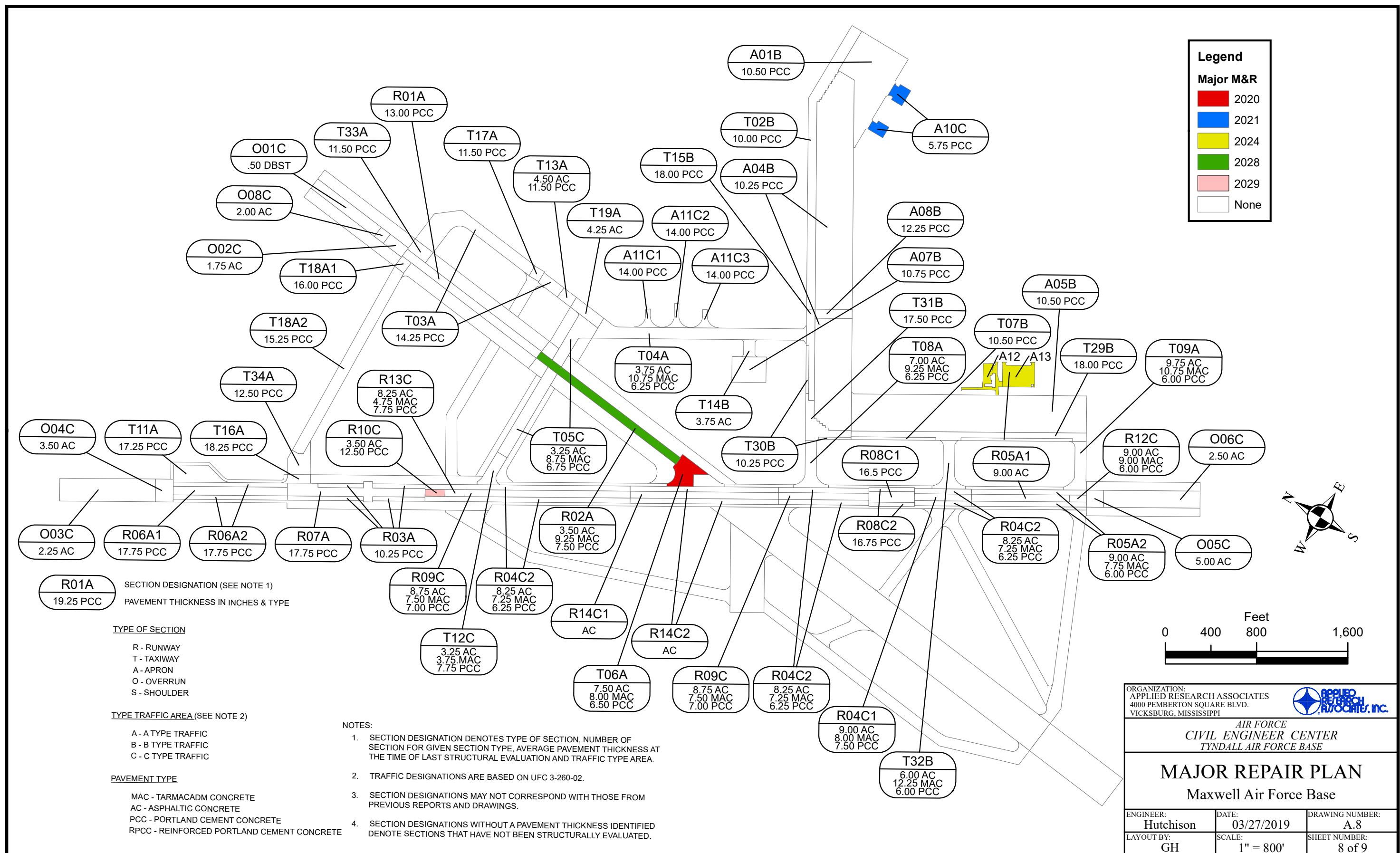


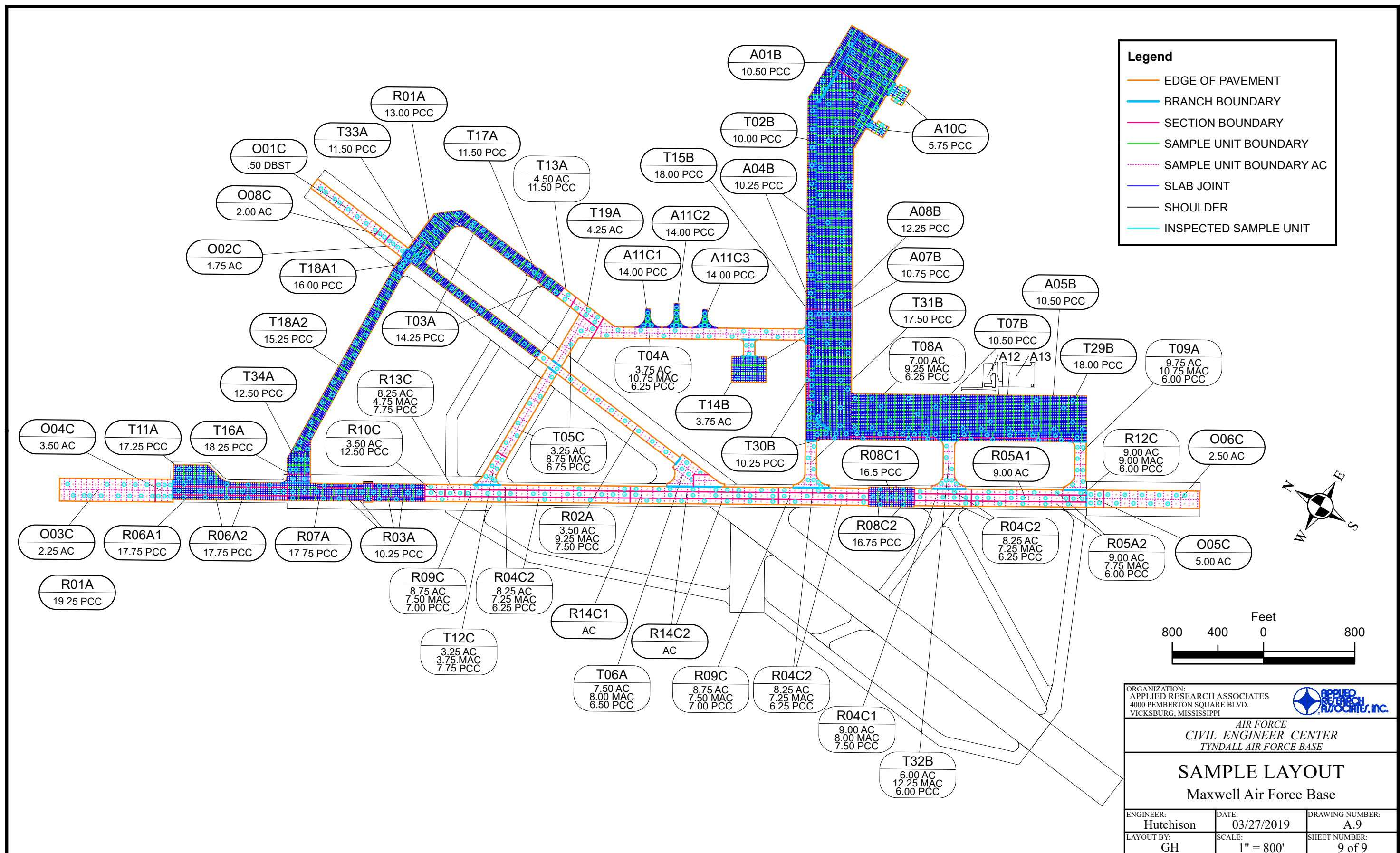












APPENDIX B
PAVEMENT INVENTORY AND CONDITION INFORMATION

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B.1 REAL PROPERTY PAVEMENT INVENTORY

Table B.1 provides the Facility Number, Real Property Unique ID (RPUID), Category Code (CATCODE), Branch Name, Section ID, Rank, and Pavement Area (ft² and yd²) for airfield pavements at Maxwell AFB.

Table B.1. Real Property Inventory.

Facility Number	RPUID	CATCODE	Branch Name	Branch ID	Rank	Branch PCI	Area (ft ²)	Area (yd ²)
Maxwell AFB Airfield Pavements								
1070	TBD09	452258	AGE Storage Lot	OAAGE	T	66	84,955	9,439
01255	474394	116664	Engine Run-up Apron	OARUNUP	S	91	82,580	9,176
03001	474380	113321	Aero Club Apron	OAAERO	T	14	36,178	4,020
03001	474380	113321	North Apron	APNORTH	P,S	87	1,589,429	176,603
03001	474380	113321	West Apron	APWEST	P	89	825,478	91,720
03017	474389	112211	Taxiway A North	TWANORTH	P	89	251,648	27,961
03018	474390	112211	Taxiway A Northwest	TWANW	P	87	253,242	28,138
03020	474392	112211	Taxiway A	TWA	P	82	191,550	21,283
03021	474393	112211	Taxiway C	TWC	P	85	51,200	5,689
03023	474395	112211	Taxiway E	TWE	P	84	48,600	5,400
03024	474396	112211	Taxiway D	TWD	S	89	53,000	5,889
03030	474403	111111	Runway 15/33	RW15/33	P	91	1,405,965	156,218
03030	474403	111111	Runway 15/33 Turn Around	TW15/33 T	P	99	57,420	6,380
03031	474404	111115	Overrun Runway 009/189	OR009/189	T	80	30,000	3,333
03038	474408	112211	Taxiway B	TWB	S	84	184,407	20,490
03040	476424	111115	Overrun Runway 15/33	OR15/33	T	78	348,485	38,721
03041	476425	111111	Runway 009/189 (Assault Strip)	RW009/189	P	84	218,433	24,270
03041	476425	111111	Runway 009/189 Turn Around	TW009/189T	P	70	72,380	8,042
TBD01	TBD01	113321	Maintenance Hangar Access Apron	OAHANG	T	85	36,084	4,009
TBD10	TBD10	852269	Parking Fuel Trucks	OAFUELS	T	88	66,531	7,392

B.2 PAVER/RPAD COMPARISON

Table B.2 is a list of the acronyms used in Table. B.3. Table B.3 is a list of discrepancies between PAVER real property and the RPAD provided.

Table B.2. Header Acronyms.

Acronym	Definition
RPUID	Real Property Unique Identifier
FAC	Facility Analysis Category
CATCODE	Category Code
RPAD	Real Property Assessment Database

Table B.3. PAVER/RPAD Comparison.

Facility Number	RPUID	FAC	CATCODE	Total Area (yd ²)		Comment and Action Items
				PAVER	RPAD	
01255	474394	1131	116664	9,178	ND	Not listed in RPAD
(Page 1 of 2)						

Table B.3. PAVER/RPAD Comparison (continued).

Facility Number	RPUID	FAC	CATCODE	Total Area (yd ²)		Comment and Action Items
				PAVER	RPAD	
03001	474380	1131	113321	272,342	264,758	Area Correction required.
03014	474387	1121	112211	29,358	27,627	Area Correction required.
03015	474388	1121	112211	28,974	28,132	Area Correction required.
03017	474389	1121	112211	27,961	22,600	Area Correction required.
03018	474390	1121	112211	28,140	28,471	Area Correction required.
03020	474392	1121	112211	21,284	3,800	Area Correction required.
03021	474393	1121	112211	5,692	5,222	Area Correction required.
03023	474395	1121	112211	5,405	4,970	Area Correction required.
03024	474396	1121	112211	5,894	5,856	Area Correction required.
03025	474397	1121	112211	20,326	22,945	Area Correction required.
03026	474398	1121	112211	11,245	11,655	Area Correction required.
03027	474399	1121	112211	21,652	19,527	Area Correction required.
03028	474400	1121	112211	18,028	13,918	Area Correction required.
03029	474401	1121	112211	5,167	5,483	Area Correction required.
03030	474403	1111	111111	275,482	213,022	Area Correction required.
03031	474404	1113	111115	11,093	60,231	Area Correction required.
03035	474405	1121	112211	22,687	35,637	Area Correction required.
03038	474408	1121	112211	20,080	23,802	Area Correction required.
03039	474409	1111	111111	112,642	129,184	Area Correction required.
03040	476424	1113	111115	38,726	54,725	Area Correction required.
03041	476425	1111	111111	32,322	106,770	Area Correction required.
1070	TBD09	8521	452258	9,439	ND	Open Storage, Air Freight - Not listed in RPAD
TBD01	TBD01	1131	113321	3,999	ND	Apron - Not listed in RPAD
TBD02	TBD02	1113	111115	26,685	ND	Overrun, Paved - Not listed in RPAD
TBD03	TBD03	1165	116642	57,477	ND	Shoulder pavement not listed in RPAD
TBD04	TBD04	1165	116642	76,247	ND	Shoulder pavement not listed in RPAD
TBD05	TBD05	1165	116642	17,981	ND	Shoulder pavement not listed in RPAD
TBD06	TBD06	1165	116642	16,593	ND	Shoulder pavement not listed in RPAD
TBD07	TBD07	1165	116642	22,187	ND	Shoulder pavement not listed in RPAD
TBD04	TBD04	1165	116642	686,223	ND	Shoulder pavement not listed in RPAD
TBD05	TBD05	1165	116642	161,831	ND	Shoulder pavement not listed in RPAD
TBD06	TBD06	1165	116642	149,333	ND	Shoulder pavement not listed in RPAD
TBD07	TBD07	1165	116642	199,682	ND	Shoulder pavement not listed in RPAD

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B.3 BRANCH USE INVENTORY

Table B.4 lists the branch use information for airfield pavements surveyed at Maxwell AFB.

Table B.4. Branch Use Information.

Branch Use	Pavement			Area-Weighted Average		Deterioration Rate (PCI pts/yr)	Rating
	Area (ft ²)	Area (%)	Sections	Age	PCI		
Apron	2,654,630	47	18	25	86	0.55	Good
Overrun	378,463	7	6	6	78	3.60	Good
Runway	1,429,389	25	18	16	91	0.58	Good
Taxiway	1,159,999	21	18	20	86	0.70	Good
Total	5,622,481	100	60	20	87	0.65	Good

B.4 PAVEMENT TYPE

Table B.5 provides the pavement surface information for Maxwell AFB airfield pavements.

Table B.5. Pavement Surface Type Information.

Surface Type	Pavement			Area-Weighted Average		Deterioration Rate (PCI pts/yr)	Rating
	Area (ft ²)	Area (%)	Sections	Age	PCI		
AC	601,160	10.7	13	7	80	3.1	Good
APC	1,437,348	25.6	17	8	85	1.8	Good
PCC	3,583,973	63.7	30	28	89	0.4	Good
Total	5,622,481	100	60	20	87	0.7	Good

B.5 INSPECTION DATA SUMMARY BY AGE GROUP

Table B.6 presents the inspection data summary by age and pavement type.

Table B.6. Inspection Data Summary by Age.

Age at Inspection	Pavement			Area-Weighted Average PCI	Deterioration Rate (PCI pts/year)	Rating
	Area (ft²)	Area (%)	Sections			
AC Pavement						
00-02	--	--	--	--	--	--
03-05	127,242	21.2	4	94	1.6	Good
06-10	443,526	73.8	8	77	3.9	Good
11-15	--	--	--	--	--	--
16-20	30,392	5.1	1	65	2.2	Fair
21-25	--	--	--	--	--	--
26-30	--	--	--	--	--	--
31-35	--	--	--	--	--	--
36-40	--	--	--	--	--	--
41-50	--	--	--	--	--	--
Over 50	--	--	--	--	--	--
All	601,160	100	13	80	3.4	Good
APC Pavement						
00-02	--	--	--	--	--	--
03-05	910,663	63.4	13	89	2.8	Good
06-10	368,985	25.7	2	83	2.8	Good
(Page 1 of 2)						

(Page 1 of 2)

Table B.6. Inspection Data Summary by Age (concluded).

Age at Inspection	Pavement			Area-Weighted Average PCI	Deterioration Rate (PCI pts/year)	Rating
	Area (ft²)	Area (%)	Sections			
APC Pavement						
11-15	--	--	--	--	--	--
16-20	--	--	--	--	--	--
21-25	107,583	7.5	1	75	1.1	Good
26-30	--	--	--	--	--	--
31-35	--	--	--	--	--	--
36-40	--	--	--	--	--	--
41-50	--	--	--	--	--	--
50+	50,117	3.5	1	59	0.6	Fair
All	1,437,348	100	17	85	1.8	Good
PCC Pavement						
00-02	--	--	--	--	--	--
03-05	--	--	--	--	--	--
06-10	--	--	--	--	--	--
11-15	35,995	1.0	3	85	1.1	Good
16-20	--	--	--	--	--	--
21-25	2,523,322	70.4	15	90	0.5	Good
26-30	--	--	--	--	--	--
31-35	877,600	24.5	10	88	0.4	Good
36-40	--	--	--	--	--	--
41-50	--	--	--	--	--	--
Over 50	147,056	4.1	2	74	0.4	Good
All	3,583,973	100	30	89	0.4	Good
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B.6 INSPECTION DATA SUMMARY BY PAVEMENT RANK

Table B.7 presents the inspection data summary by pavement rank.

Table B.7. Inspection Data Summary by Rank.

Branch Use	Pavement Rank	Pavement Type	Pavement			Area-Weighted Average		Deterioration Rate (PCI pts/yr)	Rating
			Area (ft²)	Area (%)	Sections	Age	PCI		
Apron	Primary	AC	--	--	--	--	--	--	--
		PCC	2,140,481	38.1	9	25	88	0.5	Good
	Secondary	AC	100,523	1.8	3	10	69	3.1	Fair
		PCC	413,626	7.4	6	29	79	0.7	Good
	Tertiary	AC	--	--	--	--	--	--	--
		PCC	--	--	--	--	--	--	--
Overrun	Primary	AC	--	--	--	--	--	--	--
		PCC	--	--	--	--	--	--	--
	Secondary	AC	--	--	--	--	--	--	--
		PCC	--	--	--	--	--	--	--

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Table B.7. Inspection Data Summary by Rank (concluded).

Branch Use	Pavement Rank	Pavement Type	Pavement			Area-Weighted Average		Deterioration Rate (PCI pts/yr)	Rating
			Area (ft²)	Area (%)	Sections	Age	PCI		
	Tertiary	AC	378,463	6.7	6	6	78	3.7	Good
		PCC	--	--	--	--	--	--	--
Runway	Primary	AC	92,001	1.6	3	4	97	0.8	Good
		APC	825,551	14.7	8	7	87	1.8	Good
		PCC	511,837	9.1	7	34	95	0.1	Good
	Secondary	AC	--	--	--	--	--	--	--
		PCC	--	--	--	--	--	--	--
	Tertiary	AC	--	--	--	--	--	--	--
		PCC	--	--	--	--	--	--	--
Taxiway	Primary	AC	408,200	7.3	7	13	80	1.5	Good
		PCC	518,029	9.2	8	32	90	0.3	Good
	Secondary	AC	233,770	4.2	3	5	85	3.0	Good
		PCC	--	--	--	--	--	--	--
	Tertiary	AC	--	--	--	--	--	--	--
		PCC	--	--	--	--	--	--	--
Total			5,622,481	100.0	60	20	87	0.7	Good
(Page 2 of 2)									

B.7 SECTIONS NOT INSPECTED AND EXCLUDED FROM CONDITION ANALYSIS AND WORK PLANS

ARA completed a cursory PCI on the inventoried and shoulder pavements. Table B.8 shows the sections not inspected.

Table B.8. Excluded Sections.

Branch Name	Section ID	Comment
Overrun Runway 009/189	O01C	Poor
Runway Overrun 009/189 Shoulders	S06D	Good
Runway Overrun 15/33 Shoulders	S05D	Poor
Runway 009/189 Shoulders	S02D	Fair
Runway 009/189 Shoulders	S03D	Fair
Runway 15/33 Shoulders	S01D	Good
Taxiway B Shoulders	S04D	Good

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APPENDIX C

PCI DETERIORATION RATE SUMMARY

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C.1 BRANCH AND SECTION PCI COMPARISONS

A comparison was made of the 2015 Maxwell airfield pavements PCI results and the results from the current survey of sections which could be compared. Table C.1 illustrates the comparisons, when they can be made, between results of both surveys. Unusual deterioration rates or significant changes in PCI since the last survey have been noted and explained where possible. Deterioration rate 1 in the following table is calculated as $(100 - \text{Current PCI}) / (\text{Age of Section})$. Deterioration rate 2 represents the difference in PCI from the previous inspection divided by the years since the previous inspection.

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Table C.1 PCI Deterioration Rate Summary.

Branch				Section									
Name	PCI 2015	PCI 2019	PCI Change	ID	Surface Type	Section Rank	Age	PCI 2015	PCI 2019	PCI Change	Det Rate 1 Const.	Det Rate 2 Inspect.	Comments
Aero Club Apron	14	14	0	A10C	PCC	T	74	14	14	0	-1.16	+0.00	--
AGE Storage Lot	76	66	-10	A12	AC	T	16	76	65	-11	-2.19	-2.75	--
				A13	AC	T	10	75	66	-9	-3.40	-2.32	--
Engine Run-up Apron	94	91	-3	A07B	PCC	S	34	92	91	-1	-0.26	-0.25	--
				T14B	AC	S	5	100	90	-10	-2.00	-2.49	--
Maintenance Hangar Access Apron	82	85	3	A11C1	PCC	T	15	81	83	2	-1.13	+0.50	Shattered slabs were not observed during the 2019 inspection
				A11C2	PCC	T	14	77	79	2	-1.50	+0.50	Development of medium-severity joint seal damage along with low-severity linear cracks since construction. High-severity joint seal damage was not observed during the 2019 inspection
				A11C3	PCC	T	14	90	93	3	-0.50	+0.75	High-severity joint seal damage was not observed during the 2019 inspection
North Apron	85	87	2	A01B	PCC	S	23	85	84	-1	-0.70	-0.25	--
				A04B	PCC	P	23	84	88	4	-0.52	+1.00	Corner breaks, scaling, and shrinkage cracks were not observed during the 2019 inspection
				A08B	PCC	P	23	99	99	0	-0.04	+0.00	--
				T02B	PCC	P	34	84	85	1	-0.44	+0.25	Decrease in quantity of corner breaks since the previous inspection
				T15B	PCC	P	23	97	93	-4	-0.30	-1.00	--
ND=No Data, NA=Not Applicable, Note: Det Rate 1 is the deterioration rate since initial construction. Det Rate 2 is the deterioration since the last PCI survey.													
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Table C.1 PCI Deterioration Rate Summary (continued).

Branch				Section									
Name	PCI 2015	PCI 2019	PCI Change	ID	Surface Type	Section Rank	Age	PCI 2015	PCI 2019	PCI Change	Det Rate 1 Const.	Det Rate 2 Inspect.	Comments
North Apron	85	87	2	T30B	PCC	P	23	81	81	0	-0.83	+0.00	--
				T31B	PCC	P	23	96	94	-2	-0.26	-0.50	--
West Apron	88	89	1	A05B	PCC	P	23	88	89	1	-0.48	+0.25	Faulting, pumping, and scaling were not observed during the 2019 inspection
				T07B	PCC	P	34	89	88	-1	-0.35	-0.25	--
				T29B	PCC	P	34	90	90	0	-0.29	+0.00	--
Overrun Runway 009/189	98	80	-18	O02C	AAC	T	6	98	79	-19	-3.50	-4.73	Development of low-severity L&T cracking and weathering since construction. Increase in quantity of L&T cracking and the development of weathering since the previous inspection
				O08C	AC	T	6	100	83	-17	-2.83	-4.24	Development of low-severity L&T cracking and weathering since the previous inspection
Overrun Runway 15/33	97	78	-19	O03C	AC	T	6	100	79	-21	-3.50	-5.23	Development of low-severity L&T cracking and weathering since construction and the previous inspection
				O04C	AAC	T	6	100	79	-21	-3.50	-5.23	Development of low-severity L&T cracking and weathering since construction and the previous inspection
ND=No Data, NA=Not Applicable, Note: Det Rate 1 is the deterioration rate since initial construction. Det Rate 2 is the deterioration since the last PCI survey.													
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Table C.1 PCI Deterioration Rate Summary (continued).

Branch				Section									
Name	PCI 2015	PCI 2019	PCI Change	ID	Surface Type	Section Rank	Age	PCI 2015	PCI 2019	PCI Change	Det Rate 1 Const.	Det Rate 2 Inspect.	Comments
Overrun Runway 15/33	97	78	-19	O05C	AAC	T	6	94	78	-16	-3.67	-3.99	Development of low-severity L&T cracking and weathering since construction. Development of L&T cracking since the previous inspection
				O06C	AAC	T	6	94	77	-17	-3.83	-4.24	Development of low-severity L&T cracking and weathering since construction. Development of L&T cracking since the previous inspection
Runway 009/189 (Assault Strip)	85	84	-1	R01A	PCC	P	74	88	93	5	-0.09	+1.25	High- and medium-severity joint seal damage were not observed during the 2019 inspection
				R02A	APC	P	23	83	75	-8	-1.09	-1.99	--
Runway 15/33	73	91	18	R03A	PCC	P	23	88	86	-2	-0.61	-0.50	--
				R04C1	AAC	P	3	58	99	41	-0.33	+10.25	Major M&R since the previous inspection
				R04C2	APC	P	6	93	84	-9	-2.67	-2.24	--
				R05A1	AC	P	3	61	100	39	+0.00	+9.75	Major M&R since the previous inspection
				R05A2	APC	P	6	94	80	-14	-3.33	-3.49	Development of low-severity L&T cracking since the previous inspection
				R06A1	PCC	P	23	96	99	3	-0.04	+0.75	Corner spalls, large patches, shrinkage cracks, and small patches were not observed during the 2019 inspection
ND=No Data, NA=Not Applicable, Note: Det Rate 1 is the deterioration rate since initial construction. Det Rate 2 is the deterioration since the last PCI survey.													
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Table C.1 PCI Deterioration Rate Summary (continued).

Branch				Section									
Name	PCI 2015	PCI 2019	PCI Change	ID	Surface Type	Section Rank	Age	PCI 2015	PCI 2019	PCI Change	Det Rate 1 Const.	Det Rate 2 Inspect.	Comments
Runway 15/33	73	91	18	R06A2	PCC	P	23	96	97	1	-0.13	+0.25	Corner spalls were not recorded during the 2019 inspection
				R07A	PCC	P	23	97	99	2	-0.04	+0.50	Corner spalls and joint seal damage were not recorded during the 2019 inspection
				R08C1	PCC	P	23	93	93	0	-0.30	+0.00	--
				R08C2	PCC	P	23	98	98	0	-0.09	+0.00	--
				R09C	APC	P	3	60	90	30	-3.33	+7.50	Major M&R since the previous inspection
				R10C	AC	P	6	80	76	-4	-4.00	-1.00	Development of low-severity L&T cracking and weathering since the previous inspection
				R12A	APC	P	3	94	100	6	+0.00	+1.50	Major M&R since the previous inspection
				R13C	APC	P	3	83	92	9	-2.67	+2.26	Major M&R since the previous inspection
				R14C1	APC	P	3	NA	100	NA	+0.00	NA	Major M&R since the previous inspection
				R14C2	APC	P	3	NA	100	NA	+0.00	NA	Major M&R since the previous inspection
Runway 009/189 Turn Around	41	70	29	T06A	APC	P	73	60	59	-1	-0.56	-0.25	--
Runway 009/189 Turn Around	NA	70	NA	T35A	APC	P	3	NA	96	NA	-1.33	NA	--
Runway 15/33 Turn Around	95	99	4	T11A	PCC	P	23	95	99	4	-0.04	+1.00	Joint seal replacement since the previous inspection
Taxiway A	100	82	-18	T04A	APC	P	5	100	82	-18	-3.60	-4.48	Development of low-severity L&T cracking and weathering since construction and the previous inspection
ND=No Data, NA=Not Applicable, Note: Det Rate 1 is the deterioration rate since initial construction. Det Rate 2 is the deterioration since the last PCI survey.													
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Table C.1 PCI Deterioration Rate Summary (continued).

Branch				Section									
Name	PCI 2015	PCI 2019	PCI Change	ID	Surface Type	Section Rank	Age	PCI 2015	PCI 2019	PCI Change	Det Rate 1 Const.	Det Rate 2 Inspect.	Comments
Taxiway A North	89	89	0	T16A	PCC	P	23	93	95	2	-0.22	+0.50	Joint seal replacement since the previous inspection
				T18A1	PCC	P	33	80	87	7	-0.39	+1.74	Large patches and high-severity joint seal damage were not observed during the 2019 inspection
				T18A2	PCC	P	33	91	90	-1	-0.30	-0.25	--
				T34A	PCC	P	33	81	85	4	-0.45	+1.00	Durability cracking, faulting, joint spalls, and shrinkage cracks were not observed during the 2019 inspection
Taxiway A Northwest	91	87	-4	T03A	PCC	P	33	89	89	0	-0.33	+0.00	--
				T13A	APC	P	5	100	81	-19	-3.80	-4.73	Development of low-severity joint reflection cracking and weathering since construction and the previous inspection
				T17A	PCC	P	33	86	82	-4	-0.55	-1.00	--
				T19A	AC	P	5	100	79	-21	-4.20	-5.23	Development of low-severity L&T cracking and weathering since construction and the previous inspection
				T33A	PCC	P	33	92	85	-7	-0.45	-1.74	Development of high-severity joint seal damage and increase in quantity of low-severity linear cracks since the previous inspection
ND=No Data, NA=Not Applicable, Note: Det Rate 1 is the deterioration rate since initial construction. Det Rate 2 is the deterioration since the last PCI survey.													
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Table C.1 PCI Deterioration Rate Summary (concluded).

Branch				Section									
Taxiway B	100	84	-16	T05C	APC	S	5	100	83	-17	-3.40	-4.24	Development of low-severity L&T cracking and weathering since the previous inspection
				T12C	APC	S	5	100	89	-11	-2.20	-2.74	--
Taxiway C	100	85	-15	T08A	APC	P	5	100	85	-15	-3.00	-3.74	Development of low-severity L&T cracking and weathering since the previous inspection
Taxiway D	100	89	-11	T32B	APC	S	5	100	89	-11	-2.20	-2.74	--
Taxiway E	100	84	-16	T09A	APC	P	5	100	84	-16	-3.20	-3.99	Development of low-severity L&T cracking and weathering since the previous inspection
ND=No Data, NA=Not Applicable, Note: Det Rate 1 is the deterioration rate since initial construction. Det Rate 2 is the deterioration since the last PCI survey.													
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APPENDIX D
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APPENDIX E
PAVER WORK HISTORY REPORT

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Work History Report

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: A01B		Surface: PCC
L.C.D. 1/1/1996		Use: APRON	Rank: S	Length: 573.00 (ft)	Width: 475.00 (ft)	True Area: 274393.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	10.00	☒	Random slab replacement, reseal joint
1/1/1981	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Reseal joints and random slab replace
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☒	COE apron reconstruction and extensi

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: A04B		Surface: PCC
L.C.D. 1/1/1996		Use: APRON	Rank: P	Length: 2,946.00 (ft)	Width: 325.00 (ft)	True Area: 957323.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	10.00	☒	Random slab replacement, patch spalls
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, repair crac
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE apron reconstruction and extensi

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: A08B		Surface: PCC
L.C.D. 1/1/1996		Use: APRON	Rank: P	Length: 325.00 (ft)	Width: 80.00 (ft)	True Area: 26000.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	12.00	☒	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, repair crac
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	Original construction

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: T02B		Surface: PCC
L.C.D. 1/1/1985		Use: APRON	Rank: P	Length: 2,817.00 (ft)	Width: 75.00 (ft)	True Area: 211267.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	SL-RP	Slab Replacement	0.00	10.00	☒	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/2/1945	NU-IN	New Construction - Initial	0.00	10.50	☒	COE original construction

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: T15B		Surface: PCC
L.C.D. 1/1/1996		Use: APRON	Rank: P	Length: 640.00 (ft)	Width: 75.00 (ft)	True Area: 48000.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☒	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/2/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE original construction

Work History Report

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: T30B		Surface: PCC
L.C.D. 1/1/1996	Use: APRON	Rank: P	Length: 650.00 (ft)	Width: 25.00 (ft)	True Area: 16250.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☒	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE original construction

Network: Maxwell AFB		Branch: APNORTH North Apron		Section: T31B		Surface: PCC
L.C.D. 1/1/1996	Use: APRON	Rank: P	Length: 749.00 (ft)	Width: 75.00 (ft)	True Area: 56165.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☒	Remove and replace, COE 94B0048
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, seal cracks,
1/1/1982	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 82-0025
1/2/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE original construction

Network: Maxwell AFB		Branch: APWEST West Apron		Section: A05B		Surface: PCC
L.C.D. 1/1/1996	Use: APRON	Rank: P	Length: 2,064.00 (ft)	Width: 325.00 (ft)	True Area: 670734.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	10.00	☒	Random slab replacement, patch spalls
1/1/1985	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, repair crac
1/1/1980	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, reseal joint
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE original construction

Network: Maxwell AFB		Branch: APWEST West Apron		Section: T07B		Surface: PCC
L.C.D. 1/1/1985	Use: APRON	Rank: P	Length: 1,724.00 (ft)	Width: 25.00 (ft)	True Area: 43093.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	SL-RP	Slab Replacement	0.00	10.00	☒	Random slab replacement, seal cracks,
1/1/1980	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, MAX 79-0
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE original construction

Network: Maxwell AFB		Branch: APWEST West Apron		Section: T29B		Surface: PCC
L.C.D. 1/1/1985	Use: APRON	Rank: P	Length: 2,233.00 (ft)	Width: 50.00 (ft)	True Area: 111649.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	SL-RP	Slab Replacement	0.00	10.00	☒	Random slab replacement, seal cracks,
1/1/1980	SL-PC	Slab Replacement - PCC	0.00	10.00	☐	Random slab replacement, MAX 79-0
1/1/1945	NU-IN	New Construction - Initial	0.00	10.00	☒	COE original construction

Network: Maxwell AFB		Branch: OAAERO Aero Club Apron		Section: A10C		Surface: PCC
L.C.D. 1/1/1945	Use: APRON	Rank: S	Length: 573.00 (ft)	Width: 475.00 (ft)	True Area: 36205.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1945	NU-IN	New Construction - Initial	0.00	5.75	☒	Original Construction

Work History Report

Network: Maxwell AFB		Branch: OAAGE		AGE Storage Lot		Section: A12	Surface: AC
L.C.D. 1/1/2003		Use: APRON	Rank: S	Length: 304.00 (ft)	Width: 100.00 (ft)	True Area: 30392.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/2003	M-OL	Mill and Overlay	0.00	0.00	☒	Proj. # PNQS022691	
1/1/1943	NU-IN	New Construction - Initial	0.00	0.00	☒		

Network: Maxwell AFB		Branch: OAAGE		AGE Storage Lot		Section: A13	Surface: AC
L.C.D. 1/1/2009		Use: APRON	Rank: S	Length: 545.00 (ft)	Width: 100.00 (ft)	True Area: 54563.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/2009	M-OL	Mill and Overlay	0.00	0.00	☒	Estimated date based on engineering j	
1/1/1943	NU-IN	New Construction - Initial	0.00	0.00	☒		

Network: Maxwell AFB		Branch: OAHANG		Maintenance Hang		Section: A11C1	Surface: PCC
L.C.D. 1/1/2004		Use: APRON	Rank: S	Length: 284.00 (ft)	Width: 40.00 (ft)	True Area: 11253.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/2004	NU-IN	New Construction - Initial	0.00	14.00	☒	Original construction, PNQS02-9010	

Network: Maxwell AFB		Branch: OAHANG		Maintenance Hang		Section: A11C2	Surface: PCC
L.C.D. 1/1/2005		Use: APRON	Rank: S	Length: 333.00 (ft)	Width: 40.00 (ft)	True Area: 13344.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/2005	NU-IN	New Construction - Initial	0.00	14.00	☒	PNQS95-9004	

Network: Maxwell AFB		Branch: OAHANG		Maintenance Hang		Section: A11C3	Surface: PCC
L.C.D. 1/1/2005		Use: APRON	Rank: S	Length: 284.00 (ft)	Width: 40.00 (ft)	True Area: 11398.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/2005	NU-IN	New Construction - Initial	0.00	14.00	☒	PNQS95-9004	

Network: Maxwell AFB		Branch: OARUNUP		Engine Run-up Apr		Section: A07B	Surface: PCC
L.C.D. 1/1/1985		Use: APRON	Rank: S	Length: 223.00 (ft)	Width: 300.00 (ft)	True Area: 67033.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1985	NU-IN	New Construction - Initial	0.00	10.75	☒	Original construction	

Network: Maxwell AFB		Branch: OARUNUP		Engine Run-up Apr		Section: T14B	Surface: AC
L.C.D. 12/12/2014		Use: APRON	Rank: S	Length: 156.00 (ft)	Width: 100.00 (ft)	True Area: 15568.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,593.00	2.00	☒		
1/1/1997	NU-IN	New Construction - Initial	0.00	3.25	☒		

Work History Report

Network: Maxwell AFB		Branch: OR009/189		Overrun Runway 0	Section: O02C		Surface: AC
L.C.D. 11/18/2013		Use: OVERRU	Rank: T	Length: 198.00 (ft)	Width: 300.00 (ft)	True Area: 19810.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	84,401.00	2.00	☒	Complete reconstruction with DBST	
1/1/2002	CR-AC	Complete Reconstruction - AC	0.00	0.00	☒		
1/1/1958	NU-IN	New Construction - Initial	0.00	0.00	☒		
Network: Maxwell AFB		Branch: OR009/189		Overrun Runway 0	Section: O08C		Surface: AC
L.C.D. 11/11/2013		Use: OVERRU	Rank: T	Length: 800.00 (ft)	Width: 300.00 (ft)	True Area: 10115.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
11/11/2013	NC-AC	New Construction - AC	0.00	0.00	☒		
Network: Maxwell AFB		Branch: OR15/33		Overrun Runway 1	Section: O03C		Surface: AC
L.C.D. 11/18/2013		Use: OVERRU	Rank: T	Length: 840.00 (ft)	Width: 200.00 (ft)	True Area: 168000.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
11/18/2013	CR-AC	Complete Reconstruction - AC	558,427.00	2.00	☒	Seal cracks Original construction, COE 94B0048	
1/1/2000	CS-AC	Crack Sealing - AC	0.00	0.00	☐		
1/1/1997	NU-IN	New Construction - Initial	0.00	0.50	☒		
Network: Maxwell AFB		Branch: OR15/33		Overrun Runway 1	Section: O04C		Surface: AC
L.C.D. 11/18/2013		Use: OVERRU	Rank: T	Length: 156.00 (ft)	Width: 200.00 (ft)	True Area: 31100.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	558,427.00	2.00	☒	Original construction, COE 94B0048	
1/1/1996	NU-IN	New Construction - Initial	0.00	2.50	☒		
Network: Maxwell AFB		Branch: OR15/33		Overrun Runway 1	Section: O05C		Surface: AC
L.C.D. 11/18/2013		Use: OVERRU	Rank: T	Length: 150.00 (ft)	Width: 300.00 (ft)	True Area: 22508.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	558,427.00	2.00	☒	Complete reconstruction with DBST 0.5" DBST over AC, Original construction	
1/1/2000	CS-AC	Crack Sealing - AC	0.00	0.00	☐		
1/1/1998	CR-AC	Complete Reconstruction - AC	0.00	0.00	☒		
1/1/1958	NU-IN	New Construction - Initial	0.00	2.50	☒		
Network: Maxwell AFB		Branch: OR15/33		Overrun Runway 1	Section: O06C		Surface: AC
L.C.D. 11/18/2013		Use: OVERRU	Rank: T	Length: 846.00 (ft)	Width: 300.00 (ft)	True Area: 126930.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
11/18/2013	CM-OL-2	2 in Cold Mill & Overlay	558,427.00	2.00	☒	Seal cracks Complete reconstruction with DBST Original construction	
1/1/2000	CS-AC	Crack Sealing - AC	0.00	0.00	☐		
1/1/1998	CR-AC	Complete Reconstruction - AC	0.00	0.00	☒		
1/1/1958	NU-IN	New Construction - Initial	0.00	0.50	☒		

Work History Report

Network: Maxwell AFB		Branch: RW009/189 Runway 009/189 (		Section: R01A		Surface: PCC
L.C.D. 1/1/1945		Use: RUNWAY	Rank: P	Length: 1,478.00 (ft)	Width: 75.00 (ft)	True Area: 110851.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1981	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 77-0237
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☒	COE original construction

Network: Maxwell AFB		Branch: RW009/189 Runway 009/189 (		Section: R02A		Surface: AC
L.C.D. 1/1/1996		Use: RUNWAY	Rank: P	Length: 1,537.00 (ft)	Width: 70.00 (ft)	True Area: 107583.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048
1/1/1985	ST-SS	Surface Treatment - Slurry Seal	0.00	0.00	☐	Slurry seal 70' along center line, MAX
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	Original construction

Network: Maxwell AFB		Branch: RW15/33 Runway 15/33		Section: R03A		Surface: PCC
L.C.D. 1/1/1996		Use: RUNWAY	Rank: P	Length: 1,961.00 (ft)	Width: 38.00 (ft)	True Area: 74530.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	1.50	☒	Random slab replacement, patch spalls
1/1/1983	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal PCC joints, MAX 84-0014
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☒	COE original construction

Network: Maxwell AFB		Branch: RW15/33 Runway 15/33		Section: R04C1		Surface: AC
L.C.D. 6/1/2016		Use: RUNWAY	Rank: P	Length: 500.00 (ft)	Width: 60.00 (ft)	True Area: 30000.00 (ft ²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒	Mill and overlay, COE 94B0048 Seal cracks, MAX 84-0049 Seal cracks, overlay 150' wide, MAX COE overlay Overlay COE reconstruction Original construction
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☒	
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	

Work History Report

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R04C2	Surface: AC
L.C.D. 2/11/2013	Use: RUNWAY	Rank: P	Length: 4,391.00 (ft)	Width: 90.00 (ft)	True Area: 278222.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	1,107,582.00	2.00	☒	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	COE original construction

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R05A1	Surface: AC
L.C.D. 6/1/2016	Use: RUNWAY	Rank: P	Length: 1,008.00 (ft)	Width: 60.00 (ft)	True Area: 51501.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	0.00	☒	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	COE original construction

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R05A2	Surface: AC
L.C.D. 2/11/2013	Use: RUNWAY	Rank: P	Length: 1,008.00 (ft)	Width: 90.00 (ft)	True Area: 90763.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	1,107,582.00	2.00	☒	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	0.00	☒	Mill and overlay
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	COE original construction

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R06A1	Surface: PCC
L.C.D. 1/1/1996	Use: RUNWAY	Rank: P	Length: 1,001.00 (ft)	Width: 75.00 (ft)	True Area: 75062.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	NU-IN	New Construction - Initial	0.00	17.00	☒	Original construction, COE 94B0048

Work History Report

Network: Maxwell AFB **Branch:** RW15/33 Runway 15/33 **Section:** R06A2 **Surface:** PCC
L.C.D. 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,001.00 (ft) **Width:** 75.00 (ft) **True Area:** 75062.00 (ft²)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	NU-IN	New Construction - Initial	0.00	17.00	☒	Original construction, COE 94B0048

Network: Maxwell AFB **Branch:** RW15/33 Runway 15/33 **Section:** R07A **Surface:** PCC
L.C.D. 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 1,110.00 (ft) **Width:** 98.00 (ft) **True Area:** 108832.00 (ft²)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	SL-RP	Slab Replacement	0.00	17.00	☒	Replace keel slabs, COE 94B0048
1/1/1983	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 84-0014
1/1/1945	NU-IN	New Construction - Initial	0.00	10.50	☒	COE original construction

Network: Maxwell AFB **Branch:** RW15/33 Runway 15/33 **Section:** R08C1 **Surface:** PCC
L.C.D. 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 400.00 (ft) **Width:** 94.00 (ft) **True Area:** 37500.00 (ft²)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☒	Remove AC and replace with PCC, C
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	Original construction

Network: Maxwell AFB **Branch:** RW15/33 Runway 15/33 **Section:** R08C2 **Surface:** PCC
L.C.D. 1/1/1996 **Use:** RUNWAY **Rank:** P **Length:** 400.00 (ft) **Width:** 75.00 (ft) **True Area:** 30000.00 (ft²)

Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☒	Remove AC and replace with PCC, C
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	Original construction

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Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R09C		Surface: AC
L.C.D. 6/1/2016		Use: RUNWAY	Rank: P	Length: 3,541.00 (ft)	Width: 60.00 (ft)	True Area: 134482.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒		
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay	
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048	
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049	
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX	
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay	
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay	
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction	
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	Original construction	

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R10C		Surface: AC
L.C.D. 2/11/2013		Use: RUNWAY	Rank: P	Length: 350.00 (ft)	Width: 90.00 (ft)	True Area: 10500.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	31,920.00	2.00	☒		
1/1/1999	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay	
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048	
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049	
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX	
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay	
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay	
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction	
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	Mill and overlay	

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R12A		Surface: AC
L.C.D. 6/1/2016		Use: RUNWAY	Rank: P	Length: 1,008.00 (ft)	Width: 60.00 (ft)	True Area: 9003.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒		
1/1/1996	OL-AS	Overlay - AC Structural	0.00	3.00	☒	Mill and overlay, COE 94B0048	
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	Seal cracks, MAX 84-0049	
1/1/1980	OL-AT	Overlay - AC Thin	0.00	1.50	☒	Seal cracks, overlay 150' wide, MAX	
1/1/1964	OL-AS	Overlay - AC Structural	0.00	3.00	☒	COE overlay	
1/1/1958	OL_2	2 in overlay	0.00	2.00	☒	Overlay	
1/1/1945	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction	
1/1/1943	NC-AC	New Construction - AC	0.00	0.00	☒		

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R13C		Surface: AC
L.C.D. 6/1/2016		Use: RUNWAY	Rank: P	Length: 350.00 (ft)	Width: 90.00 (ft)	True Area: 10500.00 (ft ²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒		
2/11/2013	NC-AC	New Construction - AC	0.00	0.00	☒		

Work History Report

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R14C1	Surface: AC
L.C.D. 6/1/2016		Use: RUNWAY	Rank: P	Length: 3,541.00 (ft)	Width: 60.00 (ft)	True Area: 78000.00 (ft²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒	
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	177,840.00	0.00	☒	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	0.00	☒	
1/1/1996	OL-AS	Overlay - AC Structural	0.00	0.00	☒	
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	
1/1/1980	OL-AT	Overlay - AC Thin	0.00	0.00	☒	
1/1/1964	OL-AS	Overlay - AC Structural	0.00	0.00	☒	
1/1/1958	OL_2	2 in overlay	131,040.00	0.00	☒	
1/2/1945	NC-AC	New Construction - AC	0.00	0.00	☒	

Network: Maxwell AFB		Branch: RW15/33		Runway 15/33	Section: R14C2	Surface: AC
L.C.D. 6/1/2016		Use: RUNWAY	Rank: P	Length: 4,391.00 (ft)	Width: 90.00 (ft)	True Area: 116998.00 (ft²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
6/1/2016	M-OL	Mill and Overlay	0.00	0.00	☒	
2/11/2013	CM-OL-2	2 in Cold Mill & Overlay	177,840.00	0.00	☒	
1/1/1999	OL-AS	Overlay - AC Structural	0.00	0.00	☒	
1/1/1996	OL-AS	Overlay - AC Structural	0.00	0.00	☒	
1/1/1984	CS-AC	Crack Sealing - AC	0.00	0.00	☐	
1/1/1980	OL-AT	Overlay - AC Thin	0.00	0.00	☒	
1/1/1964	OL-AS	Overlay - AC Structural	0.00	0.00	☒	
1/1/1958	OL_2	2 in overlay	131,040.00	0.00	☒	
1/2/1945	NC-AC	New Construction - AC	0.00	0.00	☒	

Network: Maxwell AFB		Branch: TW009/189T		Runway009/189 T	Section: T06A	Surface: AC
L.C.D. 1/1/1946		Use: TAXIWAY	Rank: P	Length: 342.00 (ft)	Width: 175.00 (ft)	True Area: 50117.00 (ft²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
1/1/1985	ST-SS	Surface Treatment - Slurry Seal	0.00	0.00	☐	Slurry seal, MAX 85-0022
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒	COE reconstruction project
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒	Original construction

Network: Maxwell AFB		Branch: TW009/189T		Runway009/189 T	Section: T35A	Surface: AC
L.C.D. 6/1/2016		Use: TAXIWAY	Rank: P	Length: 342.00 (ft)	Width: 175.00 (ft)	True Area: 22351.00 (ft²)
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments
6/1/2016	NC-AC	New Construction - AC	0.00	0.00	☒	

Work History Report

Network: Maxwell AFB		Branch: TW15/33 T Runway 15/33 Tur		Section: T11A		Surface: PCC	
L.C.D. 1/1/1996		Use: TAXIWAY	Rank: P	Length: 383.00 (ft)	Width: 150.00 (ft)	True Area: 57425.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
6/1/2017	JS-LC	JT SEAL DMG	0.00	0.00	☐	Original construction, COE 94B0048	
1/1/1996	NU-IN	New Construction - Initial	0.00	17.00	☒		

Network: Maxwell AFB		Branch: TWANORTH Taxiway A North		Section: T16A		Surface: PCC	
L.C.D. 1/1/1996		Use: TAXIWAY	Rank: P	Length: 76.00 (ft)	Width: 212.00 (ft)	True Area: 16046.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
6/1/2017	JS-LC	JT SEAL DMG	0.00	0.00	☐	Remove and replace, COE 94B0048	
1/1/1996	CR-PC	Complete Reconstruction - PCC	0.00	17.00	☒		
1/1/1986	SL-PC	Slab Replacement - PCC	0.00	11.50	☐	Random slab replacement and seal cra	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☒	COE original construction and extensi	

Network: Maxwell AFB		Branch: TWANORTH Taxiway A North		Section: T18A1		Surface: PCC	
L.C.D. 1/1/1986		Use: TAXIWAY	Rank: P	Length: 112.00 (ft)	Width: 100.00 (ft)	True Area: 11215.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☒	Random slab replacement and seal cra	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	10.75	☒	COE original construction and extensi	

Network: Maxwell AFB		Branch: TWANORTH Taxiway A North		Section: T18A2		Surface: PCC	
L.C.D. 1/1/1986		Use: TAXIWAY	Rank: P	Length: 1,835.00 (ft)	Width: 100.00 (ft)	True Area: 183486.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☒	Random slab replacement and seal cra	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☒	COE original construction and extensi	

Network: Maxwell AFB		Branch: TWANORTH Taxiway A North		Section: T34A		Surface: PCC	
L.C.D. 1/1/1986		Use: TAXIWAY	Rank: P	Length: 204.00 (ft)	Width: 200.00 (ft)	True Area: 40900.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☒	Random slab replacement and seal cra	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☒	COE original construction and extensi	

Network: Maxwell AFB		Branch: TWANW Taxiway A Northwest		Section: T03A		Surface: PCC	
L.C.D. 1/1/1986		Use: TAXIWAY	Rank: P	Length: 1,767.00 (ft)	Width: 100.00 (ft)	True Area: 176681.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☒	Random slab replacement, seal cracks,	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☒	COE runway extension project	

Work History Report

Network: Maxwell AFB		Branch: TWANW		Taxiway A Northwest		Section: T13A	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: P	Length: 141.00 (ft)	Width: 100.00 (ft)	True Area: 14127.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	423,597.00	2.00	☒	Mill, MAX 82-0026, Overlay, MAX 8 Overlay COE reconstruction, MACADAM lev Original construction	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☒		
1/1/1958	OL-AT	Overlay - AC Thin	0.00	1.50	☒		
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒		
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒		
Network: Maxwell AFB		Branch: TWANW		Taxiway A Northwest		Section: T17A	Surface: PCC
L.C.D. 1/1/1986		Use: TAXIWAY	Rank: P	Length: 100.00 (ft)	Width: 100.00 (ft)	True Area: 10000.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☒	Random slab replacement and seal cra	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☒	COE original construction and extensi	
Network: Maxwell AFB		Branch: TWANW		Taxiway A Northwest		Section: T19A	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: P	Length: 302.00 (ft)	Width: 100.00 (ft)	True Area: 30173.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,593.00	2.00	☒	Mill and overlay, MAX 82-0026 and Original construction, COE	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☒		
1/1/1946	NU-IN	New Construction - Initial	0.00	3.00	☒		
Network: Maxwell AFB		Branch: TWANW		Taxiway A Northwest		Section: T33A	Surface: PCC
L.C.D. 1/1/1986		Use: TAXIWAY	Rank: P	Length: 175.00 (ft)	Width: 127.00 (ft)	True Area: 22276.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
1/1/1986	SL-RP	Slab Replacement	0.00	11.50	☒	Random slab replacement, seal cracks,	
1/1/1985	JS-LC	JT SEAL DMG	0.00	0.00	☐	Reseal joints, MAX 85-0025	
1/1/1945	NU-IN	New Construction - Initial	0.00	11.50	☒	COE runway extension project	
Network: Maxwell AFB		Branch: TWA		Taxiway A		Section: T04A	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: P	Length: 1,915.00 (ft)	Width: 100.00 (ft)	True Area: 191559.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,593.00	2.00	☒	Mill, MAX 82-0026, Overlay, MAX 8 Overlay COE reconstruction, MACADAM lev Original construction	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☒		
1/1/1958	OL-AT	Overlay - AC Thin	0.00	1.50	☒		
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒		
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒		

Work History Report

Network: Maxwell AFB		Branch: TWB		Taxiway B		Section: T05C	Surface: AC
L.C.D. 12/2/2014		Use: TAXIWAY	Rank: S	Length: 862.00 (ft)	Width: 100.00 (ft)	True Area: 147120.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/2/2014	CM-OL-2	2 in Cold Mill & Overlay	450,689.00	2.00	☒	Mill and overlay, MAX 82-0026, MA Original construction	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☒		
1/1/1946	NU-IN	New Construction - Initial	0.00	3.00	☒		
Network: Maxwell AFB		Branch: TWB		Taxiway B		Section: T12C	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: S	Length: 316.00 (ft)	Width: 100.00 (ft)	True Area: 33604.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	450,689.00	0.00	☒	Mill and overlay, MAX 82-0026, MA Original construction	
1/1/1983	OL_2	2 in overlay	0.00	2.00	☒		
1/1/1946	NU-IN	New Construction - Initial	0.00	3.00	☒		
Network: Maxwell AFB		Branch: TWC		Taxiway C		Section: T08A	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: P	Length: 435.00 (ft)	Width: 100.00 (ft)	True Area: 51231.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	427,689.00	2.00	☒	Mill and overlay, MAX 82-022 and M COE reconstruction project Original construction	
1/1/1983	OL_4	4 in overlay	0.00	4.00	☒		
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒		
1/1/1943	NU-IN	New Construction - Initial	0.00	6.00	☒		
Network: Maxwell AFB		Branch: TWD		Taxiway D		Section: T32B	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: S	Length: 456.00 (ft)	Width: 100.00 (ft)	True Area: 53046.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	205,679.00	2.00	☒	Mill and overlay, MAX 82-022 and M COE reconstruction project	
1/1/1983	OL_4	4 in overlay	0.00	4.00	☒		
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒		
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☒		
Network: Maxwell AFB		Branch: TWE		Taxiway E		Section: T09A	Surface: AC
L.C.D. 12/12/2014		Use: TAXIWAY	Rank: P	Length: 436.00 (ft)	Width: 100.00 (ft)	True Area: 48642.00 (ft²)	
Work Date	Work Code	Work Description	Cost	Thickness (in)	Major M&R	Comments	
12/12/2014	CM-OL-2	2 in Cold Mill & Overlay	205,679.00	2.00	☒	Mill and overlay, MAX 82-022 and M COE reconstruction project COE original construction	
1/1/1983	OL_4	4 in overlay	0.00	4.00	☒		
1/1/1946	CR-AC	Complete Reconstruction - AC	0.00	3.00	☒		
1/2/1943	NU-IN	New Construction - Initial	0.00	6.00	☒		

APPENDIX F
PAVEMENT CONDITION INDEX RESULTS

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F.1 STANDARD PAVEMENT CONDITION INDEX

Table F.1 provides the standard pavement condition index criteria for pavements.

Table F.1. Definition of PCI Ratings.

Simplified PCI Rating	PCI Rating	PCI Range	Definition
Good	Good	86-100	GOOD: Pavement has minor or no distresses and will require only routine maintenance.
	Satisfactory	71-85	SATISFACTORY: Pavement has scattered low-severity distresses that need only routine maintenance.
Fair	Fair	56-70	FAIR: Pavement has a combination of generally low- and medium-severity distresses. M&R needs are routine to major in the near future.
Poor	Poor	41-55	POOR: Pavement has low-, medium-, and high-severity distresses that probably cause some operational problems. Near-term maintenance and repair needs may range from routine up to a requirement for reconstruction.
Poor < 40	Very Poor	26-40	VERY POOR: Pavement has predominantly medium- and high-severity distresses that cause considerable maintenance and operational problems. Near-time maintenance and repair needs will be intensive in nature.
	Serious	11-25	SERIOUS: Pavement has mainly high-severity distresses that cause operational restrictions; immediate repairs are needed.
	Failed	0-10	FAILED: Pavement deterioration has progressed to the point that safe operations are no longer possible; complete reconstruction is required.

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SECTION PCI

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Section Condition Report

Branch Name	Section ID	Area (ft²)	Surface	Index	Rating	Distress Related Cause		
						Climate	Load	Other
Aero Club Apron	A10C	36,205	PCC	14	Serious	8	84	8
AGE Storage Lot	A12	30,392	AC	65	Fair	100	0	0
AGE Storage Lot	A13	54,563	AC	66	Fair	100	0	0
Engine Run-up Apron	A07B	67,033	PCC	91	Good	95	0	5
Engine Run-up Apron	T14B	15,568	AC	90	Good	100	0	0
Maintenance Hangar Access Apron	A11C1	11,253	PCC	83	Satisfactory	59	14	27
Maintenance Hangar Access Apron	A11C2	13,344	PCC	79	Satisfactory	28	35	37
Maintenance Hangar Access Apron	A11C3	11,398	PCC	93	Good	100	0	0
North Apron	A01B	274,393	PCC	84	Satisfactory	38	31	31
North Apron	A04B	957,323	PCC	88	Good	32	20	48
North Apron	A08B	26,000	PCC	99	Good	0	0	100
North Apron	T02B	211,267	PCC	85	Satisfactory	49	28	23
North Apron	T15B	48,000	PCC	93	Good	89	0	11
North Apron	T30B	16,250	PCC	81	Satisfactory	27	51	22
North Apron	T31B	56,165	PCC	94	Good	96	0	4
West Apron	A05B	670,734	PCC	89	Good	38	15	47
West Apron	T07B	43,093	PCC	88	Good	47	24	29
West Apron	T29B	111,649	PCC	90	Good	57	7	36
Overrun Runway 009/189	O02C	19,810	AAC	79	Satisfactory	93	0	7
Overrun Runway 009/189	O08C	10,115	AC	83	Satisfactory	100	0	0
Overrun Runway 15/33	O03C	168,000	AC	79	Satisfactory	100	0	0
Overrun Runway 15/33	O04C	31,100	AAC	79	Satisfactory	100	0	0
Overrun Runway 15/33	O05C	22,508	AAC	78	Satisfactory	100	0	0
Overrun Runway 15/33	O06C	126,930	AAC	77	Satisfactory	100	0	0
Runway 009/189 (Assault Strip)	R01A	110,851	PCC	93	Good	0	13	87
Runway 009/189 (Assault Strip)	R02A	107,583	APC	75	Satisfactory	100	0	0
Runway 15/33	R03A	74,530	PCC	86	Good	0	66	34
Runway 15/33	R04C1	30,000	AAC	99	Good	100	0	0
Runway 15/33	R04C2	278,222	APC	84	Satisfactory	100	0	0
Runway 15/33	R05A1	51,501	AC	100	Good	100	0	0
Runway 15/33	R05A2	90,763	APC	80	Satisfactory	100	0	0
Runway 15/33	R06A1	75,062	PCC	99	Good	85	0	15
Runway 15/33	R06A2	75,062	PCC	97	Good	31	0	69

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Section Condition Report

Branch Name	Section ID	Area (ft²)	Surface	Index	Rating	Distress Related Cause		
						Climate	Load	Other
Runway 15/33	R07A	PCC	PCC	99	Good	0	0	100
Runway 15/33	R08C1	PCC	PCC	93	Good	0	26	74
Runway 15/33	R08C2	PCC	PCC	98	Good	0	0	100
Runway 15/33	R09C	APC	AC	90	Good	100	0	0
Runway 15/33	R10C	AC	AC	76	Satisfactory	100	0	0
Runway 15/33	R12A	APC	AC	100	Good	100	0	0
Runway 15/33	R13C	APC	AC	92	Good	100	0	0
Runway 15/33	R14C1	APC	AC	100	Good	100	0	0
Runway 15/33	R14C2	APC	AC	100	Good	100	0	0
Runway 009/189 Turn Around	T06A	APC	AC	59	Fair	100	0	0
Runway 009/189 Turn Around	T35A	APC	AC	96	Good	100	0	0
Runway 15/33 Turn Around	T11A	PCC	PCC	99	Good	0	0	100
Taxiway A	T04A	APC	AC	82	Satisfactory	100	0	0
Taxiway A North	T16A	PCC	PCC	95	Good	0	46	54
Taxiway A North	T18A1	PCC	PCC	87	Good	47	29	24
Taxiway A North	T18A2	PCC	PCC	90	Good	85	5	10
Taxiway A North	T34A	PCC	PCC	85	Satisfactory	37	31	32
Taxiway A Northwest	T03A	PCC	PCC	89	Good	58	21	21
Taxiway A Northwest	T13A	APC	AC	81	Satisfactory	100	0	0
Taxiway A Northwest	T17A	PCC	PCC	82	Satisfactory	10	21	69
Taxiway A Northwest	T19A	AC	AC	79	Satisfactory	100	0	0
Taxiway A Northwest	T33A	PCC	PCC	85	Satisfactory	63	30	7
Taxiway B	T05C	APC	AC	83	Satisfactory	100	0	0
Taxiway B	T12C	APC	AC	89	Good	100	0	0
Taxiway C	T08A	APC	AC	85	Satisfactory	100	0	0
Taxiway D	T32B	APC	AC	89	Good	100	0	0
Taxiway E	T09A	APC	AC	84	Satisfactory	100	0	0

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Section Condition Report - Summary

Age Category	Average Age at Inspection	Total Area (ft²)	Number of Sections	Arithmetic Average PCI	Standard Deviation PCI	Weighted Average PCI
03-05	4	1,037,905.00	17	90.53	7.25	89.35
06-10	6	812,511.00	10	78.10	4.66	79.62
11-15	14	35,995.00	3	85.00	5.89	84.68
16-20	16	30,392.00	1	65.00	0.00	65.00
21-25	23	2,630,905.00	16	91.81	7.10	89.03
31-35	33	877,600.00	10	87.20	2.75	88.08
50+	74	197,173.00	3	55.33	32.36	69.85
ALL	19	5,622,481.00	60	85.78	12.93	86.75

BRANCH PCI

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Branch Condition Report					
Branch ID	Number of Sections	True Area (ft²)	Use	Weighted Average PCI	Branch Rating
APNORTH	7	1,589,398.00	APRON	89.00	Good
APWEST	3	825,476.00	APRON	89.00	Good
OAAERO	1	36,205.00	APRON	14.00	Serious
OAAGE	2	84,955.00	APRON	66.00	Fair
OAHANG	3	35,995.00	APRON	85.00	Satisfactory
OARUNUP	2	82,601.00	APRON	91.00	Good
OR009/189	2	29,925.00	OVERRUN	81.00	Satisfactory
OR15/33	4	348,538.00	OVERRUN	78.00	Satisfactory
RW009/189	2	218,434.00	RUNWAY	84.00	Satisfactory
RW15/33	16	1,210,955.00	RUNWAY	90.00	Good
TW009/189T	2	72,468.00	TAXIWAY	78.00	Satisfactory
TW15/33 T	1	57,425.00	TAXIWAY	99.00	Good
TWA	1	191,559.00	TAXIWAY	82.00	Satisfactory
TWANORTH	4	251,647.00	TAXIWAY	89.00	Good
TWANW	5	253,257.00	TAXIWAY	83.00	Satisfactory
TWB	2	180,724.00	TAXIWAY	86.00	Good
TWC	1	51,231.00	TAXIWAY	85.00	Satisfactory
TWD	1	53,046.00	TAXIWAY	89.00	Good
TWE	1	48,642.00	TAXIWAY	84.00	Satisfactory

Branch Condition Report - Summary					
Use Category	Number of Sections	Total Area (ft²)	Arithmetic Average PCI	Average STD PCI	Weighted Average PCI
APRON	18	2,654,630.00	81.78	18.55	86.28
OVERRUN	6	378,463.00	79.17	1.86	78.38
RUNWAY	18	1,429,389.00	92.28	8.36	90.75
TAXIWAY	18	1,159,999.00	85.50	8.30	85.62
ALL	60	5,622,481.00	85.78	12.93	86.75

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**EXTRAPOLATED DISTRESS REPORT
(CUSTOM REPORT)**

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
Aero Club Apron	A10C	14	63	Low	7 Slabs	6.6	Linear Cracking
Aero Club Apron	A10C	14	63	Medium	21 Slabs	29.7	Linear Cracking
Aero Club Apron	A10C	14	65	High	96 Slabs	12.0	Joint Seal Damage
Aero Club Apron	A10C	14	67	Low	1 Slabs	0.8	Large Patch/Utility
Aero Club Apron	A10C	14	67	Medium	5 Slabs	10.7	Large Patch/Utility
Aero Club Apron	A10C	14	72	Low	29 Slabs	32.5	Shattered Slab
Aero Club Apron	A10C	14	72	Medium	39 Slabs	54.6	Shattered Slab
AGE Storage Lot	A12	65	48	Low	656 ft	7.8	L&T Cracking
AGE Storage Lot	A12	65	48	Medium	541 ft	14.8	L&T Cracking
AGE Storage Lot	A12	65	50	Medium	608 ft²	12.4	Patching
AGE Storage Lot	A12	65	57	Medium	29,784 ft²	20.2	Weathering
AGE Storage Lot	A13	66	48	Medium	3,459 ft	29.1	L&T Cracking
AGE Storage Lot	A13	66	57	Medium	54,563 ft²	20.3	Weathering
Engine Run-up Apron	A07B	91	65	High	26 Slabs	12.0	Joint Seal Damage
Engine Run-up Apron	A07B	91	65	Medium	154 Slabs	7.0	Joint Seal Damage
Engine Run-up Apron	A07B	91	74	Low	5 Slabs	1.0	Joint Spalling
Engine Run-up Apron	T14B	90	48	Low	144 ft	4.8	L&T Cracking
Engine Run-up Apron	T14B	90	57	Low	15,560 ft²	6.0	Weathering
Maintenance Hangar Access Apron	A11C1	83	63	Medium	2 Slabs	4.3	Linear Cracking
Maintenance Hangar Access Apron	A11C1	83	65	High	24 Slabs	12.0	Joint Seal Damage
Maintenance Hangar Access Apron	A11C1	83	65	Medium	70 Slabs	7.0	Joint Seal Damage
Maintenance Hangar Access Apron	A11C1	83	74	Low	7 Slabs	2.0	Joint Spalling
Maintenance Hangar Access Apron	A11C1	83	75	Low	6 Slabs	2.3	Corner Spalling
Maintenance Hangar Access Apron	A11C1	83	75	Medium	6 Slabs	4.2	Corner Spalling
Maintenance Hangar Access Apron	A11C2	79	63	Low	14 Slabs	8.8	Linear Cracking
Maintenance Hangar Access Apron	A11C2	79	65	Medium	133 Slabs	7.0	Joint Seal Damage
Maintenance Hangar Access Apron	A11C2	79	66	Low	6 Slabs	1.0	Small Patch
Maintenance Hangar Access Apron	A11C2	79	67	Low	1 Slabs	0.7	Large Patch/Utility
Maintenance Hangar Access Apron	A11C2	79	74	Low	8 Slabs	1.8	Joint Spalling
Maintenance Hangar Access Apron	A11C2	79	74	Medium	1 Slabs	0.8	Joint Spalling
Maintenance Hangar Access Apron	A11C2	79	75	Low	17 Slabs	5.0	Corner Spalling
Maintenance Hangar Access Apron	A11C3	93	65	Medium	114 Slabs	7.0	Joint Seal Damage
North Apron	A01B	84	63	Low	21 Slabs	2.0	Linear Cracking
North Apron	A01B	84	63	Medium	24 Slabs	5.2	Linear Cracking
North Apron	A01B	84	65	Low	53 Slabs	2.0	Joint Seal Damage

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
North Apron	A01B	84	65	Medium	1,045 Slabs	7.0	Joint Seal Damage
North Apron	A01B	84	66	Low	5 Slabs	0.2	Small Patch
North Apron	A01B	84	66	Medium	8 Slabs	0.7	Small Patch
North Apron	A01B	84	67	Low	79 Slabs	4.7	Large Patch/Utility
North Apron	A01B	84	73	N/A	21 Slabs	0.5	Shrinkage Cracking
North Apron	A01B	84	74	Low	8 Slabs	0.3	Joint Spalling
North Apron	A01B	84	74	Medium	5 Slabs	0.8	Joint Spalling
North Apron	A04B	88	63	Low	113 Slabs	3.0	Linear Cracking
North Apron	A04B	88	63	Medium	26 Slabs	2.5	Linear Cracking
North Apron	A04B	88	65	Low	1,789 Slabs	2.0	Joint Seal Damage
North Apron	A04B	88	65	Medium	821 Slabs	7.0	Joint Seal Damage
North Apron	A04B	88	66	Low	311 Slabs	1.5	Small Patch
North Apron	A04B	88	66	Medium	13 Slabs	0.7	Small Patch
North Apron	A04B	88	67	Low	258 Slabs	4.4	Large Patch/Utility
North Apron	A04B	88	67	Medium	13 Slabs	2.5	Large Patch/Utility
North Apron	A04B	88	71	Low	26 Slabs	1.0	Faulting
North Apron	A04B	88	71	Medium	7 Slabs	1.8	Faulting
North Apron	A04B	88	74	Low	20 Slabs	0.3	Joint Spalling
North Apron	A04B	88	75	Low	20 Slabs	0.4	Corner Spalling
North Apron	A04B	88	75	Medium	13 Slabs	0.9	Corner Spalling
North Apron	A08B	99	66	Low	1 Slabs	0.2	Small Patch
North Apron	A08B	99	73	N/A	3 Slabs	0.7	Shrinkage Cracking
North Apron	T02B	85	62	Low	2 Slabs	0.8	Corner Break
North Apron	T02B	85	63	High	2 Slabs	3.3	Linear Cracking
North Apron	T02B	85	63	Low	32 Slabs	3.8	Linear Cracking
North Apron	T02B	85	63	Medium	15 Slabs	4.3	Linear Cracking
North Apron	T02B	85	65	High	51 Slabs	12.0	Joint Seal Damage
North Apron	T02B	85	65	Low	666 Slabs	2.0	Joint Seal Damage
North Apron	T02B	85	65	Medium	171 Slabs	7.0	Joint Seal Damage
North Apron	T02B	85	66	Low	21 Slabs	0.6	Small Patch
North Apron	T02B	85	66	Medium	4 Slabs	0.7	Small Patch
North Apron	T02B	85	67	Low	28 Slabs	2.3	Large Patch/Utility
North Apron	T02B	85	67	Medium	2 Slabs	2.5	Large Patch/Utility
North Apron	T02B	85	71	Low	4 Slabs	1.0	Faulting
North Apron	T02B	85	71	Medium	4 Slabs	1.8	Faulting

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
North Apron	T02B	85	73	N/A	13 Slabs	0.4	Shrinkage Cracking
North Apron	T02B	85	74	Low	2 Slabs	0.3	Joint Spalling
North Apron	T02B	85	75	Low	9 Slabs	0.5	Corner Spalling
North Apron	T15B	93	65	Low	7 Slabs	2.0	Joint Seal Damage
North Apron	T15B	93	65	Medium	58 Slabs	7.0	Joint Seal Damage
North Apron	T15B	93	74	Low	1 Slabs	0.7	Joint Spalling
North Apron	T15B	93	75	Low	0 Slabs	0.4	Corner Spalling
North Apron	T30B	81	62	Low	1 Slabs	1.1	Corner Break
North Apron	T30B	81	63	Low	2 Slabs	2.8	Linear Cracking
North Apron	T30B	81	63	Medium	3 Slabs	9.4	Linear Cracking
North Apron	T30B	81	65	Medium	22 Slabs	7.0	Joint Seal Damage
North Apron	T30B	81	66	Low	4 Slabs	1.1	Small Patch
North Apron	T30B	81	67	Low	3 Slabs	2.9	Large Patch/Utility
North Apron	T30B	81	73	N/A	5 Slabs	1.7	Shrinkage Cracking
North Apron	T31B	94	65	Low	66 Slabs	2.0	Joint Seal Damage
North Apron	T31B	94	65	Medium	159 Slabs	7.0	Joint Seal Damage
North Apron	T31B	94	73	N/A	3 Slabs	0.4	Shrinkage Cracking
West Apron	A05B	89	62	Medium	5 Slabs	1.9	Corner Break
West Apron	A05B	89	63	Low	46 Slabs	1.8	Linear Cracking
West Apron	A05B	89	65	Low	2,208 Slabs	2.0	Joint Seal Damage
West Apron	A05B	89	65	Medium	111 Slabs	7.0	Joint Seal Damage
West Apron	A05B	89	66	Low	267 Slabs	1.7	Small Patch
West Apron	A05B	89	66	Medium	23 Slabs	0.7	Small Patch
West Apron	A05B	89	67	Low	134 Slabs	3.4	Large Patch/Utility
West Apron	A05B	89	67	Medium	18 Slabs	2.5	Large Patch/Utility
West Apron	A05B	89	73	N/A	60 Slabs	0.6	Shrinkage Cracking
West Apron	A05B	89	74	Low	5 Slabs	0.3	Joint Spalling
West Apron	A05B	89	74	Medium	5 Slabs	0.8	Joint Spalling
West Apron	A05B	89	75	Low	5 Slabs	0.4	Corner Spalling
West Apron	A05B	89	75	Medium	5 Slabs	0.9	Corner Spalling
West Apron	T07B	88	63	Low	1 Slabs	1.1	Linear Cracking
West Apron	T07B	88	63	Medium	1 Slabs	2.5	Linear Cracking
West Apron	T07B	88	65	Medium	172 Slabs	7.0	Joint Seal Damage
West Apron	T07B	88	66	Low	12 Slabs	1.3	Small Patch
West Apron	T07B	88	66	Medium	1 Slabs	0.7	Small Patch

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
West Apron	T07B	88	67	Low	4 Slabs	1.7	Large Patch/Utility
West Apron	T07B	88	73	N/A	1 Slabs	0.3	Shrinkage Cracking
West Apron	T07B	88	75	Low	1 Slabs	0.4	Corner Spalling
West Apron	T29B	90	62	Low	6 Slabs	1.1	Corner Break
West Apron	T29B	90	65	Low	225 Slabs	2.0	Joint Seal Damage
West Apron	T29B	90	65	Medium	191 Slabs	7.0	Joint Seal Damage
West Apron	T29B	90	67	Low	6 Slabs	1.0	Large Patch/Utility
West Apron	T29B	90	73	N/A	92 Slabs	3.1	Shrinkage Cracking
West Apron	T29B	90	74	Low	2 Slabs	0.3	Joint Spalling
West Apron	T29B	90	75	Low	12 Slabs	1.2	Corner Spalling
Overrun Runway 009/189	O02C	79	48	Low	1,058 ft	15.4	L&T Cracking
Overrun Runway 009/189	O02C	79	56	Low	42 ft²	1.5	Swelling
Overrun Runway 009/189	O02C	79	57	Low	20,000 ft²	6.0	Weathering
Overrun Runway 009/189	O08C	83	48	Low	359 ft	11.5	L&T Cracking
Overrun Runway 009/189	O08C	83	57	Low	10,000 ft²	6.0	Weathering
Overrun Runway 15/33	O03C	79	48	Low	9,767 ft	16.4	L&T Cracking
Overrun Runway 15/33	O03C	79	57	Low	168,000 ft²	6.0	Weathering
Overrun Runway 15/33	O04C	79	48	Low	1,765 ft	16.2	L&T Cracking
Overrun Runway 15/33	O04C	79	57	Low	31,100 ft²	6.0	Weathering
Overrun Runway 15/33	O05C	78	48	Low	1,362 ft	16.9	L&T Cracking
Overrun Runway 15/33	O05C	78	57	Low	22,500 ft²	6.0	Weathering
Overrun Runway 15/33	O06C	77	48	Low	8,531 ft	18.2	L&T Cracking
Overrun Runway 15/33	O06C	77	57	Low	114,197 ft²	5.9	Weathering
Runway 009/189 (Assault Strip)	R01A	93	62	Low	1 Slabs	0.8	Corner Break
Runway 009/189 (Assault Strip)	R01A	93	63	Low	1 Slabs	1.1	Linear Cracking
Runway 009/189 (Assault Strip)	R01A	93	66	Low	3 Slabs	0.2	Small Patch
Runway 009/189 (Assault Strip)	R01A	93	67	Low	12 Slabs	1.8	Large Patch/Utility
Runway 009/189 (Assault Strip)	R01A	93	67	Medium	9 Slabs	4.6	Large Patch/Utility
Runway 009/189 (Assault Strip)	R01A	93	70	Low	1 Slabs	0.3	Scaling
Runway 009/189 (Assault Strip)	R01A	93	70	Medium	1 Slabs	0.9	Scaling
Runway 009/189 (Assault Strip)	R01A	93	71	Low	3 Slabs	1.0	Faulting
Runway 009/189 (Assault Strip)	R01A	93	71	Medium	1 Slabs	1.8	Faulting
Runway 009/189 (Assault Strip)	R01A	93	73	N/A	1 Slabs	0.3	Shrinkage Cracking
Runway 009/189 (Assault Strip)	R01A	93	74	Low	7 Slabs	0.6	Joint Spalling
Runway 009/189 (Assault Strip)	R01A	93	75	Low	6 Slabs	0.6	Corner Spalling

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
Runway 009/189 (Assault Strip)	R01A	93	75	Medium	1 Slabs	0.9	Corner Spalling
Runway 009/189 (Assault Strip)	R02A	75	48	Low	6,720 ft	17.3	L&T Cracking
Runway 009/189 (Assault Strip)	R02A	75	48	Medium	510 ft	8.1	L&T Cracking
Runway 009/189 (Assault Strip)	R02A	75	57	Low	106,076 ft²	6.0	Weathering
Runway 009/189 (Assault Strip)	R02A	75	57	Medium	1,005 ft²	1.8	Weathering
Runway 15/33	R03A	86	62	Low	7 Slabs	1.9	Corner Break
Runway 15/33	R03A	86	62	Medium	1 Slabs	1.9	Corner Break
Runway 15/33	R03A	86	63	Low	14 Slabs	4.6	Linear Cracking
Runway 15/33	R03A	86	63	Medium	6 Slabs	4.8	Linear Cracking
Runway 15/33	R03A	86	66	Low	14 Slabs	1.0	Small Patch
Runway 15/33	R03A	86	67	Low	6 Slabs	1.4	Large Patch/Utility
Runway 15/33	R03A	86	71	Low	4 Slabs	1.2	Faulting
Runway 15/33	R03A	86	73	N/A	1 Slabs	0.3	Shrinkage Cracking
Runway 15/33	R03A	86	74	Low	8 Slabs	0.9	Joint Spalling
Runway 15/33	R03A	86	74	Medium	2 Slabs	0.8	Joint Spalling
Runway 15/33	R03A	86	75	Low	2 Slabs	0.4	Corner Spalling
Runway 15/33	R03A	86	75	Medium	1 Slabs	0.9	Corner Spalling
Runway 15/33	R04C1	99	48	Low	79 ft	3.5	L&T Cracking
Runway 15/33	R04C2	84	47	Low	2,055 ft	1.2	Joint Reflection Cracking
Runway 15/33	R04C2	84	48	Low	13,239 ft	10.9	L&T Cracking
Runway 15/33	R04C2	84	50	Low	790 ft²	2.0	Patching
Runway 15/33	R04C2	84	57	Low	351,107 ft²	5.8	Weathering
Runway 15/33	R05A1	100	No recorded distresses.				
Runway 15/33	R05A2	80	48	Low	5,012 ft	15.9	L&T Cracking
Runway 15/33	R05A2	80	57	Low	90,756 ft²	6.0	Weathering
Runway 15/33	R06A1	99	65	Low	48 Slabs	2.0	Joint Seal Damage
Runway 15/33	R06A1	99	74	Low	1 Slabs	0.3	Joint Spalling
Runway 15/33	R06A2	97	65	Low	23 Slabs	2.0	Joint Seal Damage
Runway 15/33	R06A2	97	67	Medium	1 Slabs	2.5	Large Patch/Utility
Runway 15/33	R06A2	97	74	Low	7 Slabs	1.1	Joint Spalling
Runway 15/33	R06A2	97	74	Medium	1 Slabs	0.8	Joint Spalling
Runway 15/33	R07A	99	74	Low	12 Slabs	1.4	Joint Spalling
Runway 15/33	R08C1	93	63	Low	2 Slabs	2.1	Linear Cracking
Runway 15/33	R08C1	93	66	Medium	1 Slabs	0.7	Small Patch
Runway 15/33	R08C1	93	70	Low	2 Slabs	0.6	Scaling

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
Runway 15/33	R08C1	93	73	N/A	11 Slabs	2.0	Shrinkage Cracking
Runway 15/33	R08C1	93	74	Low	3 Slabs	1.0	Joint Spalling
Runway 15/33	R08C1	93	74	Medium	2 Slabs	1.6	Joint Spalling
Runway 15/33	R08C2	98	67	Low	2 Slabs	1.8	Large Patch/Utility
Runway 15/33	R08C2	98	74	Low	2 Slabs	0.8	Joint Spalling
Runway 15/33	R09C	90	48	Low	2,858 ft	5.7	L&T Cracking
Runway 15/33	R09C	90	57	Low	184,153 ft²	5.8	Weathering
Runway 15/33	R10C	76	47	Low	180 ft	4.9	Joint Reflection Cracking
Runway 15/33	R10C	76	48	Low	601 ft	16.3	L&T Cracking
Runway 15/33	R10C	76	57	Low	10,500 ft²	6.0	Weathering
Runway 15/33	R12A	100	No recorded distresses.				
Runway 15/33	R13C	92	48	Low	51 ft	4.0	L&T Cracking
Runway 15/33	R13C	92	57	Low	10,500 ft²	6.0	Weathering
Runway 15/33	R14C1	100	No recorded distresses.				
Runway 15/33	R14C2	100	No recorded distresses.				
Runway 009/189 Turn Around	T06A	59	43	Low	50,145 ft²	35.6	Block Cracking
Runway 009/189 Turn Around	T06A	59	57	Low	8,358 ft²	2.5	Weathering
Runway 009/189 Turn Around	T06A	59	57	Medium	41,788 ft²	19.1	Weathering
Runway 009/189 Turn Around	T35A	96	48	Low	44 ft	3.1	L&T Cracking
Runway 009/189 Turn Around	T35A	96	50	Low	400 ft²	5.2	Patching
Runway 15/33 Turn Around	T11A	99	73	N/A	7 Slabs	1.0	Shrinkage Cracking
Runway 15/33 Turn Around	T11A	99	74	Low	1 Slabs	0.3	Joint Spalling
Taxiway A	T04A	82	48	Low	8,272 ft	13.3	L&T Cracking
Taxiway A	T04A	82	57	Low	191,550 ft²	6.0	Weathering
Taxiway A North	T16A	95	63	Low	1 Slabs	2.5	Linear Cracking
Taxiway A North	T16A	95	73	N/A	6 Slabs	1.8	Shrinkage Cracking
Taxiway A North	T16A	95	75	Low	1 Slabs	1.0	Corner Spalling
Taxiway A North	T18A1	87	63	Low	2 Slabs	4.3	Linear Cracking
Taxiway A North	T18A1	87	65	Medium	45 Slabs	7.0	Joint Seal Damage
Taxiway A North	T18A1	87	66	Low	1 Slabs	0.5	Small Patch
Taxiway A North	T18A1	87	73	N/A	1 Slabs	0.6	Shrinkage Cracking
Taxiway A North	T18A1	87	74	Low	1 Slabs	0.7	Joint Spalling
Taxiway A North	T18A1	87	74	Medium	1 Slabs	1.8	Joint Spalling
Taxiway A North	T18A2	90	63	Low	2 Slabs	1.1	Linear Cracking
Taxiway A North	T18A2	90	65	High	404 Slabs	12.0	Joint Seal Damage

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
Taxiway A North	T18A2	90	65	Medium	257 Slabs	7.0	Joint Seal Damage
Taxiway A North	T18A2	90	67	Low	2 Slabs	0.7	Large Patch/Utility
Taxiway A North	T18A2	90	71	Low	4 Slabs	1.0	Faulting
Taxiway A North	T18A2	90	73	N/A	2 Slabs	0.3	Shrinkage Cracking
Taxiway A North	T18A2	90	74	Low	2 Slabs	0.3	Joint Spalling
Taxiway A North	T34A	85	62	Low	5 Slabs	2.2	Corner Break
Taxiway A North	T34A	85	63	Low	9 Slabs	5.2	Linear Cracking
Taxiway A North	T34A	85	65	Low	23 Slabs	2.0	Joint Seal Damage
Taxiway A North	T34A	85	65	Medium	54 Slabs	7.0	Joint Seal Damage
Taxiway A North	T34A	85	66	Low	9 Slabs	1.1	Small Patch
Taxiway A North	T34A	85	67	Low	17 Slabs	6.2	Large Patch/Utility
Taxiway A North	T34A	85	75	Low	2 Slabs	0.4	Corner Spalling
Taxiway A Northwest	T03A	89	61	Low	2 Slabs	3.0	Blow-Up
Taxiway A Northwest	T03A	89	62	Low	2 Slabs	0.8	Corner Break
Taxiway A Northwest	T03A	89	62	Medium	2 Slabs	1.9	Corner Break
Taxiway A Northwest	T03A	89	63	Low	4 Slabs	1.1	Linear Cracking
Taxiway A Northwest	T03A	89	63	Medium	4 Slabs	2.5	Linear Cracking
Taxiway A Northwest	T03A	89	65	High	284 Slabs	12.0	Joint Seal Damage
Taxiway A Northwest	T03A	89	65	Low	388 Slabs	2.0	Joint Seal Damage
Taxiway A Northwest	T03A	89	66	Low	7 Slabs	0.2	Small Patch
Taxiway A Northwest	T03A	89	67	Low	20 Slabs	2.0	Large Patch/Utility
Taxiway A Northwest	T03A	89	67	Medium	2 Slabs	2.5	Large Patch/Utility
Taxiway A Northwest	T03A	89	73	N/A	4 Slabs	0.3	Shrinkage Cracking
Taxiway A Northwest	T03A	89	74	Low	14 Slabs	0.7	Joint Spalling
Taxiway A Northwest	T03A	89	75	Low	4 Slabs	0.4	Corner Spalling
Taxiway A Northwest	T13A	81	47	Low	1,111 ft	13.8	Joint Reflection Cracking
Taxiway A Northwest	T13A	81	57	Low	14,126 ft ²	6.0	Weathering
Taxiway A Northwest	T17A	82	62	Low	2 Slabs	4.1	Corner Break
Taxiway A Northwest	T17A	82	65	Low	40 Slabs	2.0	Joint Seal Damage
Taxiway A Northwest	T17A	82	67	Medium	1 Slabs	5.8	Large Patch/Utility
Taxiway A Northwest	T17A	82	73	N/A	6 Slabs	2.5	Shrinkage Cracking
Taxiway A Northwest	T17A	82	74	Low	2 Slabs	1.6	Joint Spalling
Taxiway A Northwest	T17A	82	74	Medium	1 Slabs	2.0	Joint Spalling
Taxiway A Northwest	T17A	82	75	Low	1 Slabs	1.1	Corner Spalling
Taxiway A Northwest	T19A	79	48	Low	1,700 ft	16.1	L&T Cracking

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Extrapolated Distress Report

Branch Name	Section ID	PCI	Distress Code	Severity	Quantity	PCI Deduct	Distress Description
Taxiway A Northwest	T19A	79	57	Low	30,170 ft²	6.0	Weathering
Taxiway A Northwest	T33A	85	63	Low	6 Slabs	6.1	Linear Cracking
Taxiway A Northwest	T33A	85	63	Medium	1 Slabs	2.8	Linear Cracking
Taxiway A Northwest	T33A	85	65	High	30 Slabs	12.0	Joint Seal Damage
Taxiway A Northwest	T33A	85	65	Medium	60 Slabs	7.0	Joint Seal Damage
Taxiway A Northwest	T33A	85	66	Low	2 Slabs	0.5	Small Patch
Taxiway A Northwest	T33A	85	74	Low	2 Slabs	0.7	Joint Spalling
Taxiway A Northwest	T33A	85	74	Medium	1 Slabs	0.9	Joint Spalling
Taxiway B	T05C	83	48	Low	5,610 ft	12.1	L&T Cracking
Taxiway B	T05C	83	57	Low	147,107 ft²	6.0	Weathering
Taxiway B	T12C	89	48	Low	583 ft	6.3	L&T Cracking
Taxiway B	T12C	89	50	Low	113 ft²	2.1	Patching
Taxiway B	T12C	89	57	Low	31,348 ft²	5.8	Weathering
Taxiway C	T08A	85	48	Low	1,502 ft	9.9	L&T Cracking
Taxiway C	T08A	85	57	Low	51,200 ft²	6.0	Weathering
Taxiway D	T32B	89	48	Low	732 ft	5.8	L&T Cracking
Taxiway D	T32B	89	57	Low	52,243 ft²	6.0	Weathering
Taxiway E	T09A	84	48	Low	1,682 ft	11.2	L&T Cracking
Taxiway E	T09A	84	57	Low	48,600 ft²	6.0	Weathering
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APPENDIX G

FOD POTENTIAL INFORMATION

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G.1 FOD INDEX AND FOD POTENTIAL

The FOD Potential rating is calculated from the FOD Index based on the type of aircraft using the pavement and the surface of the pavement. Aircraft with low-slung engine intakes, e.g., the F-16, are more susceptible to FOD than aircraft with high-mount engine intakes, e.g., the C-17. There are three standard aircraft categories that can be used to calculate FOD Potential: the F-16, the KC-135, and the C-17.

The FOD Index is based on pavement distresses from the PCI survey that can result in loose pieces of pavement. Table G.1 identifies FOD Index criteria. The FOD Index used in this report was generated in PAVER 7. Figure G.1 shows the FOD Index curve which is used to determine FOD Potential for the C-17, which contains the C-130 aircraft, as defined by ETL 04-9. Figure A.6 presents the FOD Potential by section. Table G.2 shows FOD-producing distresses as defined in ETL 04-9. Table G.3 gives the FOD Potential and rating for each airfield pavement section.

Table G.1. FOD Index Criteria.

FOD Index Criteria for Airfield Pavements (For Standard** Aircraft and Pavement Surface Types)								
Assessment/Rating Category	FOD Potential Rating	Definition	F-16		KC-135		C-17	
			AC	PCC	AC	PCC	AC	PCC
Good	FOD Index ≤ 45	Almost no FOD Potential to Minor Potential	0 - 32	0 - 41	0 - 44	0 - 60	0 - 59	0 - 77
Fair	$45 < \text{FOD Index} < 61$	Moderate FOD Potential	33 - 45	42 - 62	45 - 60	61 - 78	60 - 75	78 - 89
Poor	FOD Index ≥ 61	High to Severe FOD Potential	46 - 100	63 - 100	61 - 100	79 - 100	76 - 100	90 - 100

*FOD Index is calculated using the PCI survey data by considering only the distresses/severity levels capable of producing FOD.
 **For other aircraft, see AFI 32-1041, Table 8.7, to determine the appropriate FOD curve relationship to standard aircraft.

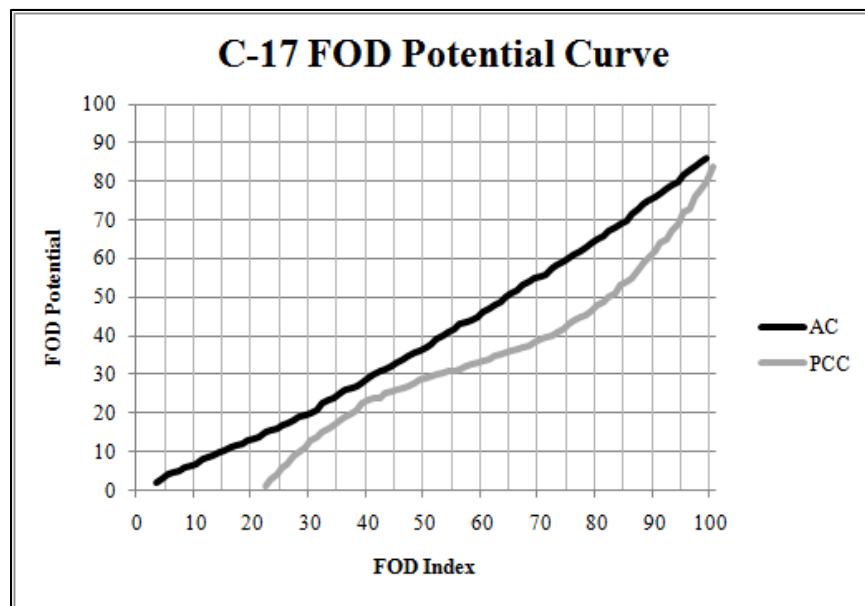


Figure G.1. C-17 Category FOD Potential Curve.

Table G.2. FOD Producing Distresses and Probable Cause.

Distresses associated with FOD Potential					
AC Distress	Severity	Cause	PCC Distress	Severity	Cause
41. Alligator cracking	L, M, H	Load related	61. Blowup	L, M, H	Climate related
43. Block cracking	L, M, H	Climate related	62. Corner break	L, M, H	Load related
46. Jet blast	NA	Other	63. LTD (Linear) cracks	L, M, H	Load related
47. Joint reflection cracking	L, M, H	Climate related	64. Durability cracks	M, H	Climate related
48. L&T cracking	L, M, H	Climate related	65. Joint seal damage	L, M, H	Climate related
49. Oil spillage	NA	Other	66 & 67. Small & Large Patches	L, M, H	Other
50. Patching	L, M, H	Other	68. Pop outs	NA	Other
52. Raveling	L, M, H	Climate related	69. Pumping	NA	Load related
54. Shoving	M, H	Other	70. Scaling	L, M, H	Other
55. Slippage cracking	NA	Other	72. Shattered slabs	L, M, H	Load related
57. Weathering	M, H	Climate related	74. Joint spalls	L, M, H	Other
			75. Corner spalls	L, M, H	Other
			76. ASR	M, H	ASR
Severity levels = low (L), medium (M), and high (H)					

Table G.3. FOD Potential and Rating by Section.

Branch Name	Section ID	Area (ft²)	Surface	Aircraft Category	Section FOD		
					Index	Potential	Rating
Aprons							
Aero Club Apron	A10C	36,205	PCC	KC-135	98	85	Poor
AGE Storage Lot	A12	30,392	AC	C-17	35	25	Good
AGE Storage Lot	A13	54,563	AC	C-17	34	24	Good
Engine Run-up Apron	A07B	67,033	PCC	C-17	32	15	Good
Engine Run-up Apron	T14B	15,568	AC	C-17	10	7	Good
Maintenance Hangar Access Apron	A11C1	11,253	PCC	C-17	36	19	Good
Maintenance Hangar Access Apron	A11C2	13,344	PCC	C-17	39	23	Good
Maintenance Hangar Access Apron	A11C3	11,398	PCC	C-17	28	10	Good
North Apron	A01B	274,393	PCC	C-17	34	17	Good
North Apron	A04B	957,323	PCC	C-17	18	0	Good
North Apron	A08B	26,000	PCC	C-17	0	0	Good
North Apron	T02B	211,267	PCC	C-17	23	3	Good
North Apron	T15B	48,000	PCC	C-17	27	9	Good
North Apron	T30B	16,250	PCC	C-17	24	4	Good
North Apron	T31B	56,165	PCC	C-17	22	1	Good
(page 1 of 3)							

Table G.3. FOD Potential and Rating by Section (continued).

Branch Name	Section ID	Area (ft²)	Surface	Aircraft Category	Section FOD		
					Index	Potential	Rating
Aprons							
West Apron	A05B	670,734	PCC	C-17	15	0	Good
West Apron	T07B	43,093	PCC	C-17	32	15	Good
West Apron	T29B	111,649	PCC	C-17	19	0	Good
Overrun Runway 009/189	O02C	19,810	AAC	C-17	20	14	Good
Overrun Runway 009/189	O08C	10,115	AC	C-17	17	12	Good
Overrun Runway 15/33	O03C	168,000	AC	C-17	21	15	Good
Overrun Runway 15/33	O04C	31,100	AAC	C-17	21	15	Good
Overrun Runway 15/33	O05C	22,508	AAC	C-17	22	15	Good
Overrun Runway 15/33	O06C	126,930	AAC	C-17	23	16	Good
Runway 009/189 (Assault Strip)	R01A	110,851	PCC	C-17	6	0	Good
Runway 009/189 (Assault Strip)	R02A	107,583	APC	C-17	25	17	Good
Runway 15/33	R03A	74,530	PCC	C-17	13	0	Good
Runway 15/33	R04C1	30,000	AAC	C-17	1	0	Good
Runway 15/33	R04C2	278,222	APC	C-17	16	11	Good
Runway 15/33	R05A1	51,501	AC	C-17	0	0	Good
Runway 15/33	R05A2	90,763	APC	C-17	20	14	Good
Runway 15/33	R06A1	75,062	PCC	C-17	2	0	Good
Runway 15/33	R06A2	75,062	PCC	C-17	3	0	Good
Runway 15/33	R07A	108,832	PCC	C-17	1	0	Good
Runway 15/33	R08C1	37,500	PCC	C-17	5	0	Good
Runway 15/33	R08C2	30,000	PCC	C-17	2	0	Good
Runway 15/33	R09C	134,482	APC	C-17	10	7	Good
Runway 15/33	R10C	10,500	AC	C-17	24	17	Good
Runway 15/33	R12A	9,003	APC	C-17	0	0	Good
Runway 15/33	R13C	10,500	APC	C-17	8	6	Good
Runway 15/33	R14C1	78,000	APC	C-17	0	0	Good
Runway 15/33	R14C2	116,998	APC	C-17	0	0	Good
Runway 009/189 Turn Around	T06A	50,117	APC	C-17	41	30	Good
Runway 009/189 Turn Around	T35A	22,351	APC	C-17	4	3	Good
Runway 15/33 Turn Around	T11A	57,425	PCC	C-17	0	0	Good
Taxiway A	T04A	191,559	APC	C-17	18	13	Good
Taxiway A North	T16A	16,046	PCC	C-17	3	0	Good
Taxiway A North	T18A1	11,215	PCC	C-17	33	16	Good
Taxiway A North	T18A2	183,486	PCC	C-17	37	20	Good
(page 2 of 3)							

Table G.3. FOD Potential and Rating by Section (concluded).

Branch Name	Section ID	Area (ft²)	Surface	Aircraft Category	Section FOD		
					Index	Potential	Rating
Aprons							
Taxiway A North	T34A	40,900	PCC	C-17	21	0	Good
Taxiway A Northwest	T03A	176,681	PCC	C-17	28	10	Good
Taxiway A Northwest	T13A	14,127	APC	C-17	19	13	Good
Taxiway A Northwest	T17A	10,000	PCC	C-17	19	0	Good
Taxiway A Northwest	T19A	30,173	AC	C-17	21	15	Good
Taxiway A Northwest	T33A	22,276	PCC	C-17	40	23	Good
Taxiway B	T05C	147,120	APC	C-17	17	12	Good
Taxiway B	T12C	33,604	APC	C-17	11	8	Good
Taxiway C	T08A	51,231	APC	C-17	15	10	Good
Taxiway D	T32B	53,046	APC	C-17	11	8	Good
Taxiway E	T09A	48,642	APC	C-17	16	11	Good
(page 3 of 3)							

APPENDIX H

STRUCTURAL RESULTS

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H.1 STRUCTURAL INDEX

ACN expresses the load imparted by an aircraft in terms of a standardized single wheel load. PCN expresses the structural capacity of a pavement in terms of the same standardized single wheel load. The ratio between ACN and PCN reflects the potential damage a given aircraft is likely to cause to a given pavement. An ACN/PCN ratio less than 1.1 indicates that the pavement will support operations of the aircraft without premature failure. An ACN/PCN ratio greater than 1.1 indicates the pavement will be overloaded and may fail prematurely. Figure H.1 provides the Structural Index rating criteria.

The ACN for each section and branch is based on the representative critical mission aircraft. ACNs for other aircraft may be used to adjust the reported Structural Indices. PCN values were obtained from the AFCEC report, Airfield Pavement Evaluation, Maxwell AFB, Alabama, January 2016, AFCEC, Tyndall AFB, Florida, for 50,000 passes of a C-17. Sections with a PCI below 40 had their AGL reduced by 20% and new PCNs calculated by AFCEC. These sections are distinguished by an asterisk in the Section column of Table H.2.

Table H.1 shows the ACN values obtained from Pavement-Transportation Computer Assisted Structural Evaluation (PCASE) software for the C-17, used to calculate the ACN/PCN ratio for each section at Maxwell AFB. The standard load of each aircraft was used in the following tables. Table D.6 shows the Structural Index by section. Figure A.7 shows the ACN/PCN value by section. The Structural Index values for the overruns are not applicable and assumed to have a structural rating of “Good”.

ACN/PCN < 1.10 (Good)
$1.10 \leq \text{ACN/PCN} \leq 1.40$ (Fair)
ACN/PCN > 1.40 (Poor)

Figure H.1. Structural Index Rating Criteria.

Table H.1. ACN Values for C-17 Aircraft.

Aircraft and Weight, pounds		
Subgrade Category	Rigid Pavement	Flexible Pavement
High Subgrade Strength (A)	36.7/R/A	35.0/F/A
Medium Subgrade Strength (B)	40.8/R/B	42.8/F/B
Low Subgrade Strength (C)	49.4/R/C	51.2/F/C
Ultra-Low Subgrade Strength (D)	63.7/R/D	58.0/F/D

Table H.2. Structural Index by Section.

Section ID	Primary Aircraft	ACN	PCN	Structural Index
A01B	C-130H	37/R/C	35/R/C/W/T	1.06
A04B	C-130H	37/R/C	36/R/C/W/T	1.03
A05B	C-130H	37/R/C	40/R/C/W/T	0.93
A07B	C-130H	34/R/B	41/R/B/W/T	0.83
A08B	C-130H	34/R/B	51/R/B/W/T	0.67
A10C	T-1A	6/R/C	15/R/C/W/T	0.40
A11C1	C-130H	34/R/B	69/R/B/W/T	0.49
A11C2	C-130H	34/R/B	69/R/B/W/T	0.49
A11C3	C-130H	34/R/B	69/R/B/W/T	0.49
A12	C-130H	ND	ND	ND
A13	C-130H	ND	ND	ND
O02C	C-130H	27/F/A	18/F/A/W/T	NA
O03C	C-130H	27/F/A	150/F/A/W/T	NA
O04C	C-130H	27/F/A	87/F/A/W/T	NA
O05C	C-130H	27/F/A	52/F/A/W/T	NA
O06C	C-130H	27/F/A	15/F/A/W/T	NA
O08C	C-130H	32/F/B	24/F/B/W/T	NA
R01A	C-130H	34/R/B	56/R/B/W/T	0.61
R02A	C-130H	27/F/A	125/F/A/W/T	0.22
R03A	C-130H	37/R/C	36/R/C/W/T	1.03
R04C1	C-130H	32/F/B	97/F/B/W/T	0.33
R04C2	C-130H	32/F/B	88/F/B/W/T	0.36
R05A1	C-130H	27/F/A	150/F/A/W/T	0.18
R05A2	C-130H	27/F/A	149/F/A/W/T	0.18
R06A1	C-130H	34/R/B	66/R/B/W/T	0.52
R06A2	C-130H	34/R/B	66/R/B/W/T	0.52
R07A	C-130H	34/R/B	71/R/B/W/T	0.48
R08C1	C-130H	34/R/B	87/R/B/W/T	0.39
R08C2	C-130H	34/R/B	80/R/B/W/T	0.43
R09C	C-130H	32/F/B	150/F/B/W/T	0.21
R10C	C-130H	27/F/A	150/F/A/W/T	0.18
R12A	C-130H	27/F/A	150/F/A/W/T	0.18
R13C	C-130H	32/F/B	150/F/B/W/T	0.21
R14C1	C-130H	32/F/B	97/F/B/W/T	0.33
R14C2	C-130H	32/F/B	88/F/B/W/T	0.36
T02B	C-130H	37/R/C	37/R/C/W/T	1.00
T03A	C-130H	37/R/C	62/R/C/W/T	0.60
ND = No Data, NA = Not Applicable				
(page 1 of 2)				

Table H.2. Structural Index by Section (concluded).

Section ID	Primary Aircraft	ACN	PCN	Structural Index
T04A	C-130H	32/F/B	80/F/B/W/T	0.40
T05C	C-130H	27/F/A	118/F/A/W/T	0.23
T06A	C-130H	27/F/A	150/F/A/W/T	0.18
T07B	C-130H	37/R/C	38/R/C/W/T	0.97
T08A	C-130H	32/F/B	97/F/B/W/T	0.33
T09A	C-130H	27/F/A	141/F/A/W/T	0.19
T11A	C-130H	34/R/B	71/R/B/W/T	0.48
T12C	C-130H	32/F/B	105/F/B/W/T	0.30
T13A	C-130H	37/R/C	101/R/C/W/T	0.37
T14B	C-130H	27/F/A	40/F/A/W/T	0.68
T15B	C-130H	34/R/B	76/R/B/W/T	0.45
T16A	C-130H	31/R/A	84/R/A/W/T	0.37
T17A	C-130H	37/R/C	37/R/C/W/T	1.00
T18A1	C-130H	37/R/C	58/R/C/W/T	0.64
T18A2	C-130H	37/R/C	60/R/C/W/T	0.62
T19A	C-130H	32/F/B	22/F/B/W/T	1.45
T29B	C-130H	34/R/B	71/R/B/W/T	0.48
T30B	C-130H	34/R/B	79/R/B/W/T	0.43
T31B	C-130H	34/R/B	53/R/B/W/T	0.64
T32B	C-130H	27/F/A	118/F/A/W/T	0.23
T33A	C-130H	37/R/C	39/R/C/W/T	0.95
T34A	C-130H	37/R/C	49/R/C/W/T	0.76
T35A	C-130H	ND	ND	ND
ND = No Data, NA = Not Applicable				
(page 2 of 2)				

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APPENDIX I

FRICTION CHARACTERISTIC RESULTS

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I.1 FRICTION INDEX

HQ AFCEC conducts tests to determine the friction characteristics of runways and compiles the results in a runway friction characteristics report for a given base. The friction values measured by approved friction testing equipment are then used to determine Friction Indices.

The most recent friction data available were from the January 2016 evaluation. Discussion of the general findings of the Friction evaluation and their implications.

Table I.1 list the average friction values applicable to a Grip Tester at 40 mph measured from the 15 end, as determined by AFCEC. Table I.2 list the average friction values applicable to a Grip Tester at 40 mph measured from the 189 end, as determined by AFCEC.

Table I.1. Runway 15/33 Friction Values.

Distance from 15 end, ft.	Average Friction Value	Friction Rating
0-500	0.77	Good
500-1,000	0.73	Good
1,000-1,500	0.76	Good
1,500-2,000	0.80	Good
2,000-2,500	0.85	Good
2,500-3,000	0.88	Good
3,000-3,500	0.87	Good
3,500-4,000	0.86	Good
4,000-4,500	0.85	Good
4,500-5,000	0.83	Good
5,000-5,500	0.80	Good
5,500-6,000	0.83	Good
6,000-6,500	0.73	Good
6,500-7,000	0.73	Good
7,000-7,500	0.78	Good
7,500-8,000	0.77	Good
8,000-8,500	0.82	Good
Overall	0.80	Good

Table I.2. Runway 009/189 Friction Values.

Distance from 189 end, ft.	Average Friction Value	Friction Rating
0-500	0.72	Good
500-1,000	0.75	Good
1,000-1,500	0.71	Good
1,500-2,000	0.86	Good
2,000-2,500	0.86	Good
2,500-3,000	0.83	Good
Overall	0.79	Good

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APPENDIX J

ENGINEERING ASSESSMENT SUMMARY TABLES

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J.1 ENGINEERING ASSESSMENT RATING DESCRIPTION

Table J.1 summarizes the criteria for determination of the EA. Table J.2 is a summary of the EA ratings by section. Ratings for the engineering assessment are further explained in Chapter 3.

Table J.1. EA Rating Criteria.

Engineering Assessment (EA) Criteria for Airfield Pavement				
Assessment/Rating Category	PCI Rating	FOD Potential Rating	Structural Index Rating	*Friction Index Rating (Runway Pavements Only)
Adequate	PCI $\geq$ 71 (GOOD)	FOD Potential Index $\leq$ 45 (GOOD)	ACN / PCN $<$ 1.10 (GOOD)	Mu Value $>$ 0.49 (GOOD)
Degraded	55 $<$ PCI $<$ 71 (FAIR)	45 $<$ FOD Potential Index $<$ 61 (FAIR)	1.10 $\leq$ ACN / PCN $\leq$ 1.40 (FAIR)	0.34 $\leq$ Mu Value $\leq$ 0.49 (FAIR)
Unsatisfactory	PCI $\leq$ 55 (POOR)	FOD Potential Index $\geq$ 61 (POOR)	ACN / PCN $>$ 1.40 (POOR)	Mu Value $<$ 0.34 (POOR)

*Friction Index Values calculated from Grip Tester Continuous Friction Measurement Equipment (CFME) testing conducted at 40 mph (65 km/h) only.

*For other testing equipment, use the values corresponding to the ratings of Good, Fair, and Poor in AFI 32-1041, Table 8.3.

Table J.2. EA Ratings by Section.

Branch Name	Section ID	PCI		FOD		Structural		Friction Rating		EA Rating
		Index	Rating	Potential	Rating	Index	Rating	Index	Rating	
Aero Club Apron	A10C	14	Poor	98	Poor	0.40	Good	NA	NA	Unsatisfactory
AGE Storage Lot	A12	65	Fair	35	Good	ND	Good	NA	NA	Degraded
AGE Storage Lot	A13	66	Fair	34	Good	ND	Good	NA	NA	Degraded
Engine Run-up Apron	A07B	91	Good	32	Good	0.83	Good	NA	NA	Adequate
Engine Run-up Apron	T14B	90	Good	10	Good	0.68	Good	NA	NA	Adequate
Maintenance Hangar Access Apron	A11C1	83	Good	36	Good	0.49	Good	NA	NA	Adequate
Maintenance Hangar Access Apron	A11C2	79	Good	39	Good	0.49	Good	NA	NA	Adequate
Maintenance Hangar Access Apron	A11C3	93	Good	28	Good	0.49	Good	NA	NA	Adequate
North Apron	A01B	84	Good	34	Good	1.06	Good	NA	NA	Adequate
North Apron	A04B	88	Good	18	Good	1.03	Good	NA	NA	Adequate
North Apron	A08B	99	Good	0	Good	0.67	Good	NA	NA	Adequate
North Apron	T02B	85	Good	23	Good	1.00	Good	NA	NA	Adequate
North Apron	T15B	93	Good	27	Good	0.45	Good	NA	NA	Adequate
North Apron	T30B	81	Good	24	Good	0.43	Good	NA	NA	Adequate
North Apron	T31B	94	Good	22	Good	0.64	Good	NA	NA	Adequate
West Apron	A05B	89	Good	15	Good	0.93	Good	NA	NA	Adequate
West Apron	T07B	88	Good	32	Good	0.97	Good	NA	NA	Adequate
West Apron	T29B	90	Good	19	Good	0.48	Good	NA	NA	Adequate

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Table J.2. EA Ratings by Section (continued).

Branch Name	Section ID	PCI		FOD		Structural		Friction Rating		EA Rating
		Index	Rating	Potential	Rating	Index	Rating	Index	Rating	
Overrun Runway 009/189	O02C	79	Good	20	Good	NA	NA	NA	NA	Adequate
Overrun Runway 009/189	O08C	83	Good	17	Good	NA	NA	NA	NA	Adequate
Overrun Runway 15/33	O03C	79	Good	21	Good	NA	NA	NA	NA	Adequate
Overrun Runway 15/33	O04C	79	Good	21	Good	NA	NA	NA	NA	Adequate
Overrun Runway 15/33	O05C	78	Good	22	Good	NA	NA	NA	NA	Adequate
Overrun Runway 15/33	O06C	77	Good	23	Good	NA	NA	NA	NA	Adequate
Runway 009/189 (Assault Strip)	R01A	93	Good	6	Good	0.61	Good	0.71	Good	Adequate
Runway 009/189 (Assault Strip)	R02A	75	Good	25	Good	0.22	Good	0.83	Good	Adequate
Runway 15/33	R03A	86	Good	13	Good	1.03	Good	NA	NA	Adequate
Runway 15/33	R04C1	99	Good	1	Good	0.33	Good	0.73	Good	Adequate
Runway 15/33	R04C2	84	Good	16	Good	0.36	Good	NA	NA	Adequate
Runway 15/33	R05A1	100	Good	0	Good	0.18	Good	0.77	Good	Adequate
Runway 15/33	R05A2	80	Good	20	Good	0.18	Good	NA	NA	Adequate
Runway 15/33	R06A1	99	Good	2	Good	0.52	Good	0.73	Good	Adequate
Runway 15/33	R06A2	97	Good	3	Good	0.52	Good	NA	NA	Adequate
Runway 15/33	R07A	99	Good	1	Good	0.48	Good	0.76	Good	Adequate
Runway 15/33	R08C1	93	Good	5	Good	0.39	Good	0.73	Good	Adequate
Runway 15/33	R08C2	98	Good	2	Good	0.43	Good	NA	NA	Adequate
Runway 15/33	R09C	90	Good	10	Good	0.21	Good	0.8	Good	Adequate
Runway 15/33	R10C	76	Good	24	Good	0.18	Good	0.85	Good	Adequate
Runway 15/33	R12A	100	Good	0	Good	0.18	Good	0.82	Good	Adequate
Runway 15/33	R13C	92	Good	8	Good	0.21	Good	0.73	Good	Adequate
Runway 15/33	R14C1	100	Good	0	Good	0.33	Good	NA	NA	Adequate
Runway 15/33	R14C2	100	Good	0	Good	0.36	Good	NA	NA	Adequate
Runway 009/189 Turn Around	T06A	59	Fair	41	Good	0.18	Good	NA	NA	Degraded
Runway 009/189 Turn Around	T35A	96	Good	4	Good	ND	Good	NA	NA	Adequate
Runway 15/33 Turn Around	T11A	99	Good	0	Good	0.48	Good	NA	NA	Adequate
Taxiway A	T04A	82	Good	18	Good	0.40	Good	NA	NA	Adequate
Taxiway A North	T16A	95	Good	3	Good	0.37	Good	NA	NA	Adequate
Taxiway A North	T18A1	87	Good	33	Good	0.64	Good	NA	NA	Adequate
Taxiway A North	T18A2	90	Good	37	Good	0.62	Good	NA	NA	Adequate
Taxiway A North	T34A	85	Good	21	Good	0.76	Good	NA	NA	Adequate
Taxiway A Northwest	T03A	89	Good	28	Good	0.60	Good	NA	NA	Adequate
Taxiway A Northwest	T13A	81	Good	19	Good	0.37	Good	NA	NA	Adequate
Taxiway A Northwest	T17A	82	Good	19	Good	1.00	Good	NA	NA	Adequate

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Table J.2. EA Ratings by Section (concluded).

Branch Name	Section ID	PCI		FOD		Structural		Friction Rating		EA Rating
		Index	Rating	Potential	Rating	Index	Rating	Index	Rating	
Taxiway A Northwest	T19A	79	Good	21	Good	1.45	Poor	NA	NA	Unsatisfactory
Taxiway A Northwest	T33A	85	Good	40	Good	0.95	Good	NA	NA	Adequate
Taxiway B	T05C	83	Good	17	Good	0.23	Good	NA	NA	Adequate
Taxiway B	T12C	89	Good	11	Good	0.30	Good	NA	NA	Adequate
Taxiway C	T08A	85	Good	15	Good	0.33	Good	NA	NA	Adequate
Taxiway D	T32B	89	Good	11	Good	0.23	Good	NA	NA	Adequate
Taxiway E	T09A	84	Good	16	Good	0.19	Good	NA	NA	Adequate
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APPENDIX K
ACTIVITY MANAGEMENT AND PREVENTATIVE MAINTENANCE
PLAN INPUTS

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K.1 PROBABILITY OF FAILURE

Probability of Failure (PoF) of a road or parking asset is determined using its respective condition, compliance, and functionality. Using the PoF allows installations to produce consistent, objective, data-driven, results to establish funding priorities. The PoF calculation, when referring to airfield pavements, is shown below.

$$PoF = (100 - PCI) \times 2$$

The Mission Dependence Index (MDI) determines an assets priority based on its category code. Table K.1 lists probability of failure information for airfield pavements surveyed at Maxwell AFB.

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Table K.1. Probability of Failure Information.

PROBABILITY OF FAILURE SUMMARY																	
Maxwell AFB																	
Branch Name	Section ID	Surface	Rank	Critical PCI	Current PCI	PCI Deterioration Rate	FOD Potential Deduction (DFOD)	Structural Index Deduction (DSI)	Friction Index Deduction (DSKID)	Adjusted PCI	2019 POF	Adjusted MDI	FY 2020 PCI BEFORE M&R	FY 2021 PCI BEFORE M&R	FY 2022 PCI BEFORE M&R	FY 2023 PCI BEFORE M&R	FY 2024 PCI BEFORE M&R
Aero Club Apron	A10C	PCC	S	55	14	0.97	9.8	0.0	0.0	4	100	86	14	13	12	11	10
AGE Storage Lot	A12	AC	S	55	65	2.94	3.5	10.0	0.0	52	97	86	63	61	58	55	52
	A13	AC	S	55	66	2.94	3.4	10.0	0.0	53	95	86	64	62	59	56	53
Engine Run-up Apron	A07B	PCC	S	55	91	0.97	3.2	0.0	0.0	88	24	86	91	90	89	88	87
	T14B	AC	S	55	90	2.86	1.0	0.0	0.0	89	22	86	89	86	83	80	77
Maintenance Hangar Access Apron	A11C1	PCC	S	55	83	0.97	3.6	0.0	0.0	79	41	86	83	82	81	80	79
	A11C2	PCC	S	55	79	0.97	3.9	0.0	0.0	75	50	86	79	78	77	76	75
	A11C3	PCC	S	55	93	0.97	2.8	0.0	0.0	90	20	86	93	92	91	90	89
North Apron	A01B	PCC	S	55	84	0.97	3.4	1.2	0.0	79	41	86	84	83	82	81	80
	A04B	PCC	P	70	88	0.33	1.8	0.6	0.0	86	29	95	88	88	87	87	87
	A08B	PCC	P	70	99	0.33	0.0	0.0	0.0	99	2	95	99	99	98	98	98
	T02B	PCC	P	70	85	0.33	2.3	0.0	0.0	83	35	95	85	85	84	84	84
	T15B	PCC	P	70	93	0.33	2.7	0.0	0.0	90	19	95	93	93	92	92	92
	T30B	PCC	P	70	81	0.33	2.4	0.0	0.0	79	43	95	81	81	80	80	80
West Apron	T31B	PCC	P	70	94	0.33	2.2	0.0	0.0	92	16	95	94	94	93	93	93
	A05B	PCC	P	70	89	0.33	1.5	0.0	0.0	88	25	95	89	89	88	88	88
	T07B	PCC	P	70	88	0.33	3.2	0.0	0.0	85	30	95	88	88	87	87	87
Overrun Runway 009/189	T29B	PCC	P	70	90	0.33	1.9	0.0	0.0	88	24	95	90	90	89	89	89
	O02C	AAC	T	55	79	2.94	2.0	10.0	0.0	67	66	63	77	75	72	69	66
Overrun Runway 15/33	O08C	AC	T	55	83	2.94	1.7	10.0	0.0	71	57	63	81	79	76	73	70
	O03C	AC	T	55	79	2.94	2.1	10.0	0.0	67	66	63	77	75	72	69	66
	O04C	AAC	T	55	79	2.94	2.1	10.0	0.0	67	66	63	77	75	72	69	66
	O05C	AAC	T	55	78	2.94	2.2	10.0	0.0	66	68	63	76	74	71	68	65
	O06C	AAC	T	55	77	2.94	2.3	10.0	0.0	65	71	63	75	73	70	67	64
Runway 009/189 (Assault Strip)	R01A	PCC	P	70	93	0.33	0.6	0.0	0.0	92	15	99	93	93	92	92	92
	R02A	APC	P	70	75	0.69	2.5	0.0	0.0	73	55	99	75	74	73	73	72
Runway 15/33	R03A	PCC	P	70	86	0.33	1.3	0.6	0.0	84	32	99	86	86	85	85	85
	R04C1	AAC	P	70	99	0.69	0.1	0.0	0.0	99	2	99	99	98	97	97	96
	R04C2	APC	P	70	84	0.69	1.6	0.0	0.0	82	35	99	84	83	82	82	81
	R05A1	AC	P	70	100	0.69	0.0	0.0	0.0	100	0	99	100	99	98	98	97
	R05A2	APC	P	70	80	0.69	2.0	0.0	0.0	78	44	99	80	79	78	78	77
	R06A1	PCC	P	70	99	0.33	0.2	0.0	0.0	99	2	99	99	99	98	98	98
	R06A2	PCC	P	70	97	0.33	0.3	0.0	0.0	97	7	99	97	97	96	96	96
	R07A	PCC	P	70	99	0.33	0.1	0.0	0.0	99	2	99	99	99	98	98	98
	R08C1	PCC	P	70	93	0.33	0.5	0.0	0.0	93	15	99	93	93	92	92	92
	R08C2	PCC	P	70	98	0.33	0.2	0.0	0.0	98	4	99	98	98	97	97	97
2019 Adjusted PCI and POF calculations are IAW with the AFI 32-1401																	
(Page 1 of 2)																	

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Table K.1. Probability of Failure Information (concluded).

PROBABILITY OF FAILURE SUMMARY																	
Maxwell AFB																	
Branch Name	Section ID	Surface	Rank	Critical PCI	Current PCI	PCI Deterioration Rate	FOD Potential Deduction (DFOD)	Structural Index Deduction (DSI)	Friction Index Deduction (DSKID)	Adjusted PCI	2019 POF	Adjusted MDI	FY 2020 PCI BEFORE M&R	FY 2021 PCI BEFORE M&R	FY 2022 PCI BEFORE M&R	FY 2023 PCI BEFORE M&R	FY 2024 PCI BEFORE M&R
Runway 15/33	R09C	APC	P	70	90	0.69	1.0	0.0	0.0	89	22	99	90	89	88	88	87
	R10C	AC	P	70	76	0.69	2.4	0.0	0.0	74	53	99	76	75	74	74	73
	R12A	APC	P	70	100	0.69	0.0	0.0	0.0	100	0	99	100	99	98	98	97
	R13C	APC	P	70	92	0.69	0.8	0.0	0.0	91	18	99	92	91	90	90	89
	R14C1	APC	P	70	100	0.69	0.0	0.0	0.0	100	0	99	100	99	98	98	97
	R14C2	APC	P	70	100	0.69	0.0	0.0	0.0	100	0	99	100	99	98	98	97
Runway 009/189 Turn Around	T06A	APC	P	70	59	0.69	4.1	0.0	0.0	55	90	95	59	58	57	57	56
	T35A	APC	P	70	96	0.69	0.4	10.0	0.0	86	29	95	96	95	94	94	93
Runway 15/33 Turn Around	T11A	PCC	P	70	99	0.33	0.0	0.0	0.0	99	2	95	99	99	98	98	98
Taxiway A	T04A	APC	P	70	82	0.69	1.8	0.0	0.0	80	40	95	82	81	80	80	79
Taxiway A North	T16A	PCC	P	70	95	0.33	0.3	0.0	0.0	95	11	95	95	95	94	94	94
	T18A1	PCC	P	70	87	0.33	3.3	0.0	0.0	84	33	95	87	87	86	86	86
	T18A2	PCC	P	70	90	0.33	3.7	0.0	0.0	86	27	95	90	90	89	89	89
	T34A	PCC	P	70	85	0.33	2.1	0.0	0.0	83	34	95	85	85	84	84	84
Taxiway A Northwest	T03A	PCC	P	70	89	0.33	2.8	0.0	0.0	86	28	95	89	89	88	88	88
	T13A	APC	P	70	81	0.69	1.9	0.0	0.0	79	42	95	81	80	79	79	78
	T17A	PCC	P	70	82	0.33	1.9	0.0	0.0	80	40	95	82	82	81	81	81
	T19A	AC	P	70	79	0.69	2.1	9.0	0.0	68	64	95	79	78	77	77	76
	T33A	PCC	P	70	85	0.33	4.0	0.0	0.0	81	38	95	85	85	84	84	84
Taxiway B	T05C	APC	S	55	83	2.86	1.7	0.0	0.0	81	37	86	82	79	76	73	70
	T12C	APC	S	55	89	2.86	1.1	0.0	0.0	88	24	86	88	85	82	79	76
Taxiway C	T08A	APC	P	70	85	0.69	1.5	0.0	0.0	84	33	95	85	84	83	83	82
Taxiway D	T32B	APC	S	55	89	2.86	1.1	0.0	0.0	88	24	86	88	85	82	79	76
Taxiway E	T09A	APC	P	70	84	0.69	1.6	0.0	0.0	82	35	95	84	83	82	82	81
2019 Adjusted PCI and POF calculations are IAW with the AFI 32-1401																	
(Page 2 of 2)																	

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APPENDIX L

BUDGET DETAILS

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L.1 PAVEMENT PRIORITIZATION AND CRITICAL PCI

Pavement priority is assigned based on the use of a pavement section. Primary pavements are mission-essential. These pavements include runways, parallel taxiways, main parking aprons, arm-disarm pads, alert aircraft pavements, and overruns (when used as a taxiway or take-off roll). Secondary pavements are also mission-essential, but only receive occasional use. These pavements including ladder taxiways, overflow parking areas, and overruns (when used to test aircraft arresting gear). Tertiary pavements include pavements used by towed or light aircraft, such as maintenance hangars, aero club parking, washracks, and overruns (when not used as a taxiway or to test aircraft arresting gear). Graphic representations of the pavement priority rankings for Maxwell AFB may be viewed on Figures A.13-16 in Appendix A.

The U.S. Air Force uses a concept of critical PCI (or pavement maintenance minimum values) to determine the suitability of a pavement section's condition. The critical PCI is the PCI value at which Major M&R on the section is required. Primary airfield pavements are assigned a critical PCI of 70, secondary and tertiary airfield pavements are assigned a critical value of 55.

L.2 MAINTENANCE AND REPAIR POLICIES

PAVER applies four maintenance policies, which define the purpose and type of M&R to be used. The maintenance policies are:

- Localized Stopgap M&R (Safety M&R)
- Localized Preventive M&R
- Global Preventive M&R
- Major M&R

Each of the four policies is discussed in the following paragraphs. Each policy assumes different maintenance goals for the network. Two of the policies, Stopgap and Major M&R, have conflicting goals, are mutually exclusive, and are never applied to a given section concurrently.

Stopgap (Safety) M&R is used when a pavement is below the critical PCI and there is not enough funding to perform Major M&R but M&R is required to keep the pavement operational. Repairs under a Stopgap maintenance policy are adequate to correct a given distress but are not the ideal solution. A Stopgap M&R policy is designed to keep a facility operational but not to improve the overall pavement condition. The pavements will continue to degrade as this maintenance is only applied until adequate funding is available for Major M&R.

Localized Preventive M&R is applied to pavements to slow the rate of deterioration. This policy is utilized when the section PCI is above the critical PCI. When applied correctly, Preventive M&R will extend the serviceable life of the pavement.

Global Preventive M&R is used to retard large-scale pavement degradation. Treatments recommended under this type of policy are applied to entire sections and facilities without regard for distresses present. Instead, treatments are recommended based on the elapsed time since the last construction or maintenance date. For example, a Global Preventive policy might prescribe a fog seal every 2 years and a thin overlay every 10 years following Major M&R on all AC pavements.

Major M&R addresses severe defects in the pavement system. These defects may result from material deficiencies, traffic overloads, environmental factors, or from pavement reaching the end of its life cycle. Major M&R is assigned to a pavement section when the condition is at or below the critical PCI with any distress types, or because the section is above the critical PCI but has structural related distresses. These defects usually require replacement of a part or all of a pavement section. Repairs typically consist of significant slab replacements in PCC pavements, overlays in AC pavements, or complete reconstruction.

L.3 PRIORITIZATION OF M&R POLICIES

PAVER performs a cost analysis to determine the best treatment to apply to a section of pavement in order to improve the condition of the entire network. The three main parameters for deciding where to spend money are:

- How much money is available?
- What is the priority of the section requiring M&R?
- What is the condition of the section requiring M&R?

PAVER determines how much money is available from a user-defined budget table, if one is provided, that specifies the money available for each year of the program plan. If a budget table is not provided, PAVER will perform an iterative procedure in order to fulfill a user-defined requirement, such as the elimination of existing backlog or condition stabilization, and maintain an even amount of funding throughout each year of the work plan. PAVER sorts the list of pavement sections according to PCI, and then allocates funds in the order listed in Table L.1. If a pavement section receives priority 4 or 5 funding, any priority 1, 2, or 3 funding also given to that section is reallocated. Allocation of funds in this manner ensures that all pavements receive at least Stopgap M&R before any pavement receives Major M&R. No single pavement section may receive both Localized and Major M&R funding; however, different sections within a branch in the same M&R work plan can receive both Localized and Major M&R funding. This effectively halts serious pavement degradation.

Table L.1. PAVER Funding Priority.

Funding Priority	Policy
1	Stopgap M&R
2	Localized Preventive M&R
(Page 1 of 2)	

Table L.1. PAVER Funding Priority (concluded).

Funding Priority	Policy
3	Global Preventive M&R
4	Major M&R for pavement above the critical PCI having structural defects
5	Major M&R for pavement below the critical PCI
(Page 2 of 2)	

L.4 PRIORITIZATION OF PAVEMENT

Table L.2, below, provides the pavement priority matrix used by PAVER. A runway will have a higher branch use priority than a taxiway which will have a higher branch priority than an apron. Therefore, if a taxiway and an apron have the same pavement rank and PCI, the taxiway will have a higher priority in the work plan.

Table L.2. Pavement Priority Matrix.

Branch Use Priority	Section Rank		
	Primary	Secondary	Tertiary
High	1	3	6
Medium	2	5	8
Low	4	7	9

L.5 M&R POLICY TABLES

Tables L.3 through L.6 present the M&R policy tables that were applied to Maxwell AFB.

Table L.3. AC Stopgap M&R Policies.

Distress Types	Severity Levels	Options for Repair
41. Alligator Cracking (Series of interconnecting cracks caused by fatigue failure under repeated traffic loading.)	High	Full-depth patch
43. Block Cracking (Interconnecting cracks that divide the pavement into approximately rectangular pieces.)	High	Seal cracks
44. Corrugation (Closely spaced ridges and valleys occurring at fairly regular intervals along the pavement.)	High	Full-depth patch
45. Depression (Localized surface areas having elevations slightly lower than those of the surrounding pavement.)	High	Full-depth patch
(Page 1 of 2)		

Table L.3. AC Stopgap M&R Policies (concluded).

Distress Types	Severity Levels	Options for Repair
47. Joint Reflection Cracking (Cracks from AC pavement over a PCC pavement, longitudinal & transverse PCC slabs.)	High	Seal cracks
48. Longitudinal & Transverse Cracking (Cracks parallel to the pavement's centerline or extending across the pavement at right angles.)	High	Seal cracks
50. Patching and Utility Cut Patching (A defect, no matter how well it is performing.)	High	Full-depth patch
53. Rutting (A surface depression in the wheel path.)	High	Full-depth patch
54. Shoving (Pavement growth of PCC that shoves the AC pavements, causing them to swell and crack.)	High	Full-depth patch
55. Slippage Cracking (Cracks by sliding/deforming of pavement surface.)	No Severity	Partial-depth patch
56. Swelling (An upward bulge in the pavement's surface.)	High	Full-depth patch
(Page 2 of 2)		

Table L.4. PCC Stopgap M&R Policies.

Distress Type & Description	Severity Levels	Options for Repair
61. Blowup (Occurs at a transverse crack or joint that is not wide enough to permit expansion.)	Medium	Full-depth patch
	High	Slab replacement
62. Corner Break (Crack that intersects the joints at a distance less than or equal to one half of the slab length.)	High	Full-depth patch
63. Longitudinal, Transverse & Diagonal Cracks (Cracks dividing the slab into two or three pieces.)	High	Full-depth patch
64. Durability ("D") Cracks (Cracking caused by the concrete's inability to withstand environmental factors.)	High	Slab replacement
66. Patches (small) (Where original pavement has been removed and replaced by a filler material.)	High	Partial-depth patch
67. Patches (large-over 5 ft²) and Utility Cut (Where original pavement has been replaced or because of placement of underground utilities.)	High	Full-depth patch
70. Scaling (A network of shallow, fine, or hairline cracks that extend only through the upper surface of the slab.)	High	Full-depth patch
71. Settlement or Faults (Difference of elevation at a joint or crack caused by upheaval or consolidation.)	High	Slab replacement
(Page 1 of 2)		

Table L.4. PCC Stopgap M&R Policies (concluded).

Distress Type & Description	Severity Levels	Options for Repair
72. Shattered Slab, Intersecting Cracks (Cracks that break the slab into four or more pieces due to overloading or inadequate support, or both.)	High	Slab replacement
74. Spalls (transverse and longitudinal joint) (Breakdown of the slab edges within 2 ft of the side of the joint.)	High	Partial-depth patch
75. Spalls (Corner) (Raveling or breakdown of the slab within approximately 2 ft of the corner.)	High	Partial-depth patch
76. ASR (Chemical reaction between alkalis and certain reactive silica minerals which form a gel)	High	Slab Replacement
(Page 2 of 2)		

Table L.5. AC Preventive M&R Policies.

Distress Types	Severity Levels	Options for Repair
41. Alligator Cracking (Series of interconnecting cracks caused by fatigue failure under repeated traffic loading.)	Low	Crack seal
	Medium	Full-depth patch
	High	Full-depth patch
42. Bleeding (A film of bituminous material on the surface that creates a shiny, glass like, reflecting surface.)	No Severity	Do nothing
43. Block Cracking (Interconnecting cracks that divide the pavement into approximately rectangular pieces.)	Low	Do nothing
	Medium	Seal cracks
	High	Partial-depth patch
44. Corrugation (Closely spaced ridges and valleys occurring at fairly regular intervals along the pavement.)	Low	Do nothing
	Medium	Grinding
	High	Partial-depth patch
45. Depression (Localized surface areas having elevations slightly lower than those of the surrounding pavement.)	Low	Do nothing
	Medium	Full-depth patch
	High	Full-depth patch
46. Jet Blast (Erosion causes darkened areas on the pavement surface where bituminous binder has been burned.)	No Severity	Do nothing
47. Joint Reflection Cracking (Cracks from AC pavement over a PCC pavement, longitudinal & transverse PCC slabs.)	Low	Do nothing
	Medium	Seal cracks
	High	Seal cracks
48. Longitudinal & Transverse Cracking (Cracks parallel to the pavement's centerline or extending across the pavement at right angles.)	Low	Do nothing
	Medium	Seal cracks
	High	Seal cracks
49. Oil Spillage (Deterioration or softening of the pavement surface caused by spillage of oil, fuel, or other solvents.)	No Severity	Partial-depth patch
50. Patching and Utility Cut Patching (A defect, no matter how well it is performing.)	Low	Do nothing
	Medium	Full-depth patch
	High	Full-depth patch
51. Polished Aggregate (Polishing of aggregate caused by repeated traffic.)	No Severity	Do nothing
(Page 1 of 2)		

Table L.5. AC Preventive M&R Policies (concluded).

Distress Types	Severity Levels	Options for Repair
52. Raveling (The dislodging of coarse aggregate particles from the pavement surface.)	Low	Do nothing
	Medium	Do nothing
	High	Partial-depth patch
53. Rutting (A surface depression in the wheel path.)	Low	Do nothing
	Medium	Full-depth patch
	High	Full-depth patch
54. Shoving (Pavement growth of PCC that shoves the AC pavements, causing them to swell and crack.)	Low	Do nothing
	Medium	Grinding
	High	Full-depth patch
55. Slippage Cracking (Cracks by sliding/deforming of pavement surface.)	No Severity	Partial-depth patch
56. Swelling (An upward bulge in the pavement's surface.)	Low	Do nothing
	Medium	Full-depth patch
	High	Full-depth patch
(Page 2 of 2)		

Table L.6. PCC Preventive M&R Policies.

Distress Type & Description	Severity Levels	Options for Repair
61. Blowup (Occurs at a transverse crack or joint that is not wide enough to permit expansion.)	Low	Partial-depth patch
	Medium	Full-depth patch
	High	Slab replacement
62. Corner Break (Crack that intersects the joints at a distance less than or equal to one half of the slab length.)	Low	Do nothing
	Medium	Full-depth patch
	High	Full-depth patch
63. Longitudinal, Transverse & Diagonal Cracks (Cracks dividing the slab into two or three pieces.)	Low	Do nothing
	Medium	Seal cracks
	High	Partial-depth patch
64. Durability ("D") Cracks (Cracking caused by the concrete's inability to withstand environmental factors.)	Low	Do nothing
	Medium	Full-depth patch
	High	Slab replacement
65. Joint Seal Damage (Any condition that allows water infiltration or enables soil/rocks to accumulate in the joints.)	Low	Do nothing
	Medium	Seal joints
	High	Seal joints
66. Patches (small) (Where original pavement has been removed and replaced by a filler material.)	Low	Do nothing
	Medium	Partial-depth patch
	High	Partial-depth patch
67. Patches (large-over 5 ft²) and Utility Cut (Where original pavement has been replaced or because of placement of underground utilities.)	Low	Do nothing
	Medium	Partial-depth patch
	High	Full-depth patch
68. Popouts (Small piece of pavement that breaks loose from the surface due to freeze-thaw/expansive aggregates.)	No Severity	Do nothing
69. Pumping (Ejection of material by water through joints or cracks caused by deflection of the slab under load.)	No Severity	Under-seal
70. Scaling (A network of shallow, fine, or hairline cracks that extend only through the upper surface of the slab.)	Low	Do nothing
	Medium	Partial-depth patch
	High	Slab replacement
(Page 1 of 2)		

Table L.6. PCC Preventive M&R Policies (concluded).

Distress Type & Description	Severity Levels	Options for Repair
71. Settlement or Faults (Difference of elevation at a joint or crack caused by upheaval or consolidation.)	Low	Do nothing
	Medium	Slab grinding
	High	Slab grinding
72. Shattered Slab, Intersecting Cracks (Cracks that break the slab into four or more pieces due to overloading or inadequate support, or both.)	Low	Seal cracks
	Medium	Slab replacement
	High	Slab replacement
73. Shrinkage Cracks (Hairline cracks that are usually only a few feet long and do not extend across the entire slab.)	No Severity	Do nothing
74. Spalls (transverse and longitudinal joint) (Breakdown of the slab edges within 2 ft of the side of the joint.)	Low	Seal cracks
	Medium	Partial-depth patch
	High	Partial-depth patch
75. Spalls (Corner) (Raveling or breakdown of the slab within approximately 2 ft of the corner.)	Low	Seal cracks
	Medium	Partial-depth patch
	High	Partial-depth patch
76. ASR (Chemical reaction between alkalis and certain reactive silica minerals which form a gel)	Low	Do nothing
	Medium	Partial-depth patch
	High	Slab replacement

(Page 2 of 2)

L.6 M&R COSTS

M&R costs are determined by one of two methods:

- Based on the method of treatment (consequence analysis of localized distress maintenance, used only for first year after survey)
- Based on the PCI of the pavement (cost-by-condition, used to develop ten-year work plans)

L.6.1 Cost-By-Work Type M&R Costs

Distress details are known only for the first year of Stopgap M&R and Localized Preventive M&R; therefore, PAVER estimates the cost of repairs based on the selected method of treatment only for the first year of Stopgap or Localized Preventive M&R. The quantity and type of treatment are determined by matching each distress to an entry in the treatment method table (Stopgap and Preventive M&R). The quantity of the treatment is then multiplied by its unit cost to estimate M&R costs. The Stopgap and Localized Preventive M&R policies assume the same type of maintenance actions occur in each policy.

The work items in Tables L.3 through L.6 have an associated unit cost. Costs were determined from the *2019 RSMeans Heavy Construction Cost Data*. Data entries in this publication are listed by activities (e.g., Sawcut Pavement, 3" Depth), not M&R treatments (Full-depth AC Patch, Reconstruct AC). Costs for M&R treatments were developed by listing activities involved in a particular M&R treatment and determining costs for each activity. Costs for each activity were adjusted to provide costs in

consistent units, typically square or linear feet. This often involved assumptions regarding a particular repair, e.g., patch size or depth of repair. The costs of all component activities of an M&R treatment were summed to determine the cost of the treatment.

Table L.7 illustrates the method used to calculate unit costs for an M&R item, in this case full-depth PCC patching, from the *2019 RSMeans Heavy Construction Cost Data*. The field "crew" is taken from the *2019 RSMeans Heavy Construction Cost Data*. "Bare" and "Loaded" are in-house and contract costs, respectively. The conversion field is used to convert each item to units of square feet, and often involves making assumptions about repair sizes. For example, a large patch was assumed to be 5 ft x 5 ft in size in the corner of a slab. A 10-ft saw cut would be required, resulting in a conversion ratio of 10 ft/25 ft², or 0.4 ft/ft².

Table L.7. Sample M&R Item Unit Cost Calculation (full-depth PCC patching).

Item	Description	Crew	Bare (\$/unit)	Loaded (\$/unit)	Unit	Conversion
03 81 13.50-0400	Sawcut PCC, 3"	B89	\$1.35	\$1.80	ft	0.4 ft/ft ²
03 81 13.50-0420	Sawcut PCC, per add'l inch (x12)	B89	\$0.45	\$0.60	ft	0.4 in-ft/ft ²
02 41 13.17-5700	Remove w/air tools, <6" (x2.5)	B39	\$1.61	\$2.34	ft ²	1 ft ² /ft ²
32 13 13.23-0750	Tie bars, grouted	B23	\$65.55	\$81.50	ea.	0.64 ea/ft ²
32 13 13.24-0800	PCC pavement, 15" (x2) for small area	B26	\$136.24	\$176.22	yd ²	0.111 yd ² /ft ²
Total	Full Depth PCC Patch		\$48.68	\$61.61	ft²	

Unit costs were developed for each repair method that might be appropriate for Maxwell AFB using the methodology described in the above example. These costs, used for the consequence analysis of pavements in Section 4.4.2, are shown in Table L.8, along with the U.S. Air Force Average Cost per Unit. The code shown in the left-most column is the PAVER code for each repair method. The PAVER analyses require that every treatment method listed below have an associated unit cost. The total M&R cost is estimated by multiplying the area of the section undergoing M&R by the unit cost of M&R.

Table L.8. M&R Cost by Treatment.

Code	Name	Amount	With Cost Factor	Work Unit
PA-AS	Patching - AC Shallow	\$12.86	\$10.96	ft ²
PA-AD	Patching - AC Deep	\$15.19	\$12.94	ft ²
CS-AC	Crack Sealing - AC	\$1.19	\$1.01	ft
CS-PC	Crack Sealing - PCC	\$4.20	\$3.58	ft
PA-PP	Patching - PCC Partial Depth	\$10.50	\$8.95	ft ²
PA-PF	Patching - PCC Full Depth	\$61.61	\$52.49	ft ²
Area cost factor at Maxwell AFB (found in <i>2019 RSMeans</i>) = 0.852				
(Page 1 of 2)				

Table L.8. M&R Cost by Treatment (concluded).

Code	Name	Amount	With Cost Factor	Work Unit
SL-PC	Slab Replacement - PCC	\$22.52	\$19.19	ft ²
JS-LC	Joint Seal (Localized)	\$3.50	\$2.98	ft
UN-PC	Undersealing - PCC	\$3.48	\$2.96	ft
GR-PP	Grinding (Localized)	\$5.06	\$4.31	ft
MO-AC	Mill and Overlay 2" AC	\$1.55	\$1.32	ft ²
OL-AC	Overlay 2" AC	\$1.24	\$1.06	ft ²
CR-AC	Total reconstruct AC	\$8.38	\$7.14	ft ²
CR-PCC	Total reconstruct PCC	\$31.56	\$26.89	ft ²
Area cost factor at Maxwell AFB (found in 2019 RSMeans) = 0.852				
(Page 2 of 2)				

L.6.2 Cost-By-Condition M&R Work Plans and Costs

Each ten-year work plan (budget scenario) is developed with a different objective and based on the condition of the pavement. For example, the backlog elimination scenario is intended to eliminate the Major M&R backlog and often significantly improve the condition of the pavement; whereas, the Stopgap M&R scenario is designed to maintain an operable pavement network but not improve its overall condition. To support the overall objective of each budget scenario, different proportions of each work type policy are included in each scenario. Specific M&R actions are not given in the budget scenario; each scenario gives an approximate amount to be spent on each policy type.

While PAVER can predict the PCI of a pavement at some point in the future, it does not predict which distresses will be present on that pavement; therefore, the cost of all M&R policies for multi-year work plans is based on the projected PCI of the pavement and the cost-versus-PCI relationship. PAVER estimates the cost of M&R using a cost-by-condition relationship and the projected PCI. The total M&R cost is calculated by multiplying the area of the section undergoing M&R by the unit cost of M&R obtained from the cost-by-condition relationship. Figure L.1 shows that when PCI decreases, applicable M&R treatments become more involved, and therefore more costly. This concept is important because pavements often remain in serviceable condition for a long period of time, followed by a period of rapid deterioration.

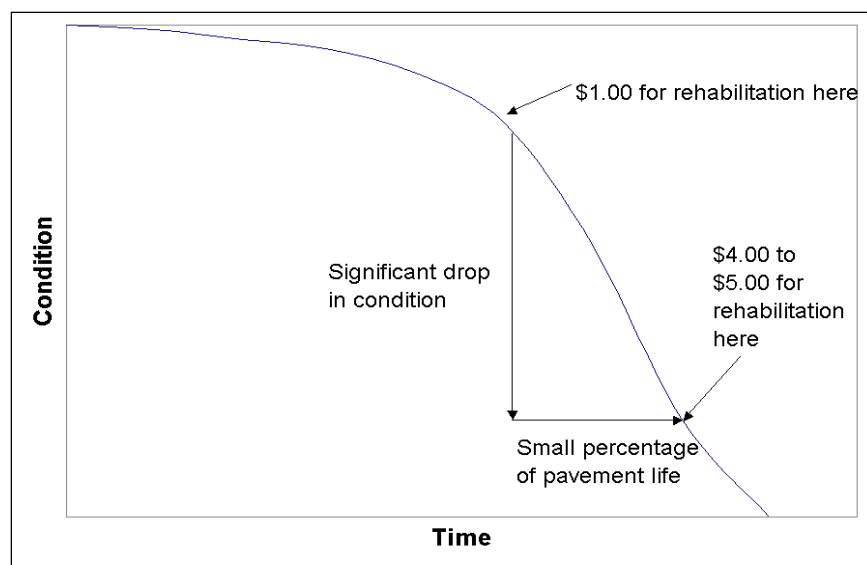


Figure L.1. Significant Drop in Pavement Condition Near End of Pavement Life Cycle.

Major M&R cost-by-condition curves were developed assuming that pavements with a PCI below 40 would be completely reconstructed. PCC pavements with a PCI between 40 and 90 would undergo selective slab replacement (20% slab replacement at 50 PCI, 10% at 60 PCI, and 5% at 70-90 PCI) as well as spot repair of some areas. AC pavements with a PCI between 40 and 70 were assumed to undergo mill and overlay with spot repair of some areas. Costs were determined by adding the cost of a mill and overlay to the Preventive M&R cost-by-condition to account for any spot repair. AC pavements with structural defects but a PCI above 70 were assumed to undergo an overlay with Preventive M&R of some areas. Costs were determined by adding the cost of an overlay to the Preventive M&R cost-by-condition curve to account for any spot repair.

Stopgap and Preventive M&R cost-by-condition curves were obtained by using PAVER to calculate the cost per area of local M&R for approximately 9,000 previously inspected sections at Air Force bases in Europe, Asia, and North America. The use of sections from numerous bases allowed the cost-by-condition relationship to be based on an average distress condition using a wide range of distress types and pavement conditions. The sections were combined into a single PAVER database. The curves were developed by applying updated and installation specific costs for specific repairs to a consequence analysis of localized distress maintenance policies. Once the consequence of localized distress maintenance was performed, the unit cost to repair each section at its respective PCI was plotted and a best-fit polynomial regression was used to determine the unit cost to repair a pavement section at its respective PCI value.

Separate cost curves were developed for AC and PCC pavements. The unit costs, with the localized cost factor applied, for AC pavement are shown in Figure L.2 while the unit costs, with the localized cost factor applied, for PCC pavement are shown in Figure L.3. The points shown on these cost-by-condition figures were input into PAVER. The AC and PCC Preventive M&R cost/ft² values were adjusted in PAVER to create more

accurate multi-year budget scenarios. Typically, an AC pavement section may require Preventive M&R every three years; therefore, a cost factor of 1/3 was applied. Typically, a PCC pavement section may require Preventive M&R every five years; therefore, a cost factor 1/5 was applied.

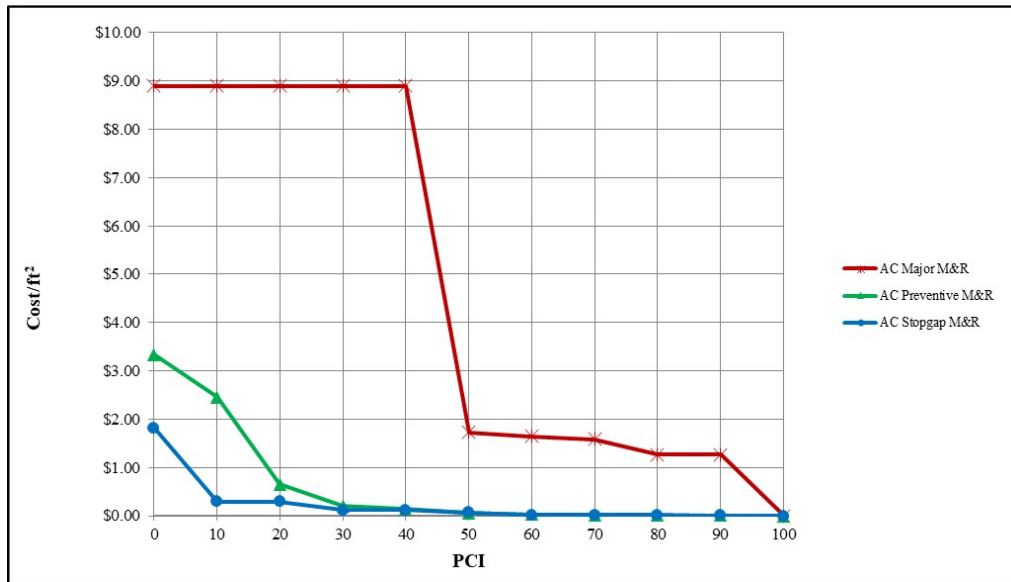


Figure L.2. AC M&R Cost-by-Condition.

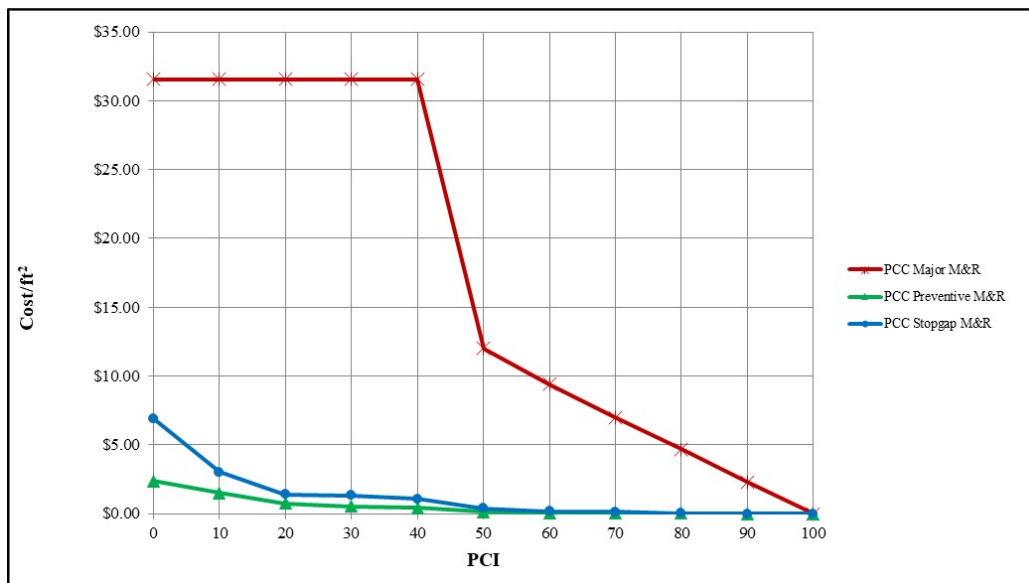


Figure L.3. PCC M&R Cost-by-Condition.

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APPENDIX M

FAMILY MODELS FOR CONDITION DETERIORATION PREDICTION

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M.1 FAMILY CURVES

PAVER uses PCI survey results, pavement history data, user-defined maintenance policies, and user-defined cost guidelines for planning and budgeting future M&R needs. PAVER creates a "family curve" or "deterioration curve" that models how pavements at the airfield have deteriorated in the past. The deterioration curve is based on the PCI rating and age of each section of pavement.

Family curves are typically developed using pavement use (apron, overrun, runway, or taxiway), pavement surface type (AC or PCC), pavement rank (primary, secondary, or tertiary), and traffic area as defined in UFC 3-260-02:

- Type A Traffic Areas: Pavements that receive channelized traffic and full design weight of the aircraft.
- Type B Traffic Areas: Pavements where traffic is nearly uniform over the full width of the pavement facility, but receives the full design weight of the aircraft.
- Type C Traffic Areas: Pavements in which the volume of traffic is low or applied weight of the operating aircraft is less than the design weight.
- Type D Traffic Areas: Pavements in which the traffic volume is extremely low, and/or the applied weight of operating aircraft is considerably lower than the design weight.

PAVER plots the PCI value of each section against its age within each respective pavement family. Once the pavement sections have been plotted, PAVER calculates the deterioration rate of the pavement family. The deterioration rates for this report are expressed linearly using the best-fit curve. This process is repeated for each pavement family in the network. Future pavement conditions are predicted by interpolation and extrapolation of the family curve, as shown conceptually in Figure M.1.

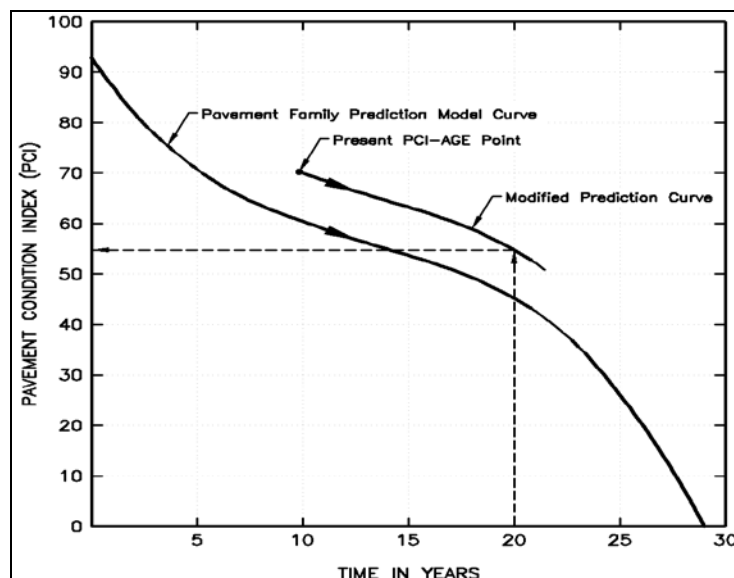


Figure M.1. Predicting Pavement Condition.

Five family curves were developed for the airfield pavements at Maxwell AFB and are shown in Figures M.2 through M.6. These family curves were used by PAVER to estimate the future condition of airfield sections. Each section of pavement has been assigned to its

respective family and any future degradation can be applied for the development of work plans. The deterioration rate is represented by the green line. The red lines are the outlier boundaries and are calculated to a 95% confidence interval. Outliers are defined as data that fall more than one standard deviation away from the deterioration curve. PAVER ignores the outliers, creating a more accurate curve.

It should be acknowledged that the deterioration rates for Maxwell AFB account for localized M&R, which increases the PCI of a pavement section but does not constitute a new construction date for the deterioration models. If no localized M&R had been performed on any pavements, then it is fair to assume that the average deterioration rates would be much greater.

The primary AC pavement at Maxwell AFB is currently deteriorating at a rate of 0.69 PCI points per year. The area-weighted average age of these pavements was nine years old at the time of the inspection. Sections in this family are predicted to deteriorate below their critical PCI value at forty-three years of age.

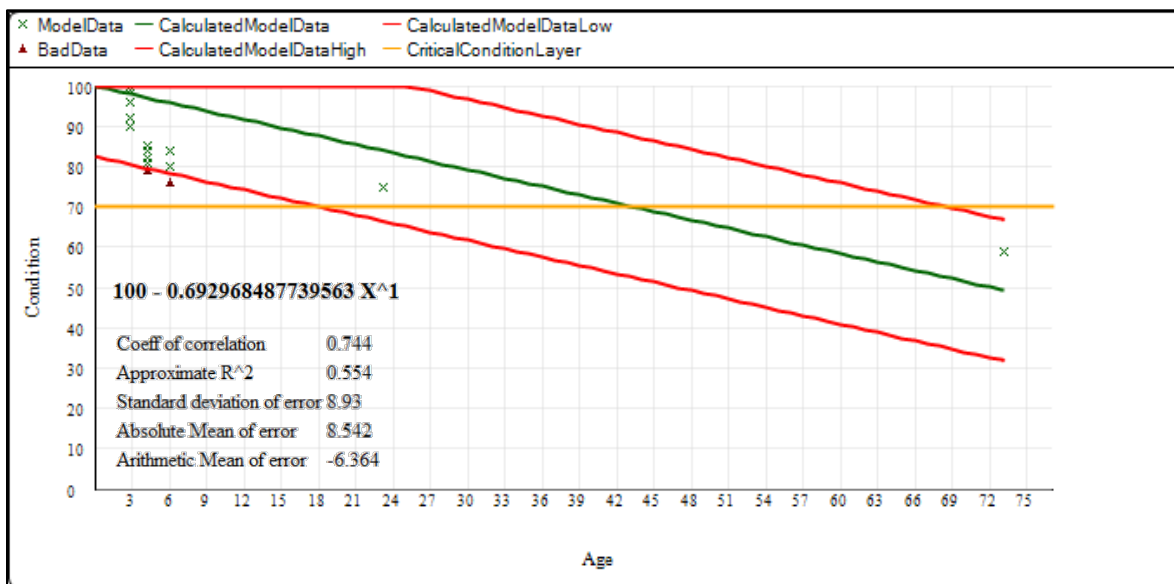


Figure M.2. Family Plot for Primary AC Pavement (0.69 PCI points per year).

The secondary AC pavement at Maxwell AFB is currently deteriorating at a rate of 2.85 PCI points per year. The area-weighted average age of these pavements was seven years old at the time of the inspection. Sections in this family are predicted to deteriorate below their critical PCI value at sixteen years of age.

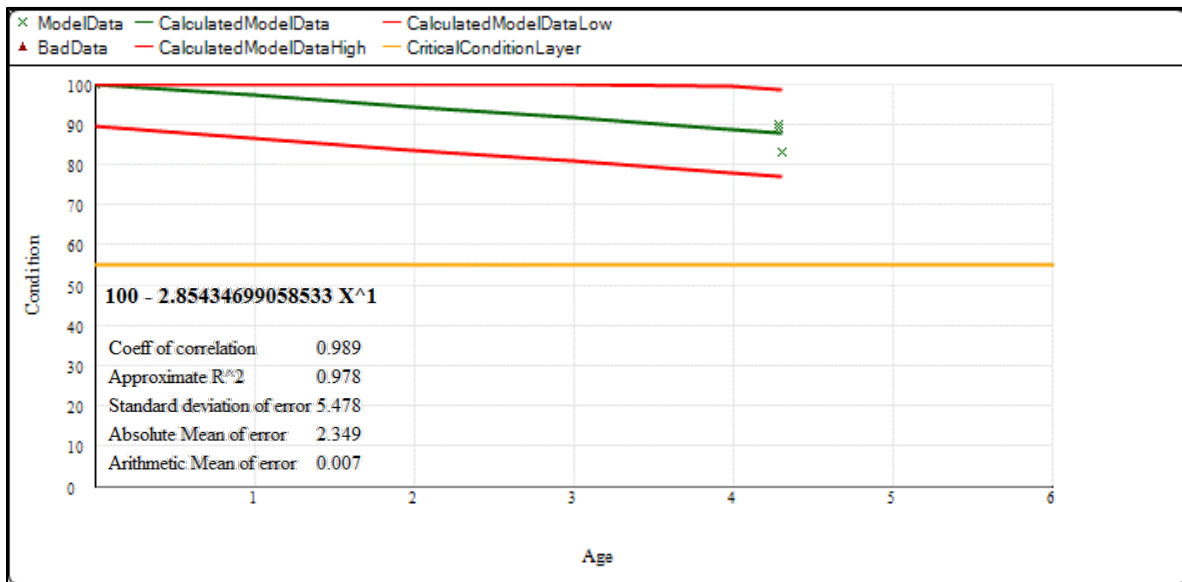


Figure M.3. Family Plot for Secondary AC Pavement (2.85 PCI points per year).

The tertiary AC pavement at Maxwell AFB is currently deteriorating at a rate of 2.93 PCI points per year. The area-weighted average age of these pavements was six years old at the time of the inspection. Sections in this family are predicted to deteriorate below their critical PCI value at fifteen years of age.

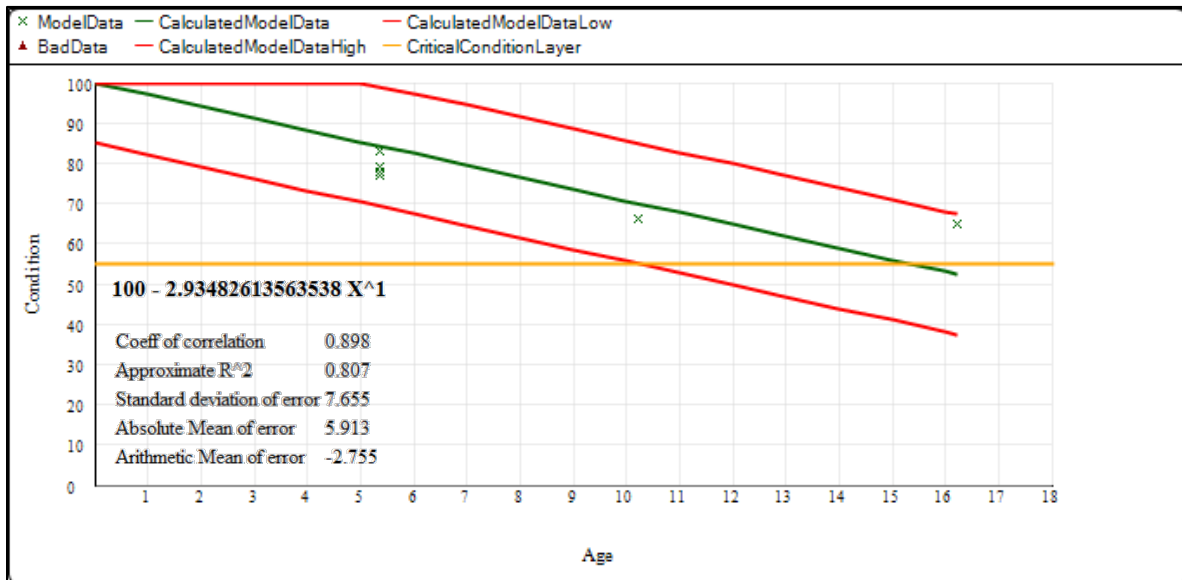


Figure M.4. Family Plot for Tertiary AC Pavement (2.93 PCI points per year).

The primary PCC pavement at Maxwell AFB is currently deteriorating at a rate of 0.33 PCI points per year. The area-weighted average age of these pavements was twenty-seven years old at the time of the inspection. Sections in this family are predicted to deteriorate below their critical PCI value at ninety-one years of age.

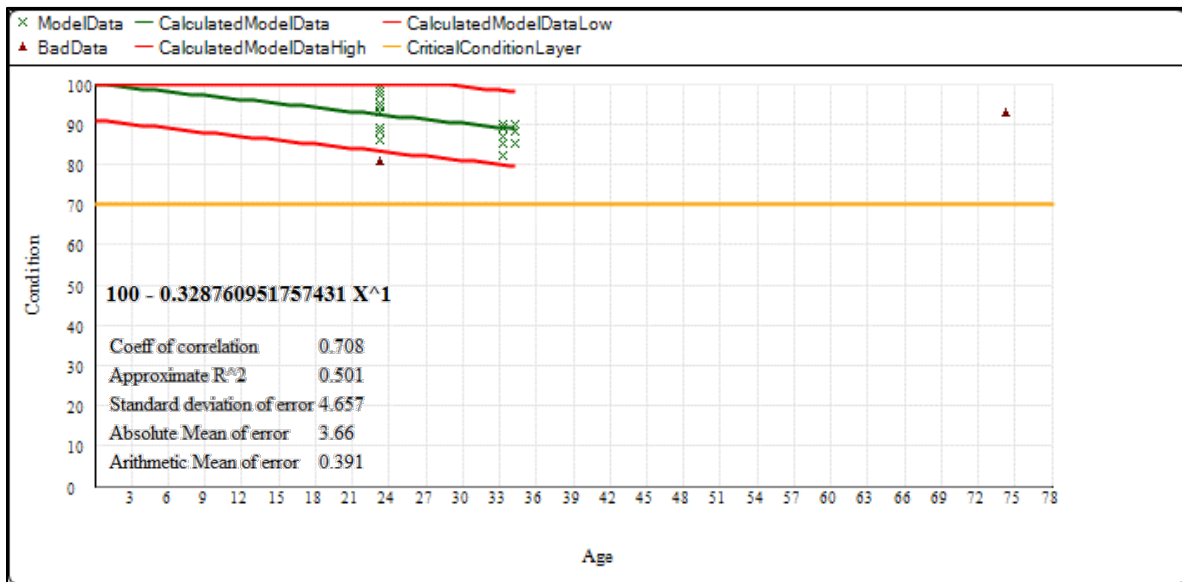


Figure M.5. Family Plot for Primary PCC Airfield Pavement (0.33 PCI points per year).

The secondary and tertiary PCC pavement at Maxwell AFB is currently deteriorating at a rate of 0.97 PCI points per year. The area-weighted average age of these pavements was twenty-eight years old at the time of the inspection. Sections in this family are predicted to deteriorate below their critical PCI value at forty-six years of age.

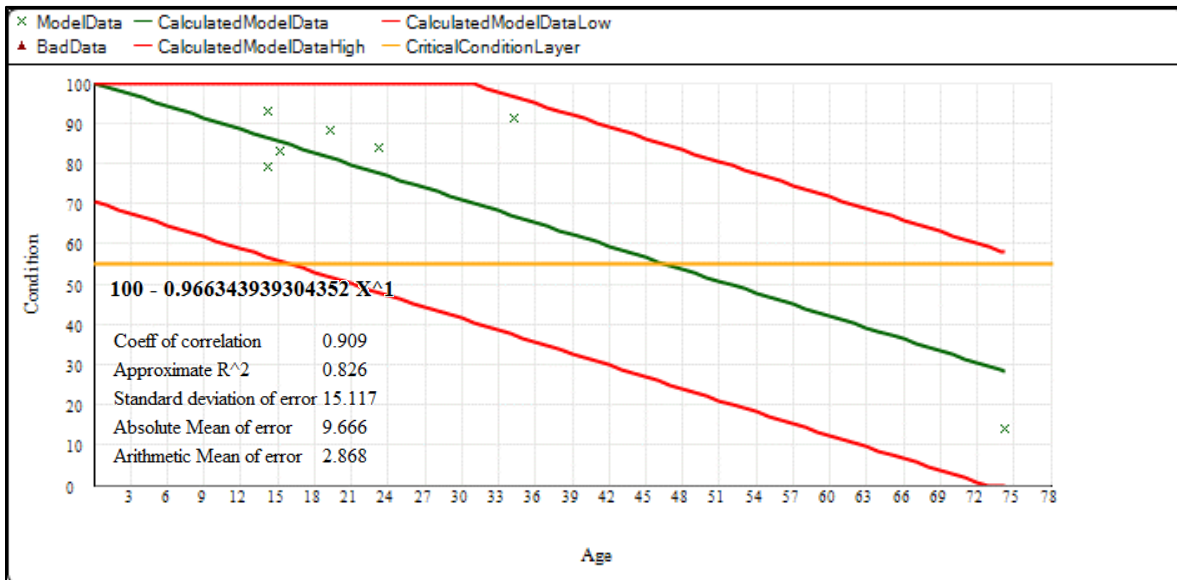


Figure M.6. Family Plot for Secondary and Tertiary PCC Airfield Pavement (0.97 PCI points per year).